Date

电电

第一讲 电路定律

一. 基本电量

1. 电流 $i=\frac{dq}{dt}$

(1) 真实电流信号可分解为直流分量与交流分量之和

$i(t)=I_{DC}+i_{AC}(t)$

$I_{DC}=\overline{i(t)}=\lim_{T\to+\infty}\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} i(t)dt$ { if 有周期, T=周期; else $T\to\infty$ }

$i_{AC}(t)=i(t)-\overline{i(t)}$ (验证, $\overline{i_{AC}(t)}=0$)

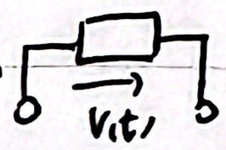
(2) KCL: $\sum i_k=0$

2. 电动势 $\mathcal{E}=\frac{dE}{dq}$ $1V=1J/C$

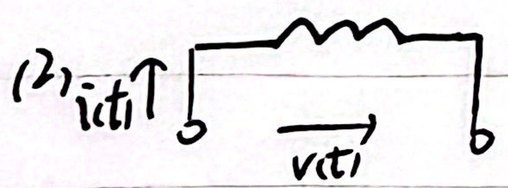
(1) $V_{AB}=\frac{W_{AB}}{q}$ $V_{AB}=\varphi_A-\varphi_B$

(2) KVL: $\sum v_k=0$

3. 功率 $P=\frac{dW}{dt}=\frac{dW}{dq}\cdot\frac{dq}{dt}=v\cdot i$

(1) 

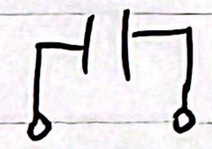
$v(t)=R\cdot i(t)$ $p(t)=v(t)\,i(t)=i^2(t)\cdot R\geq 0$ $\therefore\int p\geq 0$ $\therefore$ 耗能元件 无源

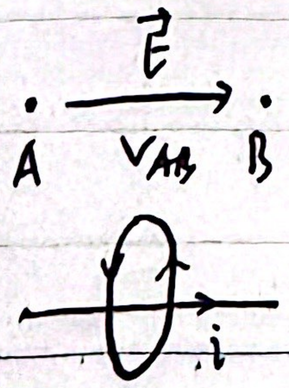
(2) 

$v(t)=L\frac{di(t)}{dt}$ $p(t)=i(t)L\frac{di(t)}{dt}$

$\int_{t_1}^{t_2}p(t)\,dt=\frac{1}{2}L[i^2(t_2)-i^2(t_1)]$ { if $i(t)=0$, 无源; else. 有源 }

无损元件定义储能为 $\frac{1}{2}Li^2(t)$

(3)  与(2)对偶

4. 

$V_{AB}=\int_A^B\vec{E}\cdot d\vec{l}$

$i=\oint_l\vec{H}\cdot d\vec{l}$

电压是电场的空间离散化抽象

电流是磁场的空间离散化抽象

KOKUYO

No.

Date

三. "数"

1. SI 词头

10^{-15} f 飞

10^{6} M 兆

10^{9} G 吉

10^{12} T 太

2. dB 数

P: $10 \lg \frac{P}{\text{单位功率}}$

V: $20 \lg \frac{V}{\cdot\cdot}$

I: $20 \lg \frac{I}{\cdot\cdot}$

P相对: $10 \lg \frac{P_0}{P_i}$

V相对: $20 \lg \frac{V_0}{V_i}$

$\lg 2 = 0.3$

3. 复数

$S = A + jB = A_m e^{j\theta} = A_m \angle \theta$

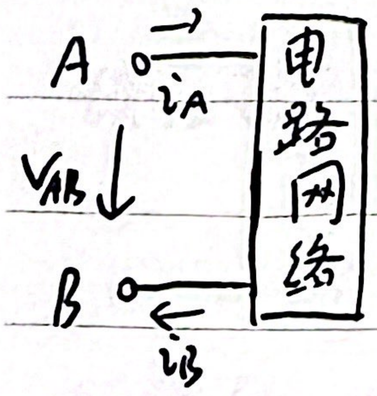
乘 $e^{j\alpha} \Leftrightarrow \alpha$ 度相移 $(\alpha = \frac{\pi}{2},\ e^{j\alpha} = j)$

KOKUYO

第二讲 电阻与电源

一. 基本概念

1. 端口与网络



(一般规定 v, i 的参考方向如图)

对从一个端点流入电流 i, 从另一个端点流出电流大小与之相等.

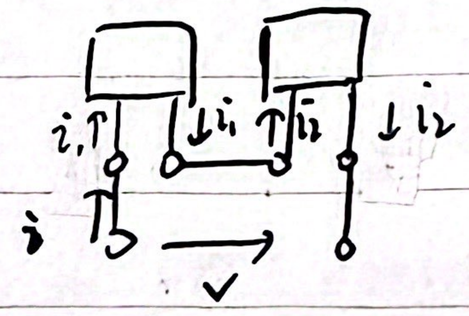
则这两个端点可构成一个端口

一个端口就是一条支路

电磁场问题可抽象为电路问题的准静态条件: $d_{AB} \ll \lambda$

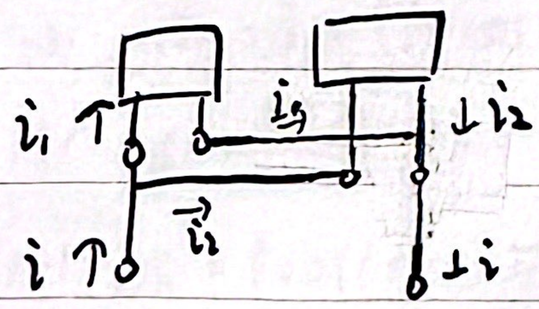
↳ n个端口需n个方程来描述欧姆定律

2. 端口连接关系

(1)  串联

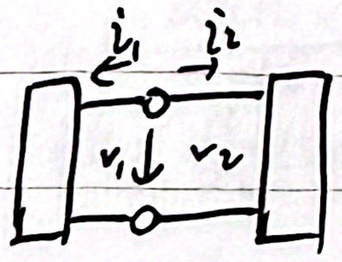
$i = i_1 = i_2$

$v = v_1 + v_2$

(2)  并联

$i = i_1 + i_2$

$v = v_1 = v_2$

(3)  对接

$i_1 = -i_2$

$v_1 = v_2$

可视为串联后总端口短路 / 并开 o?

3. 有/无源

有/无能力 (见第一讲)

电容. 电感一般视为无源. 但有初始 v/i 的容/感 有源.

Date

3. 线性/非线性

定义法：$v(t)=f(i(t))$ 看f线性 or 非线性

⇓

$$f(\alpha i_1(t)+\beta i_2(t))=\alpha f(i_1(t))+\beta f(i_2(t))=\alpha v_1(t)+\beta v_2(t)$$

（i-v为过原点的直线）.

线性电阻：R与v.i无关

容：C

感：L

非线性：e.g. 二极管

4. 时变/时不变

时变 ⇔ v-i关系式中，系数参量随t变化，且该变化与u.i的变化独立

（对course不一定 v-i, v_{in}-v_{out} 等也可）（不要理解系数，理解为除v.i之外的所有参量比较好）

how 理解独立？v,i不也是v(t),i(t)吗？

答：参考多元微积分中"显含"的概念，顾名思义，"独立"大概可理解为不直接相关

例：(1) $v(t)=300i(t)$ 线.不

(2) $v(t)=300(1+\cos 2\pi t)i(t)$ 线.变

(3) $v(t)=300i^3(t)$ 非.不

(4) $v(t)=300(1+\cos 2\pi t)i^3(t)$ 非.变

(5) $i(t)=\frac{d(C(t)v(t))}{dt}$ 线.变

(6) $i(t)=\frac{d}{dt}(v(t)+v^2(t))$ 非.不

(1) $f(i)=300i$ $f(\alpha i_1+\beta i_2)=\alpha f(i_1)+\beta f(i_2)$ ∴线

300与t无关.不.

(2) $f(i)=300(1+\cos 2\pi t)i$ $f(\alpha i_1+\beta i_2)=300(1+\cos 2\pi t)(\alpha i_1+\beta i_2)=\alpha f(i_1)+\beta f(i_2)$ ∴线

$300(1+\cos 2\pi t)$ 与t有关且与v.i独立 ∴变

(3) $f(i)=300i^3$ $f(\alpha i_1+\beta i_2)=300(\alpha i_1+\beta i_2)^3\neq\alpha 300i_1^3+\beta 300i_2^3$ ∴非

300与t无关 ∴不.

(4) 略 (5) $f(v)=\frac{d}{dt}(Cv)$ $f(\alpha v_1+\beta v_2)=\frac{d}{dt}C(\alpha v_1+\beta v_2)=\alpha\frac{d}{dt}Cv_1+\beta\frac{d}{dt}Cv_2$ ∴线

$C(t)$ 与t有关且与v.i独立 ∴变

(6) 略

KOKUYO

Date

例1：(1) $v_{out}(t)=10+\int_0^t i_{in}(t)dt$ ……不，有源

(2) $v_{out}=\begin{cases} v_{in} & v_c>0 \\ 0 & v_c<0 \end{cases}$

(开关)

(3) $v_{out}=\begin{cases} v_s & v_{in}>0 \\ 0 & v_{in}<0 \end{cases}$

(4) $v_{out}=\begin{cases} v_{in} & v_{in}>0 \\ 0 & v_{in}\le 0 \end{cases}$

(1) 按定义，$f(i)=10+\int_0^t i$ 非线 （计忽略10，按不严格定义，可以为线）

参量与t无关∴不

当 $i_{in}(t)=0$ 时，$\frac{v_{out}^2}{R}>0$ ∴有源 （in fact，可视作电容：$i=\frac{dCv}{dt}\Rightarrow v=10+\frac{1}{C}\int i$ ∴10为一有源）

(2) $v_{out}=S_w(v_c)\cdot v_{in}$ $\left(S_w(x)=\begin{cases}1 & x>0\\ 0 & x<0\end{cases}\right)$

① $f(v_{in})=S_w(v_c)\,v_{in}$ $f(\alpha v_{in1}+\beta v_{in2})=\alpha f(v_{in1})+\beta f(v_{in2})$ ∴线

② $S_w(v_c)$ 与t有关且与 v_{in}、v_{out} 独立 ∴变 （这里v_c由默认文流了）

(3) $v_{out}=S_w(v_{in})\cdot v_s$

① $f(v_{in})=S_w(v_{in})\cdot v_s$ $f(\alpha v_{in1}+\beta v_{in2})=S_w(\alpha v_{in1}+\beta v_{in2})v_s\neq \alpha S_w(v_{in1})v_s+\beta S_w(v_{in2})v_s$ ∴非

② i) 若 v_s 交流，则 v_s 与t有关且与 v_{in}、v_{out} 独立 ∴变

ii) 若 v_s 直流，则 v_s 与t无关 ∴不

(4) $v_{out}=S_w(v_{in})\,v_{in}$

① $f(v_{in})=S_w(v_{in})\cdot v_{in}$ $f(\alpha v_{in1}+\beta v_{in2})\neq \alpha f(v_{in1})+\beta f(v_{in2})$ ∴非

② 无"与t有关且与 v_{in} 独立"的参数 ∴不

时变/不变：in fact，一个更直接的理解是：不同时刻给激励，响应变不变 （e.g. 不同时刻画出的v-i图象是否相同）

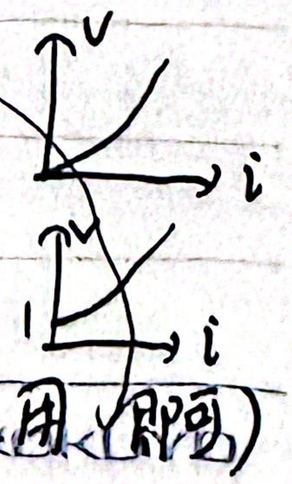
变：时变
不变：时不变

(验证略)

随便出个题：$v(t)=t+i^2(t)$ （量纲不调）

不同t，v-i关系变] 比如：t=0时，v-i关系为 $v=i^2$

t=1时，v-i关系为 $v=1+i^2$



∴时变

（但，计求导，$\frac{dv}{dt}=1+2i$ ，仅为不好解释 ∴不管求导（即阿）→不含t

Date

第三讲 电路定理

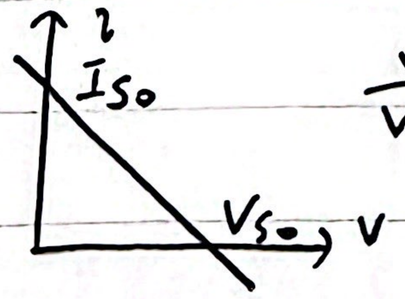
一. 电阻与电源

1. 理想电压源

2. 理想电流源

3. 带内阻电源

(伏曲为直)



$$\frac{V}{V_{S0}} + \frac{i}{I_{S0}} = 1$$

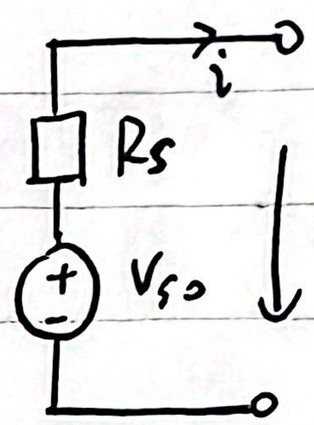
① 流控形式

$$V = V_{S0} - i\left(\frac{V_{S0}}{I_{S0}}\right) = R_S$$

② 压控形式

$$i = I_{S0} - V\cdot\left(\frac{I_{S0}}{V_{S0}}\right) = G_S$$

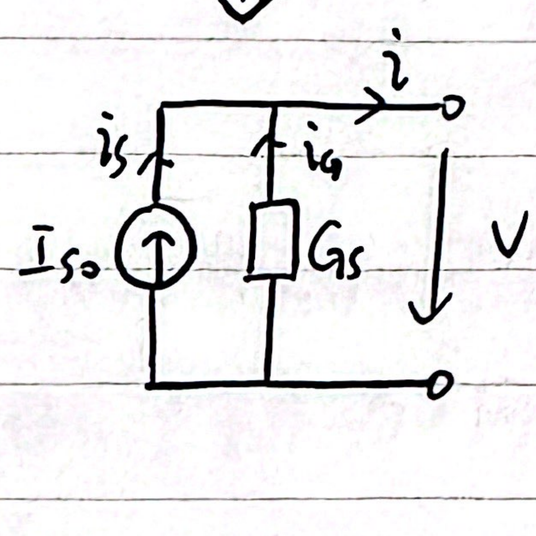
⇕



戴维南电压源

$V = V_{S0} - iR_S$

⇕



诺顿电流源

$i = I_{S0} - V\cdot G_S$

驱动负载功率分析：

$$P_L = i^2 R_L = \left(\frac{V_S}{R_S + R_L}\right)^2 R_L$$

$$\therefore P_{L有效} \le \frac{V_{Srms}^2}{4R_S}$$

当且仅当 $R_L = R_S$ 时"="

$$P_L = i^2 R_L = \left(\frac{G_L}{G_L + G_S} I_{S0}\right)^2 R_L$$

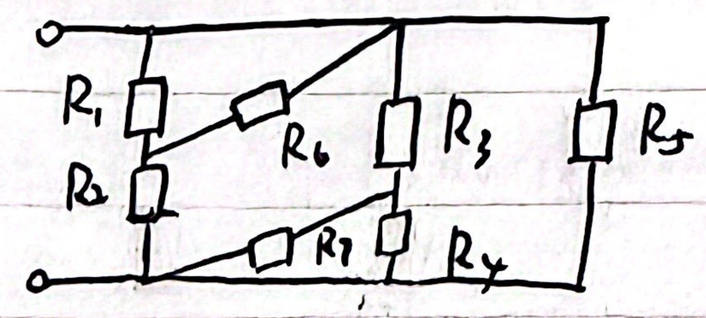
$$\therefore P_{L有效} \le \frac{1}{4} I_{Srms}^2 \cdot G_S$$

当且仅当 $G_L = G_S$ 时"="

⇒ 对偶原理 (略)

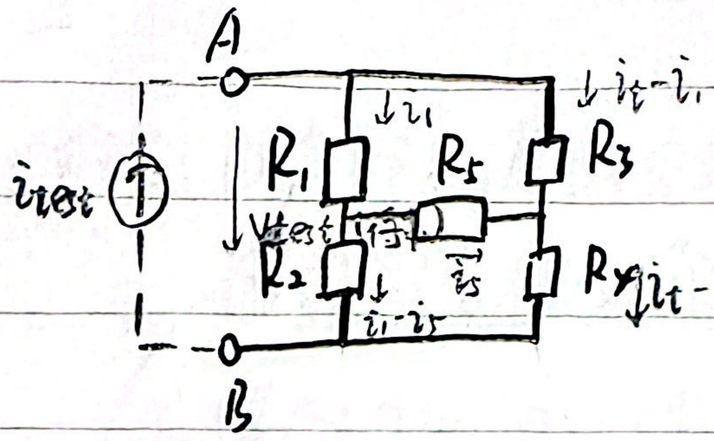
二. 单端口等效

例1 (简单串并联)



$$\Leftrightarrow (R_1 // R_6 + R_2) // (R_3 + R_4 // R_7) // R_5$$

例2：（非简单串并联）



用加压求流/加流求压的方法求等效

如 加流求压：（解出 $\frac{V_{test}}{i_{test}}$ 即可）

$$\begin{cases} i_1 R_1 + (i_1 - i_5) R_2 = V_t \\ -i_1 R_1 + (i_t - i_1) R_3 - i_5 R_5 = 0 \\ (i_1 - i_5) R_2 + i_5 R_5 + (i_t - i_1 + i_5) R_4 = 0 \end{cases}$$

3个方程，4个未知数（i_1, i_5, i_t, V_t），可解出 i_1, i_5 和 $\frac{V_t}{i_t}$

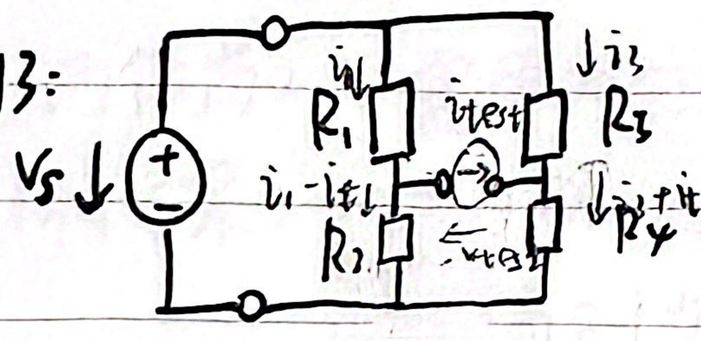
$\Rightarrow R_{AB} = \frac{V_t}{i_t} = \frac{\cdots}{\cdots}$　　$i_1 = \cdots$

$i_5 = \frac{R_2R_3 - R_1R_4}{\cdots} i_t$　　$\Rightarrow$ 电桥平衡时，$i_5 = 0$，$V_5 = 0$
（$R_2R_3 - R_1R_4 = 0$）　　↓ 开路　　↓ 短路

$\Rightarrow$ 2.1 替代定理：

对于一个电路网络可分割为两个单端口网络的对接关系，且这无其他作用关系，那么对某单端口网络的端口V或I确定，则可用理压V或I源替代

例3：



（含源端口的电路等效）

加流求压　　$\begin{cases} \cdots \\ \cdots \\ \cdots \end{cases} \Rightarrow \begin{cases} i_1 = \cdots \\ i_3 = \cdots \\ V_t = \cdots \end{cases}$

$$V_t = \left(\frac{R_1}{R_1+R_2} - \frac{R_3}{R_3+R_4}\right) V_S + \left(\frac{R_1R_2}{R_1+R_2} + \frac{R_3R_4}{R_3+R_4}\right) i_t$$

（两个项的线性叠加）

$\Rightarrow$ 2.2 叠加定理：

对于线性电路，计算多个独立源同时激励，则总响应 = 分响应之和

分响应：单独一个源起作用，将其他源置零（恒压源短路，恒流源开路）

由此可说明

进一步分析：$V_t = \underbrace{\left(\frac{R_1}{R_1+R_2} - \frac{R_3}{R_3+R_4}\right)V_S}_{V} + \underbrace{\left(\frac{R_1R_2}{R_1+R_2} + \frac{R_3R_4}{R_3+R_4}\right)i_t}_{R\cdot i}$ ⇒可等效成戴~电压

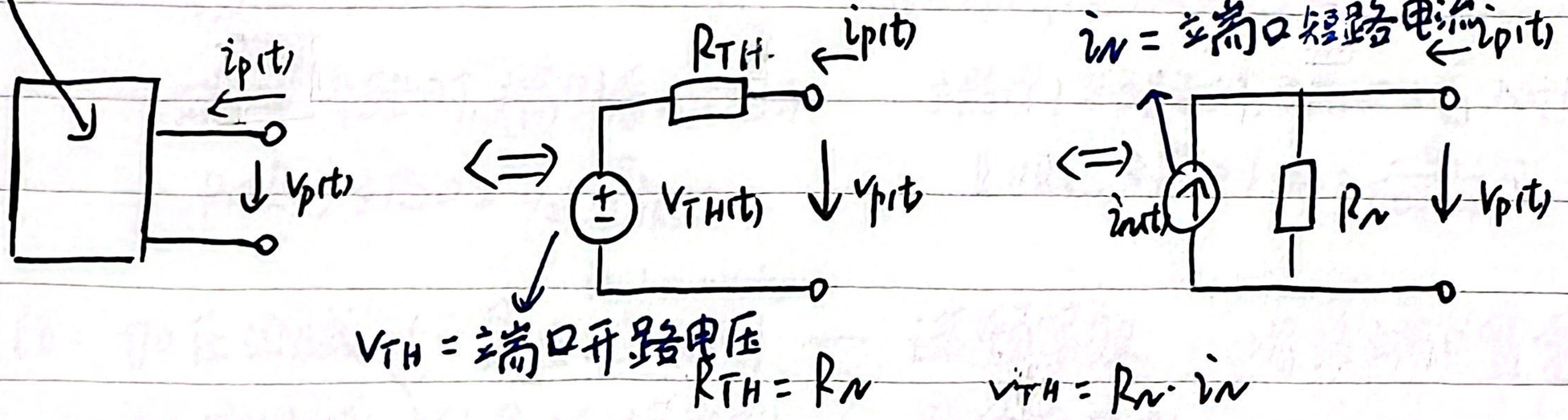
结论：①内部无源的单端口线性电阻网络 ⇔ 线性电阻

②内部含独立源的单端口线性电阻网络 ⇔ 戴维南电压源 ⇔ 诺顿电流源
（加流求压）（加压求流）

求等效电压源：端口开路

求等效内阻：内部独立源置零（但受控源不置零）

（这些是叠加定理应用的必然结论）



（可用此分析例3，略）

三. 二端口网络等效

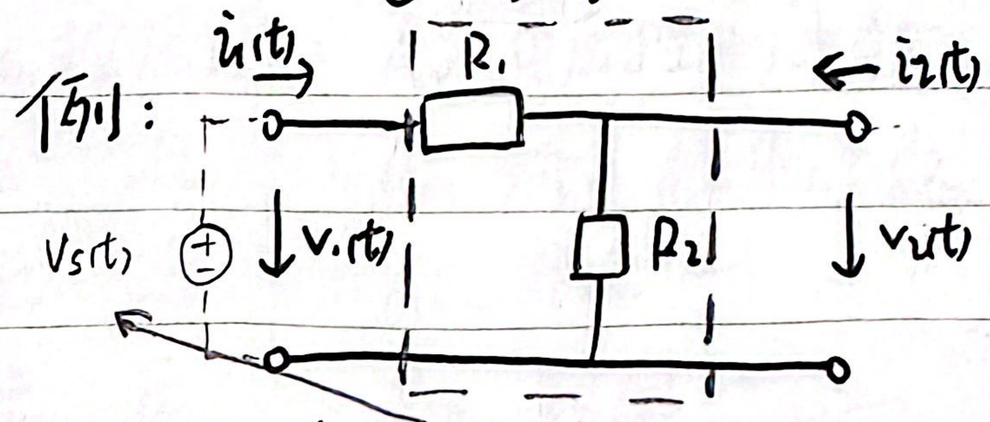
同样可以通过加压求流/加流求压获得二端口网络的等效电路

∵两个端口 ∴需要两个测试源

6种方案：①②③④：$C_2^1 \times C_2^1 = 4$

⑤：端口1加压.加流 ⑥：端口2～

例：



e.g. step1：端口1加压$v_S(t)$ ⇒ $\begin{cases} v_1(t) = v_S(t) & i_1(t) = \dfrac{v_S(t)}{R_1+R_2} \\ v_2(t) = \dfrac{R_2}{R_1+R_2}v_S(t) = \dfrac{R_2}{R_1+R_2}v_1(t) & i_2(t) = 0 \end{cases}$

端口2开路

step2：端口2加流$i_S(t)$ ⇒ $\begin{cases} i_2(t) = i_S(t) & v_2(t) = \dfrac{i_2(t)}{G_1+G_2} \text{（并联）} \\ i_1(t) = -\dfrac{G_1}{G_1+G_2}i_2(t) & v_1(t) = 0 \end{cases}$

端口1短路

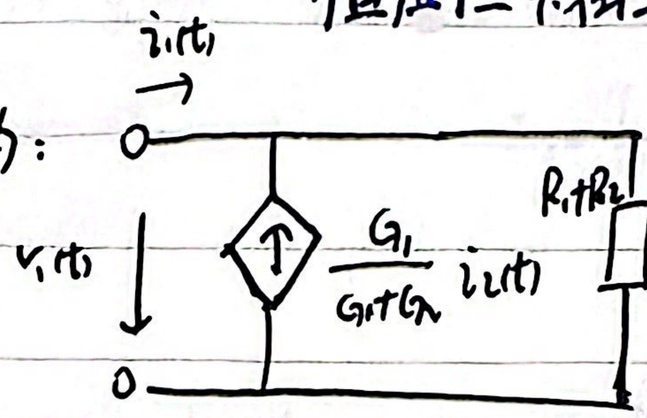
恒流但不独立，是流控流源

用叠加定理，得 $\begin{cases} V_1(t) = V_S(t) + 0 = V_S(t) \\ V_2(t) = \frac{R_2}{R_1+R_2} V_1(t) + \frac{1}{G_1+G_2} i_2(t) \end{cases}$ $\quad i_1(t) = \frac{1}{R_1+R_2} V_1(t) - \frac{G_1}{G_1+G_2} i_2(t)$

恒压但不独立，是压控压源

$i_2(t) = i_S(t) + 0 = i_S(t)$

∴ 等效为：

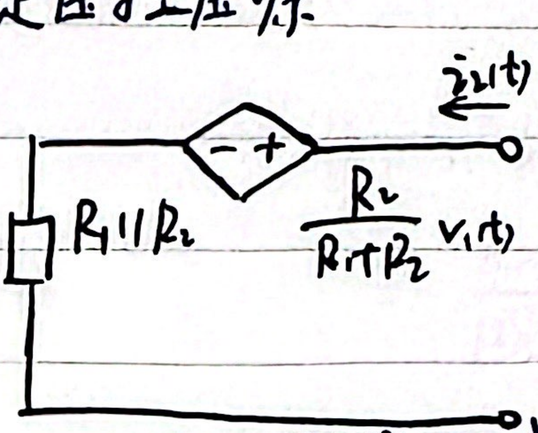


端口1加压测试，获得

诺顿等效

端口2开路时，端口1看入电阻为 R_1+R_2，端口2 $V_2 = \frac{R_2}{R_1+R_2} V_1$

$i = I_{S0} - V \cdot G_S$



端口2加流测试，获得

戴维南等效

端口1短路时，端口2看入电阻为 $R_1 \| R_2$，端口1 $i_1 = \frac{-G_1}{G_1+G_2} i_2$

总结：加压测试（∴是压控形式）— 诺顿等效　将该端口置零时，短路开

加流测试（∴是流控形式）— 戴维南等效

$V = V_{S0} - iR_S$

实际上，写出1侧端口的 i，1侧端口的 V，即可获得等效电路

以 i_1, i_2 作自变量

V_1, i_2 作自变量

如上例，$i_1 = \frac{1}{R_1+R_2} V_1 - \frac{G_1}{G_1+G_2} i_2$ ①　$V_2 = \frac{R_2}{R_1+R_2} V_1 + \frac{1}{G_1+G_2} i_2$ ②　即可

找准因变量很重要！∵因变量与戴/诺等效直接对应

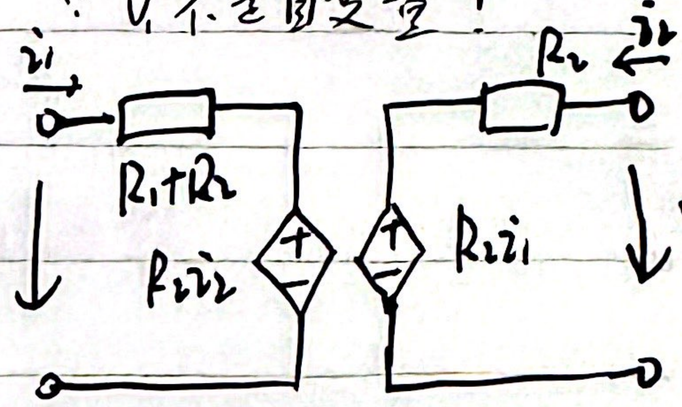
问：how 获得戴-戴等效？

答：可直接从①②出发，以 V_1, V_2 作因变量，i_1, i_2 为自变量，即得戴戴等效

由①，$V_1 = (R_1+R_y) i_1 + R_2 i_2$　$V_2 = R_2 i_1 + R_2 i_2$

不能再写 $V_2 = \frac{R_2}{R_1+R_2} V_1 + \frac{1}{G_1+G_2} i_2$ ∵ V_1 不是自变量！

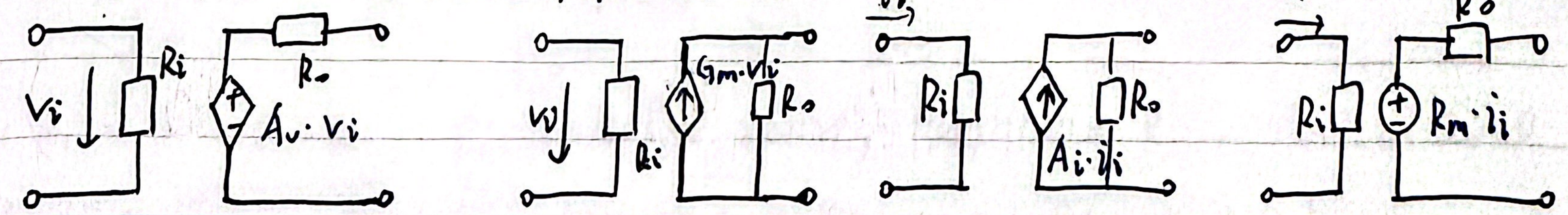
图：



4种理想受控源

- 压控压源：A_v 电压控制系数
- 压　流：G_m 跨导控制系数
- 流　流：A_i 电流～
- 流　压：R_m 跨阻～

⇒ 以受控源为核心的4种基本单向放大器



VCVS 电压放大器　VCCS 跨导放大器　CCCS 电流放大器　CCVS 跨阻放大器

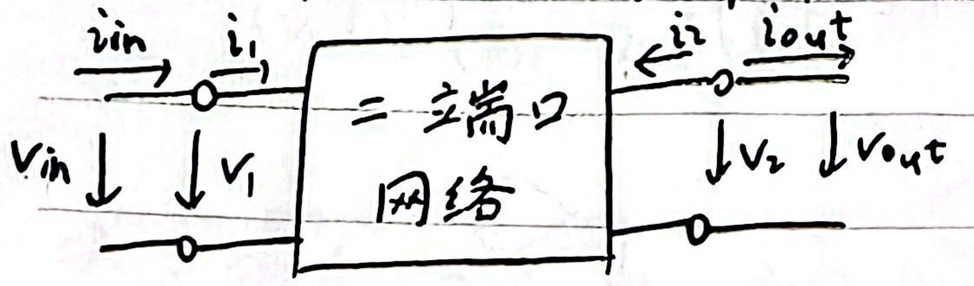
SKUYO

Date

第四讲 方程列写

一. 二端口网络方程的六种表述方式

1. 二端口网络的戴维南定理



一般默认端口1为输入，端口2为输出

$-i_2 = i_{out}$

2. 二端口网络方程的6种表述

v_1, i_1, v_2, i_2 选2个作自变量，2个作因变量，共 $C_4^2 = 6$ 种方式：

① $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$

阻抗参量矩阵　z_{21}：跨阻增益

$\vec{V} = Z \cdot \vec{I}$

② $\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$

导纳　y_{21}：跨导增益

$\vec{I} = Y \vec{V}$

③ $\begin{pmatrix} v_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ v_2 \end{pmatrix}$

混合　h_{21}：电流增益

④ $\begin{pmatrix} i_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \begin{pmatrix} v_1 \\ i_2 \end{pmatrix}$

逆混　g_{21}：电压增益

⑤ $\begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix}$

传输

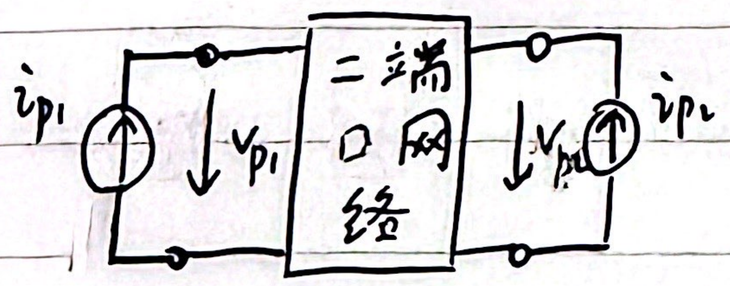
⑥ $\begin{pmatrix} v_2 \\ i_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} v_1 \\ -i_1 \end{pmatrix}$

无物理上的电路与之对应

网络参量测量：加压加流测量

① 获得z参量：两端口同时加独立变化的i

用i表示v



② y：都加v，表示i　　±　　±

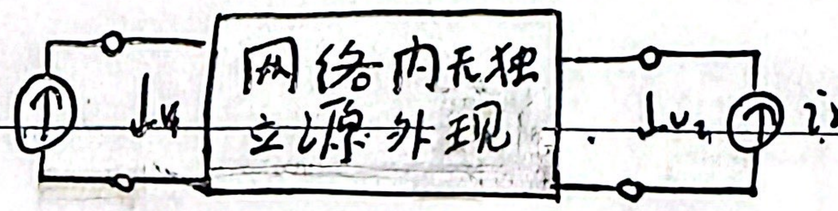
③ h：端1加i，2加v　　↑　　±

④ g　　1加v，2加i　　±　　↑

~~①~~ ~~④~~ 对应等效电路同前，（流控一戴；压控一诺）略

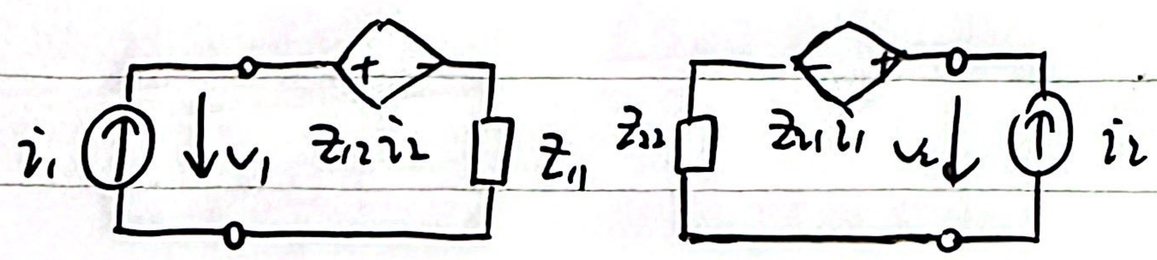
KOKUYO

分析：e.g. Z：



$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} \Leftrightarrow \begin{cases} V_1 = Z_{11} i_1 + Z_{12} i_2 \\ V_2 = Z_{21} i_1 + Z_{22} i_2 \end{cases}$$

$\Rightarrow$



$Z_{11} = \left.\frac{V_1}{i_1}\right|_{i_2=0}$ 当端口2开路时，端口1看入阻抗

$Z_{12} = \left.\frac{V_1}{i_2}\right|_{i_1=0}$ 当端口2电流对端口1开路电压的线性跨阻控制系数

"跨阻反馈系数"

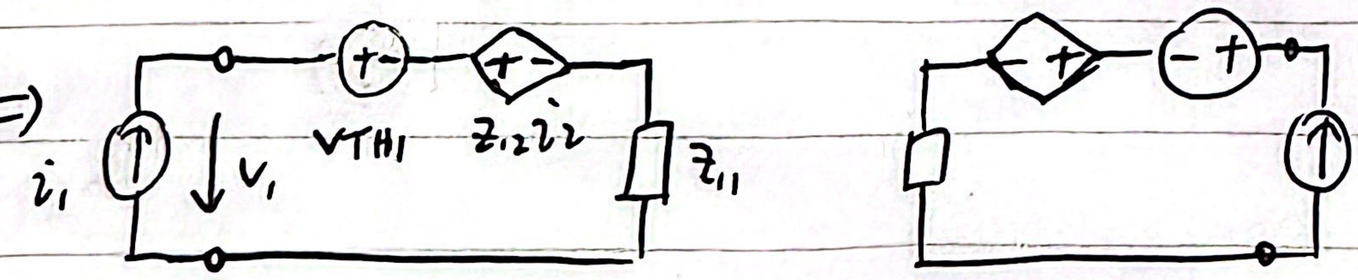
$Z_{21} = \left.\frac{V_2}{i_1}\right|_{i_2=0}$ .. 1 .. 2 .. .. 跨阻控制系数"

$Z_{22} = \left.\frac{V_2}{i_2}\right|_{i_1=0}$ ~

若内有独立源，由叠加定理，

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} + \begin{pmatrix} V_{TH1} \\ V_{TH2} \end{pmatrix}$$

$\Rightarrow$



$$\vec{V} = Z\vec{i} + \vec{V_{TH}}$$

$V_{TH1} = V_1 \big|_{i_1=i_2=0}$ 1.2 开路，1的开路电压

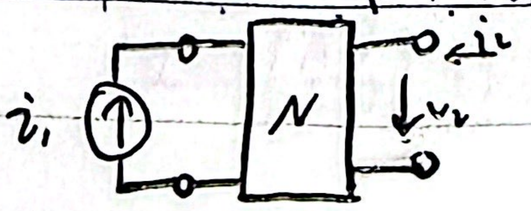
$V_{TH2} = V_2 \big|_{i_1=i_2=0}$

$Z_{11} = \left.\frac{V_1}{i_1}\right|_{i_2=0, V_{TH1}=0}$ 内部独立源置零，端口2开路，端口1看入阻抗

$Z_{21} = \left.\frac{V_2}{i_1}\right|_{i_2=0, V_{TH2}=0}$ 内独置零，1对2V的(开路)线性跨阻控制系数

Date

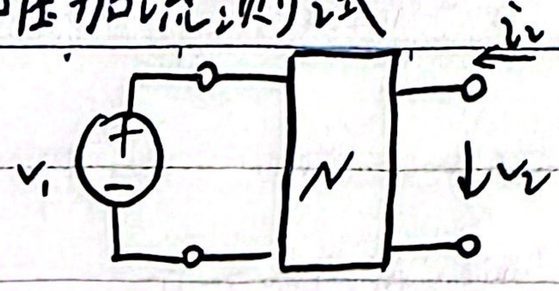
传输参量 (Z_{21}. h_{21}. y_{21}. g_{21})：单端加压加流测试



1 加i，2 测 开路 v 和 短路 i

$Z_{21} = \frac{v_2}{i_1}\Big|_{i_2=0}$ 跨阻传递系数

$h_{21} = \frac{i_2}{i_1}\Big|_{v_2=0}$ 电流传递系数

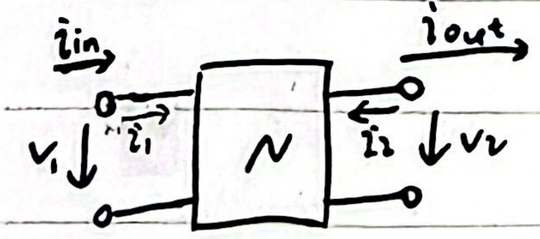


1 加 v，测 2 ～

$y_{21} = \frac{i_2}{v_1}\Big|_{v_2=0}$ 跨导～

$g_{21} = \frac{v_2}{v_1}\Big|_{i_2=0}$ 电压～

本征±增益



$$\begin{cases} A_{vo} = \frac{v_{out}}{v_{in}}\Big|_{i_{out}=0} = g_{21} & \text{本征电压±增益} \\ G_{mo} = \frac{i_{out}}{v_{in}}\Big|_{v_{out}=0} = -y_{21} & \sim \text{跨导} \sim \\ R_{mo} = \frac{v_{out}}{i_{in}}\Big|_{i_{out}=0} = z_{21} & \sim \text{跨阻} \sim \\ A_{io} = \frac{i_{out}}{i_{in}}\Big|_{v_{out}=0} = -h_{21} & \sim \text{电流} \sim \end{cases}$$

$$\underline{\begin{pmatrix} v_{in} \\ i_{in} \end{pmatrix} = \begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} v_2 \\ -i_2 \end{pmatrix}}$$

(传输参量 A.B.C.D 是本征增益的倒数)

$$= \begin{pmatrix} \frac{1}{g_{21}} & \frac{1}{-y_{21}} \\ \frac{1}{z_{21}} & \frac{1}{-h_{21}} \end{pmatrix}\begin{pmatrix} v_2 \\ -i_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{A_{vo}} & \frac{1}{G_{mo}} \\ \frac{1}{R_{mo}} & \frac{1}{A_{io}} \end{pmatrix}\begin{pmatrix} v_{out} \\ i_{out} \end{pmatrix}$$

虽然表示成这种形式，但本质上是以 in 作自变量，out 作因变量 (此倒数形式体现为这一点)

e.g. $v_1 = Av_2 - Bi_2$ $\quad A = \frac{v_1}{v_2}\Big|_{i_2=0}$ 2 开路，只能 1 作激励 $\quad \frac{1}{\frac{v_2}{v_1}\Big|_{i_2=0}} = \frac{1}{g_{21}}$

逆传参量：$\begin{pmatrix} v_2 \\ i_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} v_1 \\ -i_1 \end{pmatrix} \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} A & -B \\ -C & D \end{pmatrix}^{-1}$

∵默认 1 入 2 出 ∴逆传参量几乎无用

Z . Y . H . G . $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ 知一求五

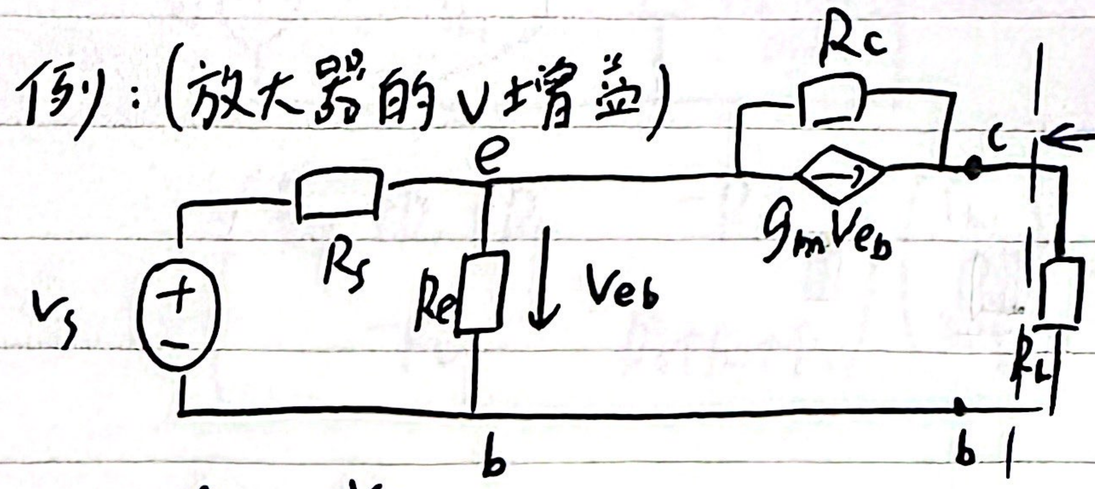
(注：有些参量可能不对 1 分母参量 =0 的))

KOKUYO

串串乙相加，并并乙相加，串并人相加，并串9相加，

级联$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$作矩阵乘法

例：(放大器的V增益)



$$A_V = \frac{V_L}{V_S}$$

用诺顿源作等效分析

(in. R_N)

求出短路in

(对于置零) 此时 再加 i_{test} 得 v_{test} 得 $R_N = \frac{v_t}{i_t}$

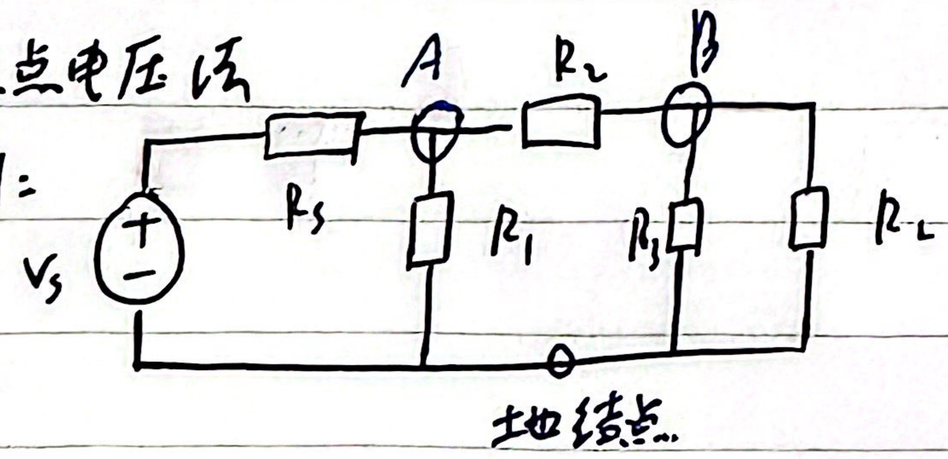
$\Rightarrow V_L = i_N \cdot (G_L + G_N)$

$\Rightarrow A_V = \frac{V_L}{V_S}$ 求出

二. 完备方程列写方法

1. 结点电压法

例1：



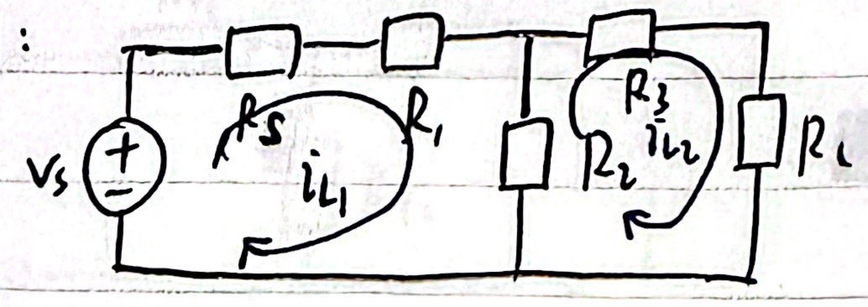
$$\begin{pmatrix} G_s + G_1 + G_2 & -G_2 \\ -G_2 & G_2 + G_3 + G_L \end{pmatrix} \begin{pmatrix} V_A \\ V_B \end{pmatrix} = \begin{pmatrix} G_s V_s \\ 0 \end{pmatrix}$$

例2：

Date

2. 回路电流法

例1：



$$\begin{pmatrix} R_S+R_1+R_2 & -R_L \\ -R_L & R_2+R_3+R_L \end{pmatrix} \begin{pmatrix} i_{L1} \\ i_{L2} \end{pmatrix} = \begin{pmatrix} V_S \\ 0 \end{pmatrix}$$

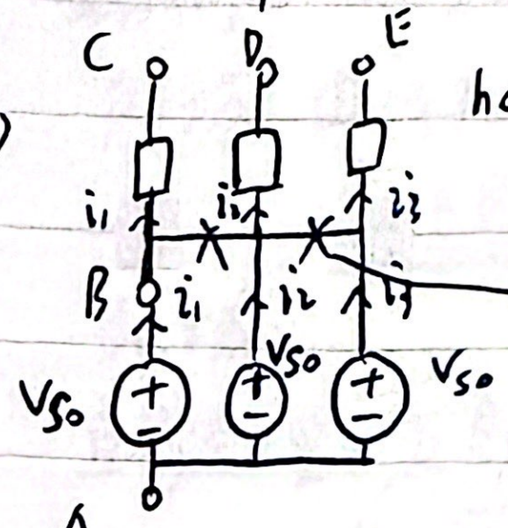
例2：

第五讲 网络分析

(课前)

非独立节点处理：

(1)

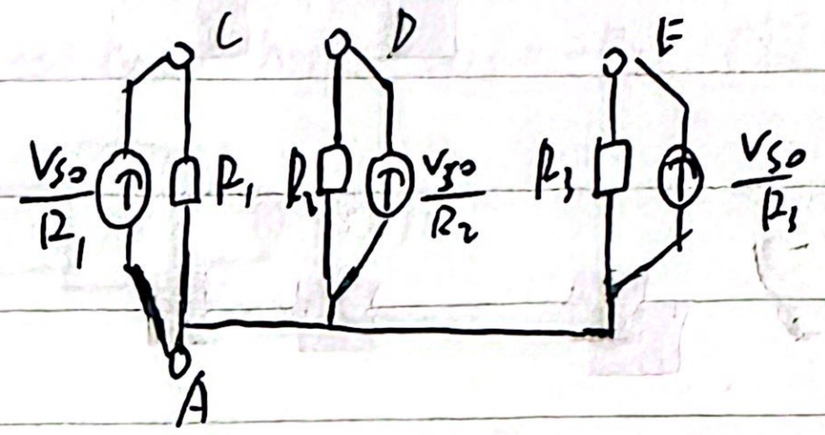


how 消除B结点？

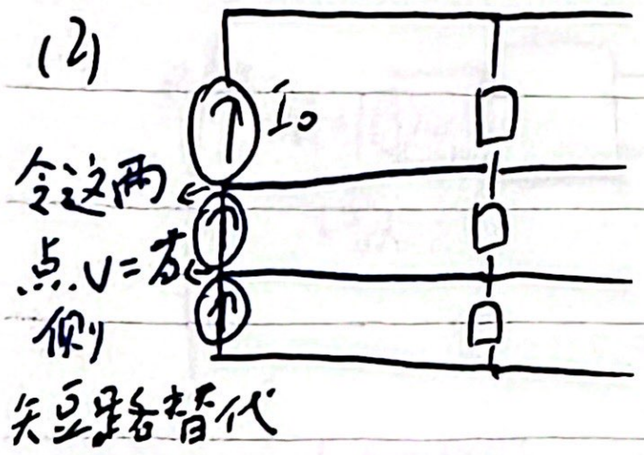
再并2个 ⊕V_{S0}，∵ i 随意 ∴ 含3个 ⊕ 恰提供 R_1 R_2 R_3 所需的 i，i=0

⇒ 开路替代 (∵ i=0) ⇒ 3个戴源

⇒ 再诺顿等效

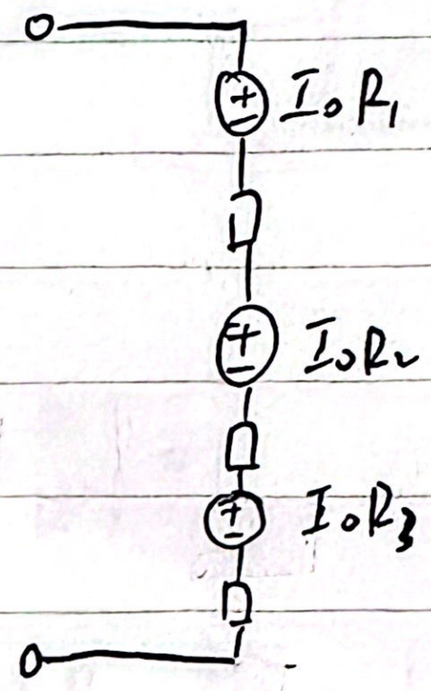


(2)



⇒ 先somo替代

⇒ 3个诺源 ⇒ 戴等效



一. 不同属性网络的参量特点

1. 线性 / 非线性

↓

扣除独立源影响后，网络端口 v与i 间关系为线性方程

2. 时变 / 时不变

3. 阻性 / 动态

↓

端口 v. i 关系用代数方程可完全描述

No.

Date

4. 互易 / 非互易

激励与响应互换

互易网络一般是针对线性网络定义的

时域：由线性时不变 R、C ($v_0=0$)、L ($i_0=0$)、传输线等互易元件构成的网络是互易网络（中）

频域：

互易定理：互易 $\Leftrightarrow$ $Z_{12}=Z_{21}$ $\Leftrightarrow$ $Y_{12}=Y_{21}$ $\Leftrightarrow$ $h_{12}=-h_{21}$ $\Leftrightarrow$ $g_{12}=-g_{21}$ （设 Z、Y、h、g 都有）

本征跨阻增益 ～导～ 本征跨导增益

本征i增益 ～反向～ 本征v增益

e.g. $V_S=1V$ 激励 → $i_{out}=2A$

$i_{out}=2A$ → $V_S=1V$ 激励

则互易 ($Y_{12}=Y_{21}$)

e.g. $V_S=1$ 激励 → $V_{out}=0.5$

$i_{out}=0.5$ ← $I_S=1$

注：不能加④

∵原来为开路，∴加源时要加①

不能改变电路的时续

$g_{21}=-g_{12}$

$\Leftrightarrow \begin{vmatrix} A & B \\ C & D \end{vmatrix}=1 \Leftrightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix}=1$

$\left(\begin{pmatrix} a & b \\ c & a \end{pmatrix} = \begin{pmatrix} A & -B \\ -C & D \end{pmatrix}^{-1} = \frac{1}{\begin{vmatrix} A & B \\ C & D \end{vmatrix}} \cdot \begin{pmatrix} D & -B \\ -C & A \end{pmatrix} = \begin{pmatrix} D & B \\ C & A \end{pmatrix} \Rightarrow \begin{cases} a=D \\ \vdots \end{cases} \checkmark \right)$

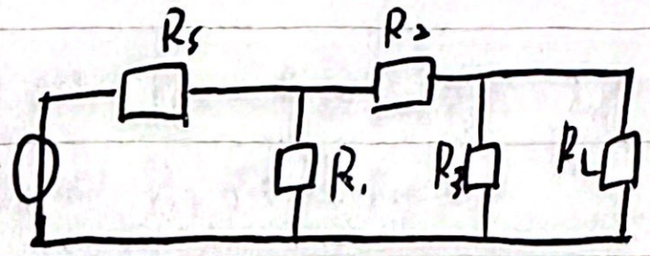
5. 对称 / 非对称

（线性端口网络 对称 $\overset{中}{\Rightarrow}$ 互易）

对称 $\Leftrightarrow$ $Z_{11}=Z_{22}$ 且 $Z_{12}=Z_{21}$ $\Leftarrow$ $Y_{11}=Y_{22}$ 且 $Y_{12}=Y_{21}$ $\Leftrightarrow$ $\begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix}=1$ 且 $h_{12}=-h_{21}$

$\Leftrightarrow \begin{vmatrix} A & B \\ C & D \end{vmatrix}=1$ 且 $A=D$

例：



△

KOKUYO

6. 有源 / 无源

计 二端口吸收总功率恒 ≥ 0，则无源

以 y 已知为例：

$$y=\begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} \xlongequal{\text{记为}} \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}$$

$$p = v_1 i_1 + v_2 i_2 = v_1(G_{11}v_1 + G_{12}v_2) + v_2(G_{21}v_1 + G_{22}v_2)$$

$$= G_{11}v_1^2 + (G_{12}+G_{21})v_1 v_2 + G_{22}v_2^2$$

s.t. 以下任一种则有源 ① $G_{11}<0$ ② $G_{22}<0$ ③ $\Delta = (G_{12}+G_{21})^2 - 4G_{11}G_{22} > 0$

$$\Rightarrow p=\begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \quad (p=z,y,h,g)$$

① $p_{11}<0$ ② $p_{22}<0$ ③ $(p_{12}+p_{21})^2 - 4p_{11}p_{22} > 0$ s.t. 任一种则有源

负阻有源性　　受控源有源性

(用途：放大器、振荡器)

复数域内统一判据：$p^T + p^*$ 非半正定：有源

$p^T + p^*$ 半正定：无源

7. 无损 / 有损 (针对无源网络而言)

端口总吸收功率恒 $=0$

无损线性二端口阻性网络：$p_{11}=0$ 且 $p_{22}=0$ 且 $p_{12}=-p_{21}$

…… 时不变动态 …… ：$p^T + p^* = 0$

8. 双向 / 单向

二端口网络中，只有端口A对端口B的作用关系(等效为受控源)，无B对A的作用关系

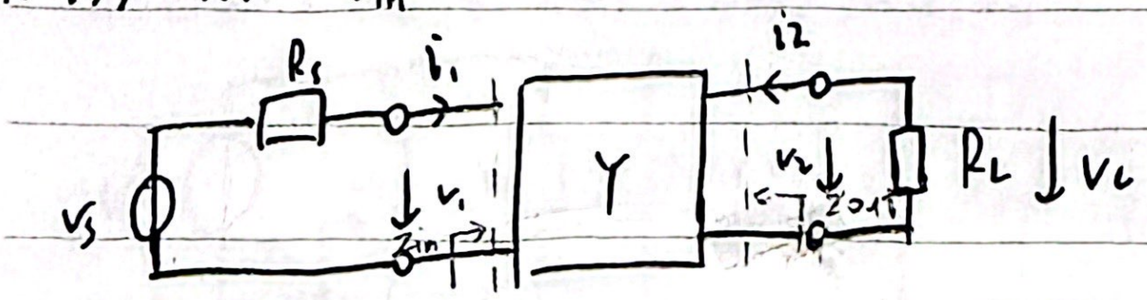
单向 $\Leftrightarrow$ $p_{12}=0$ 且 $p_{21}\neq 0$ $(p=z,y,h,g)$ $\Leftrightarrow \begin{vmatrix} A & B \\ C & D \end{vmatrix} = 0$ $(\Rightarrow a,b,c,d$ 不 $\exists)$

双向 $\Leftrightarrow$ $p_{12}\neq 0$ 且 $p_{21}\neq 0$

用于理想放大器。

二. 传递函数

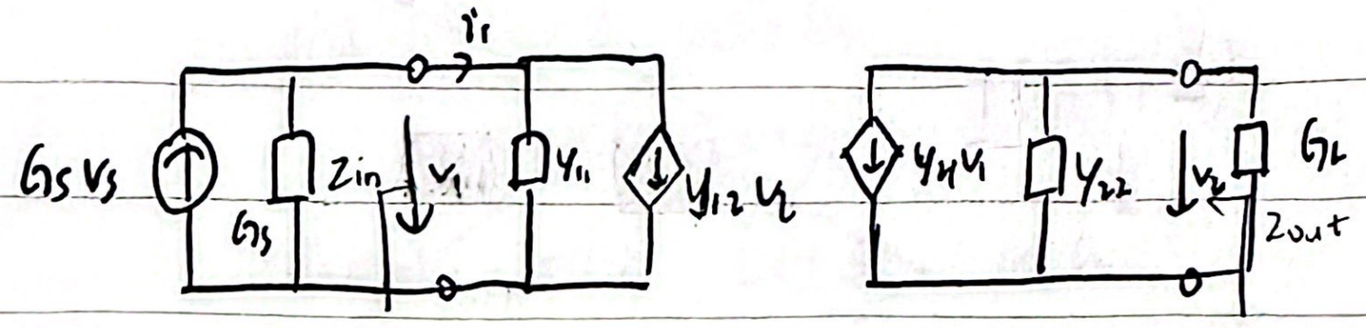
1. 例：$H_V = \frac{v_{out}}{v_{in}}$



$H_V = \frac{v_2}{v_s}$　　假设y已测得，求 H_v

法一) $\begin{cases} i_1 = y_{11}v_1 + y_{12}v_2 \\ i_2 = y_{21}v_1 + y_{22}v_2 \\ v_s - v_1 = i_1 R_s \\ v_2 = -i_2 R_L \end{cases}$ ✓

法二) 等效分析　(∵已知y(导纳) ∴→诺顿源，较合适)



$\begin{cases} v_1(G_s + y_{11}) = G_s v_s - y_{12}v_2 \\ v_2(G_L + y_{22}) = -y_{21}v_1 \end{cases}$

法三) 倒推　略

最终解得 $H_V = \frac{v_2}{v_s} = \frac{y_{21}G_s}{y_{12}y_{21} - (G_s + y_{11})(G_L + y_{22})}$

极端检查：$y_{12} = 0$，$H_V = \frac{-y_{21}G_s}{(G_s + y_{11})(G_L + y_{22})}$

why 把 $y_{12}=0$（单向网络）当作极端情况？　∵省去直接写结果

∵不含2→1的反馈 ∴总=分之积　$V_2 = \frac{-y_{21}}{y_{22} + G_L} \cdot \frac{G_s}{G_s + y_{11}} v_s$

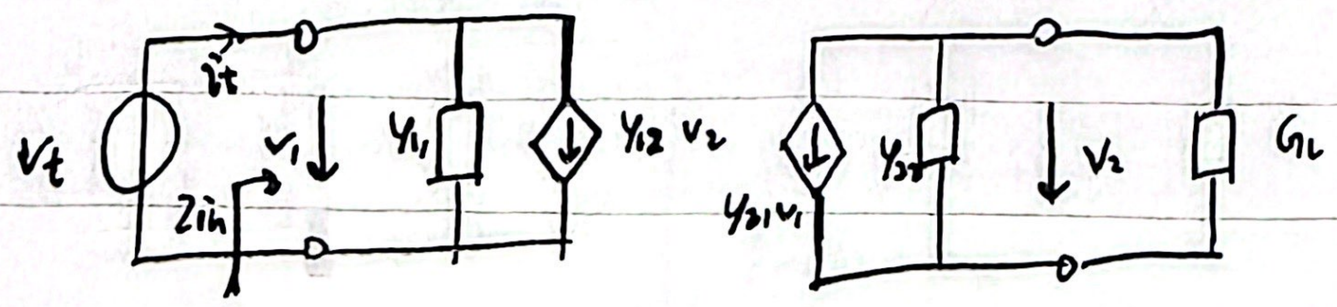
⇒双向也可直接写结果！

No.

Date

2. 求 Z_{in}, Z_{out} (见前图)

∵已知 y ∴求输入导纳 y_{in} 更好 $Z_{in} = y_{in}^{-1}$



$$y_{in} = \frac{i_t}{V_t} \qquad \begin{cases} V_2(G_L + y_{22}) = -y_{21}V_t \\ i_t = y_{11}V_1 + y_{12}V_2 \end{cases} \qquad (V_t = V_1)$$

$$\Rightarrow \frac{i_t}{V_t} = y_{11} - \frac{y_{12}y_{21}}{y_{22}+G_L}$$

总结规律：$y_{in} = y_{11} - \dfrac{y_{12}y_{21}}{y_{22}+G_L} = g_{11} - \dfrac{g_{12}g_{21}}{g_{22}+R_L}$ （单向网络，这项 $\to 0$）

求 Z：$y \leftrightarrow z$, $G_L \leftrightarrow R_L$

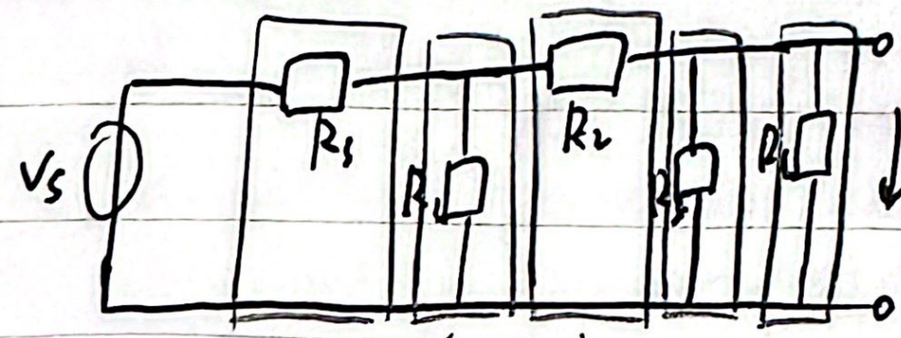
$g \leftrightarrow h$ $R_L \leftrightarrow g_L$

求 out：$1 \leftrightarrow 2$, $L \leftrightarrow S$ $A \leftrightarrow G$

(见作业)

阻抗变换网络一定是双向网络 ($P_{12}P_{21} \neq 0$ $P = y, z, g, h$)

ABCD 适用于梯形网络传递函数的求解

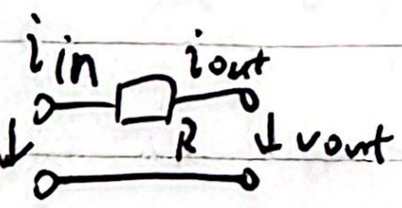


$H_v = \dfrac{V_L}{V_S} = A_{vo} = \dfrac{1}{A}$ ∴求出 ABCD 即求出传递函数

串 并 串 并 并

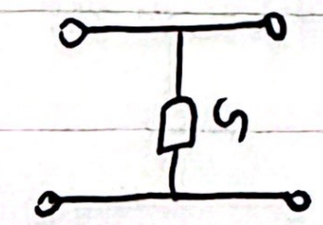
$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}_{R_S} \begin{pmatrix} A & B \\ C & D \end{pmatrix}_{R_1} \begin{pmatrix} A & B \\ C & D \end{pmatrix}_{R_2} \begin{pmatrix} A & B \\ C & D \end{pmatrix}_{R_3 \| R_L}$$

求串臂阻抗的 $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$：



$$\begin{pmatrix} V_{in} \\ i_{in} \end{pmatrix} = \begin{pmatrix} 1 & R \\ 0 & 1 \end{pmatrix} \begin{pmatrix} V_{out} \\ i_{out} \end{pmatrix} \to \begin{pmatrix} 1 & Z \\ 0 & 1 \end{pmatrix}$$

并臂导纳的 $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$

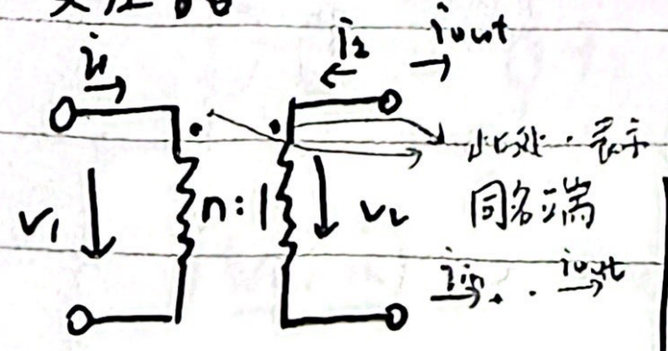


$$\begin{pmatrix} V_{in} \\ i_{in} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ G & 1 \end{pmatrix} \begin{pmatrix} V_{out} \\ i_{out} \end{pmatrix} \to \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix}$$

KOKUYO

三. 典型网络

1. 变压器

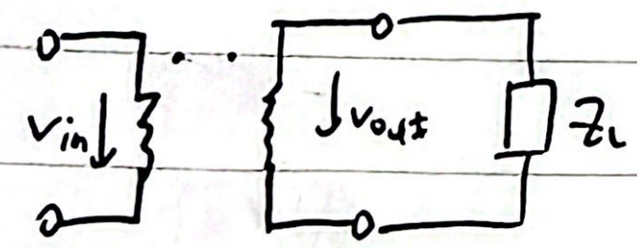


(无损)

$$\begin{cases} v_{out} = \frac{1}{n} v_{in} \\ i_{out} = n\, i_{in} \\ p_{out} = p_{in} \end{cases}$$

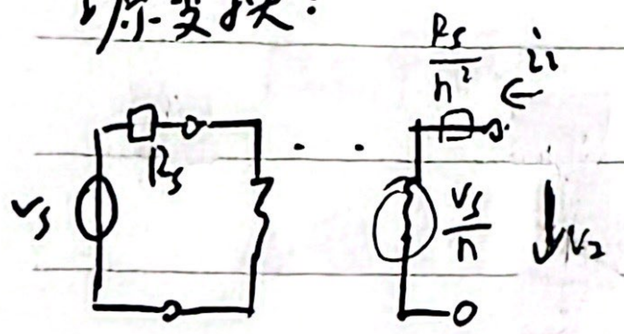
$$\begin{pmatrix} v_1 \\ i_1 \end{pmatrix} = \begin{pmatrix} n & 0 \\ 0 & \frac{1}{n} \end{pmatrix} \begin{pmatrix} v_2 \\ -i_2 \end{pmatrix}$$

阻抗变换功能：



$$Z_{in} = \frac{v_{in}}{i_{in}} = n^2 Z_L$$

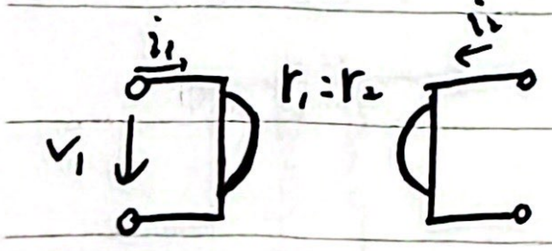
源变换：



$$v_2 = \frac{v_S}{n} + \frac{R_S}{n^2} i_2 \quad (\text{易验证 } i_2 v_2 = i_1 v_1)$$

功能：作最大功率传输匹配网络 (取 $n = \sqrt{\frac{R_S}{R_L}} \Rightarrow Z_{in} = n^2 R_L = R_S \Rightarrow P_{max}$)

2. 回旋器



$$v_1 = -r_1 i_2$$
$$v_2 = r_2 i_1$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 & -r_1 \\ r_2 & 0 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

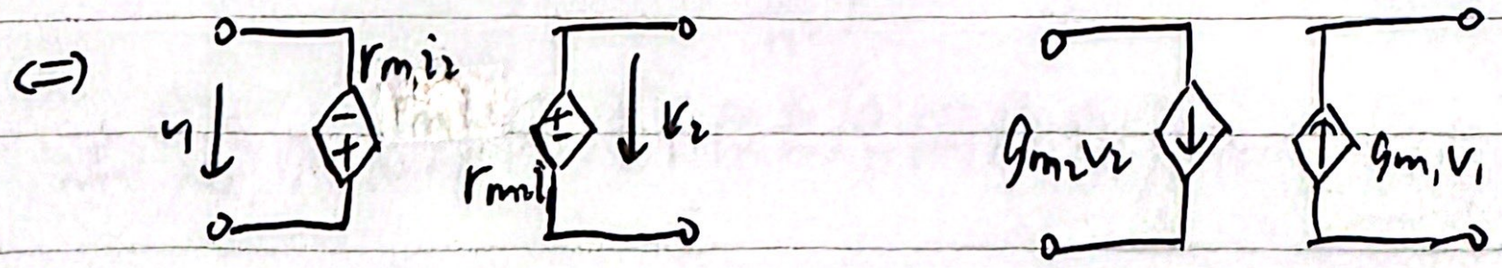
$P_\Sigma = v_1 i_1 + v_2 i_2 = (r_2 - r_1) i_1 i_2$

$r_2 \neq r_1$ 时，P_Σ 可 < 0 的可能性，有记忆

$r_2 = r_1$ 无记忆(且无损)(称为理想回旋器)

$$Z = \begin{pmatrix} 0 & -r_{m1} \\ r_{m2} & 0 \end{pmatrix} \quad \text{非互易} \quad Y = \begin{pmatrix} 0 & g_{m2} \\ -g_{m1} & 0 \end{pmatrix}$$

h, g 不可



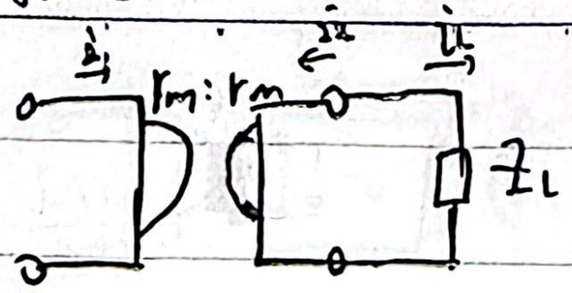
Z　　Y

双向网络

No.

Date

阻抗变换：



$v_1 = -r_m i_2$

$v_2 = r_m i_1$

$f(v_1, i_1) = 0 \Leftrightarrow f(v_1, -i_2) = 0 \Leftrightarrow f(r_m i_1, g_m v_1) = 0$

对偶变换

端口2

开路

恒压源

C

RLC串联

端口1

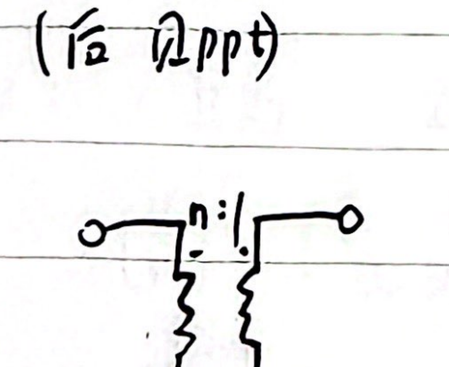
短路

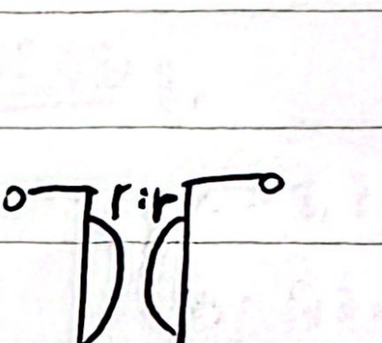
恒流源

L

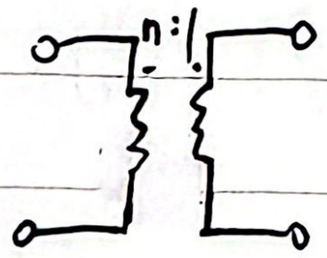
GCL并联

∿型负导 $\rightleftharpoons$ S型负阻



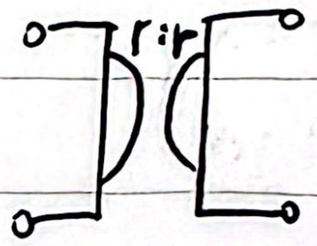


(后见ppt)



同属阻变换

互易 无损



对偶变换

非互易 无损

注：等效电容 $= \frac{C}{n^2}$

$\because v = \frac{1}{C}\int i\,dt$

$= \frac{n^2}{C}\int i\,dt = \frac{1}{\frac{C}{n^2}}\int i\,dt$

注：添加无损网络不改变原网络有门多位

KOKUYO

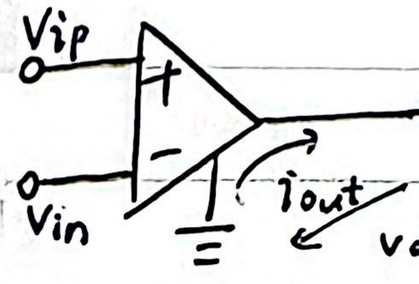
Date

第六讲 电容电感

一.(续)典型线性网络) 运算放大器

同相输入端 V_{ip}

反相输入端 V_{in}



输出端

$V_o = A_{vo} V_{id} = A_{vo}(V_{ip} - V_{in})$

(A_{vo} 极大,可抽象为 $+\infty$)

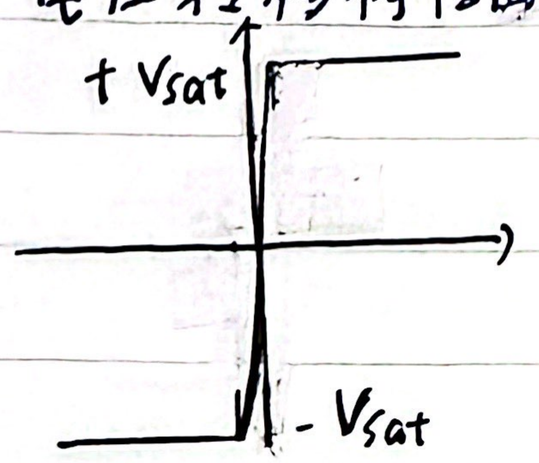
(习惯不画,默认存在)

$V_{id} = V_{ip} - V_{in}$

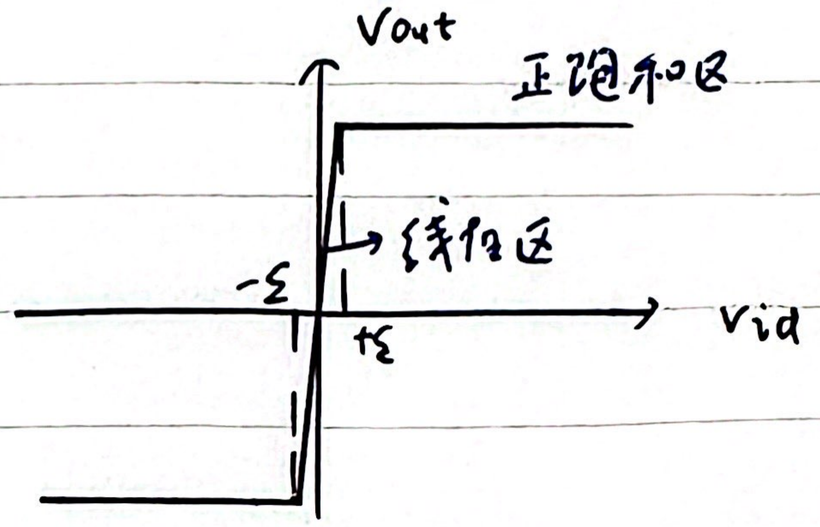
(差模输入)

(实际电路图见ppt)

电压转移特性曲线:

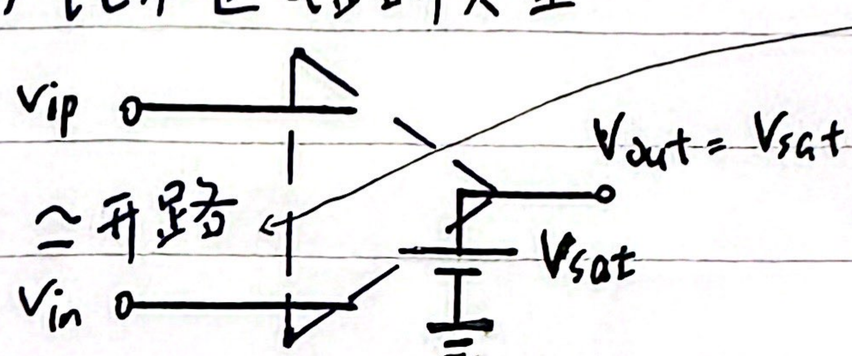


分段折线近似



$$V_{out} = \begin{cases} -V_{sat} & V_{id} < -\varepsilon \\ A_{vo} V_{id} & -\varepsilon < V_{id} < \varepsilon \\ V_{sat} & V_{id} > \varepsilon \end{cases} \qquad \varepsilon = \frac{V_{sat}}{A_{vo}}$$

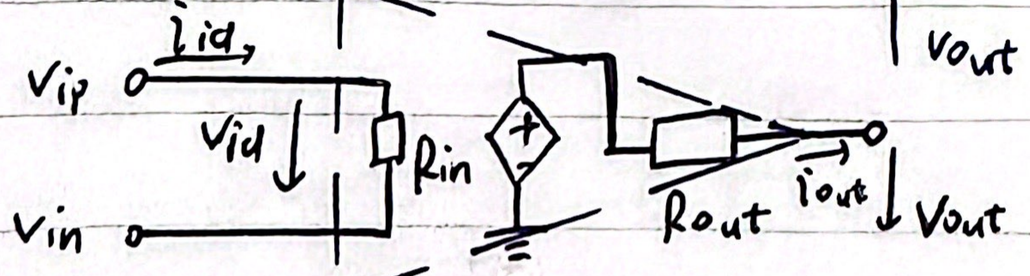
(1) 饱和区电路模型



$$\begin{cases} i_{id} = 0 \\ V_{out} = V_{sat} \end{cases} \quad (V_{id} > \varepsilon \text{ 时})$$

负饱和区与之类似

(2) 线性区电路模型



$$\begin{cases} i_{id} = \dfrac{V_{id}}{R_{in}} \\ V_{out} = A_{vo} V_{id} - R_{out} i_{out} \end{cases} \quad (-\varepsilon < V_{id} < \varepsilon \text{ 时})$$

(in fact, R_{in} 极大 ∴≃开路
R_{out} 很小 ∴可≃0)

$$\begin{pmatrix} i_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{R_{in}} & 0 \\ A_{vo} & R_{out} \end{pmatrix} \begin{pmatrix} v_1 \\ i_2 \end{pmatrix}$$

No.

Date

理想运放：具有无穷大增益的运放

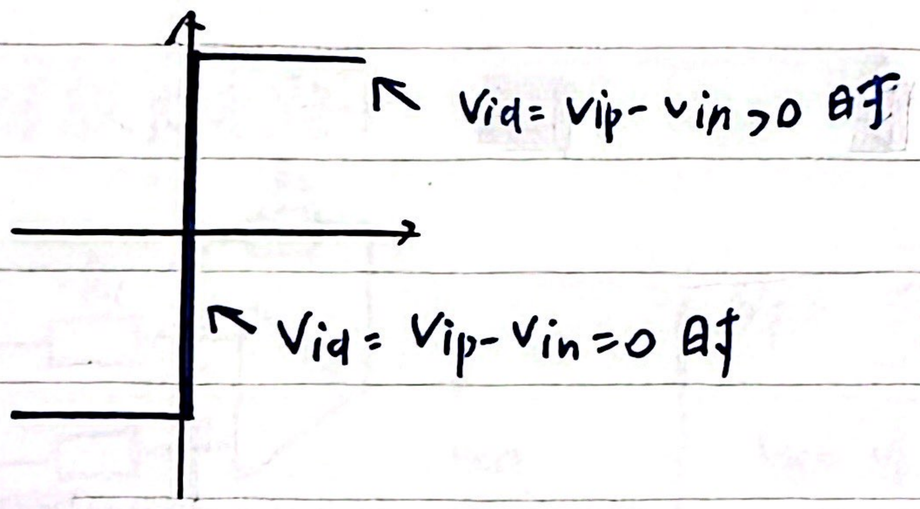
$$\begin{pmatrix} V_{id} \\ i_{id} \end{pmatrix} = \begin{pmatrix} \frac{1}{A_{vo}} & \frac{1}{G_{mo}} \\ \frac{1}{R_{mo}} & \frac{1}{A_{io}} \end{pmatrix} \begin{pmatrix} v_{out} \\ i_{out} \end{pmatrix} = \begin{pmatrix} \frac{1}{A_{vo}} & \frac{R_{out}}{A_{vo}} \\ \frac{1}{A_{vo}R_{in}} & \frac{R_{out}}{A_{vo}R_{in}} \end{pmatrix} \begin{pmatrix} v_{out} \\ i_{out} \end{pmatrix}$$

$A_{vo} \to +\infty$ $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \quad \\ \quad \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Rightarrow$ $v_{id} = 0$ — 虚短

$i_{id} = 0$ — 虚断

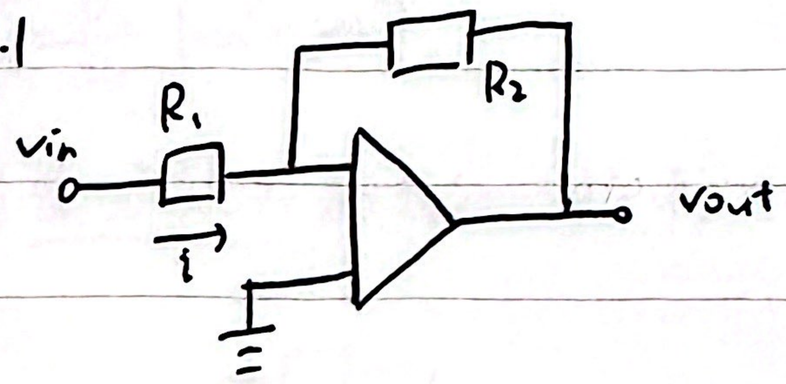
$\Rightarrow$



用负反馈 连接确保运放工作在线性区 (具体见 PPt)

$\Rightarrow$ 用虚短、虚断做运放分析

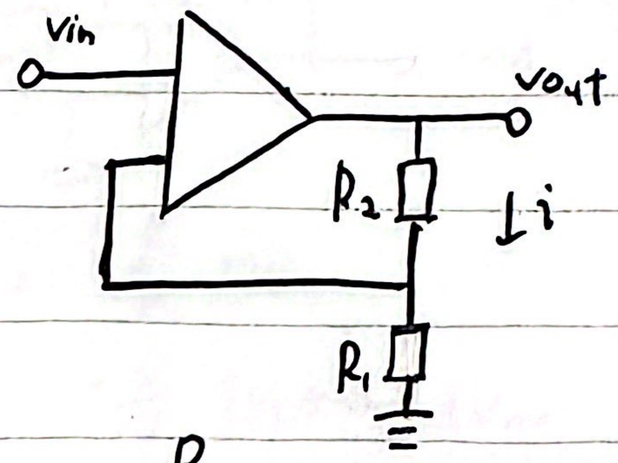
e.g.1



$$\frac{v_{in} - 0}{R_1} = i = \frac{0 - v_{out}}{R_2}$$

$\Rightarrow A_v = \frac{v_{out}}{v_{in}} = -\frac{R_2}{R_1}$ 反相电压放大电路

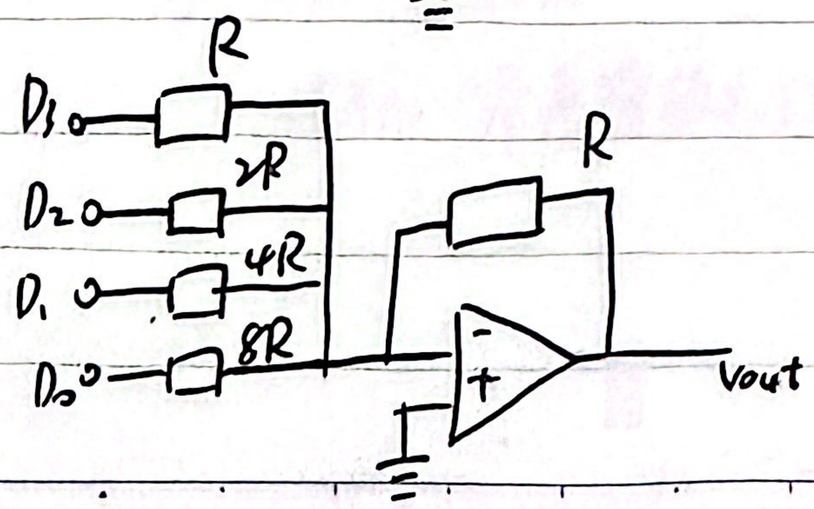
e.g.2



$$\frac{v_{in}}{R_1} = i = \frac{v_{out} - v_{in}}{R_2}$$

$\Rightarrow A_v = 1 + \frac{R_2}{R_1}$ (同相~)

e.g.3



$$v_{out} = -\left(\frac{D_3}{R} + \frac{D_2}{2R} + \frac{D_1}{4R} + \frac{D_0}{8R}\right)R$$

$$= -2^3 \sum_{k=0}^{3} D_k 2^k$$

实现了二进制加权平和 (DAC)

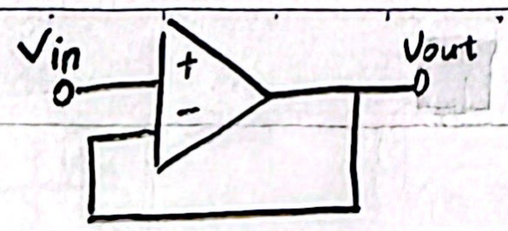
KOKUYO

No.

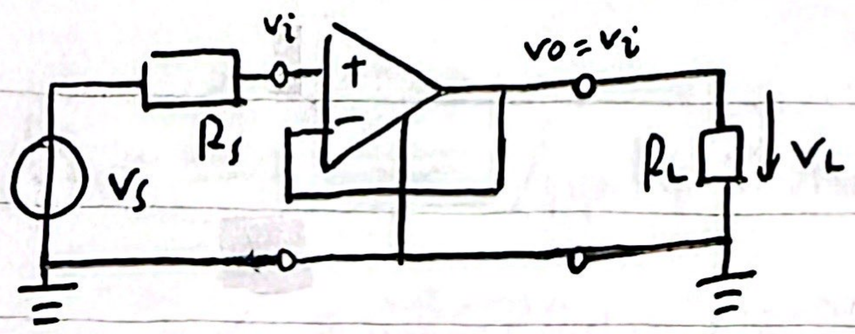
Date

e.9.4 电压跟随器

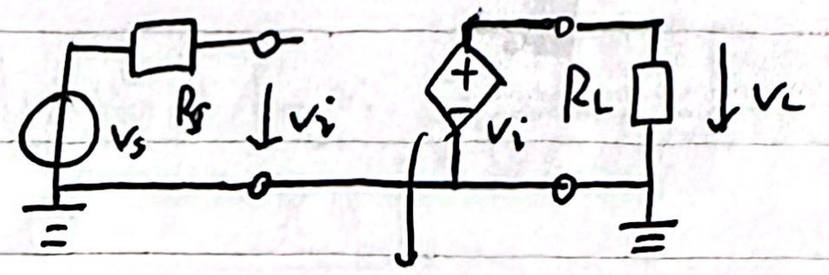
e.9.2 图 $\xrightarrow{R_2=0}$



$V_{out} = V_{in}$



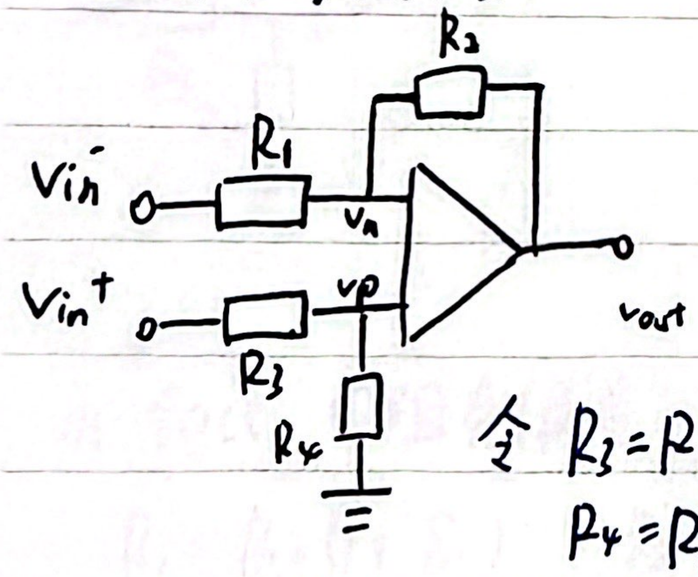
(=)



土增益=1的压控压源

好处：① ② ③ (见ppt)

差分放大电路：



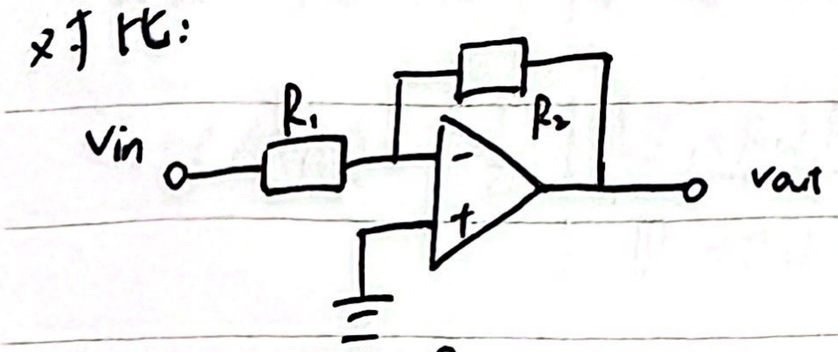
令 $R_3=R_1$
$R_4=R_2$

$$\begin{cases} v_p = \dfrac{R_4}{R_3+R_4} v_{in}^+ \\ v_n = v_p \\ \dfrac{v_{in}^- - v_n}{R_1} = \dfrac{v_n - v_{out}}{R_2} \end{cases}$$

$$\Rightarrow V_{out} = \frac{R_2}{R_1}(v_{in}^+ - v_{in}^-)$$

用差分放大的好处：抑制共模噪声

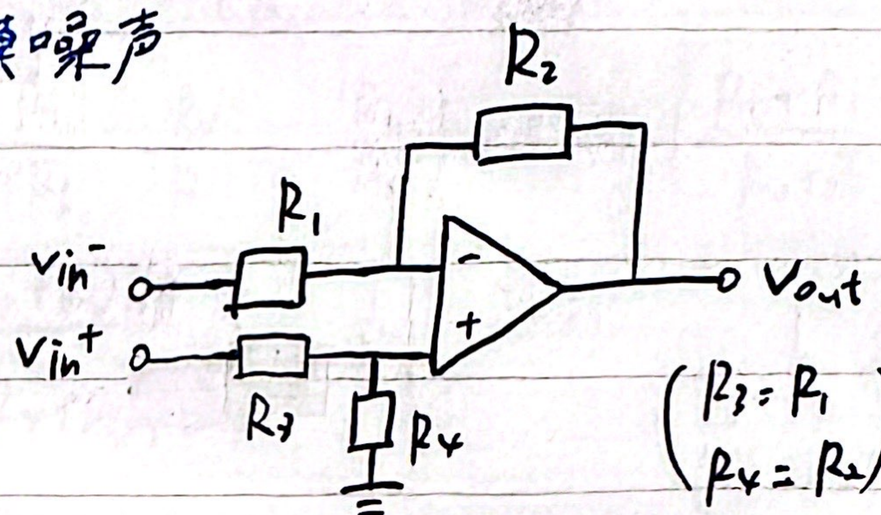
对比：



$$v_{out} = -\frac{R_2}{R_1} v_{in}$$

$$v_{out} = -\frac{R_2}{R_1}(v_s + \underline{v_{ns}}) + v_{no}$$

对输入噪声有放大作用



$(R_3 = R_1, R_4 = R_2)$

$$v_{out} = \frac{R_2}{R_1}(v_{in}^+ - v_{in}^-)$$

减没]

$$v_{out} = \frac{R_2}{R_1}[(v_s^+ + \underline{v_{ns}}) - (v_s^- + \underline{v_{ns}})] + v_{no}$$

$$= \frac{R_2}{R_1}(v_s^+ - v_s^-) + v_{no}$$

↓

可再改进，把输出噪声也减没

No.

Date

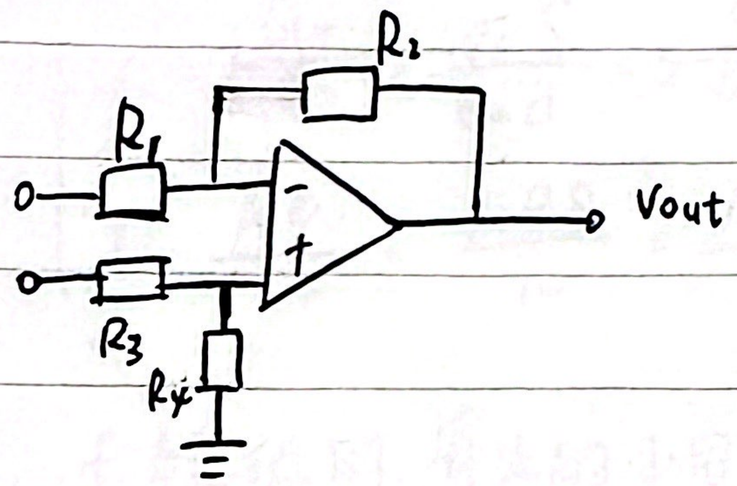
对双端输入信号，定义

差模信号 $V_{id} = V_{ip} - V_{in}$

共模信号 $V_{ic} = \frac{V_{ip} + V_{in}}{2}$ $\Rightarrow$ $\begin{cases} V_{ip} = V_{ic} + 0.5\,V_{id} \\ V_{in} = V_{ic} - 0.5\,V_{id} \end{cases}$ （双端输入信号可分解为共模信号与差模信号的叠加）

$V_{out} \xrightarrow{\text{叠加定理}} A_{vp} V_{ip} + A_{vn} V_{in} = A_{vd} V_{id} + A_{vc} V_{ic} \xrightarrow[\text{只放大差模，不放大共模}]{\text{理想差分放大器}} A_{vd} V_{id}$

定义 共模抑制比 $CMRR = \left|\frac{A_{vd}}{A_{vc}}\right| \xrightarrow{\text{理想差分放大器}} +\infty$



共模抑制能力 from 对称性：要求 $\frac{R_3}{R_4} = \frac{R_1}{R_2} \Rightarrow A_v^+ = \frac{R_2}{R_1}$ $\Leftarrow = -A_v^-$

$V_{out} = A_v^+ V_{in}^+ + A_v^- V_{in}^-$

$= \frac{R_2}{R_1}(V_{in}^+ - V_{in}^-) = A_{vd} V_{id} + A_{vc} V_{ic}$

$= A_{vd} V_{id}$

（可见只有对称才能做到 $CMRR = +\infty$，即 $\left|\frac{A_{vd}}{A_{vc}}\right| = \left|\frac{R_2/R_1}{0}\right| = +\infty$）

in fact，电阻有偏差

$R_1 = R_{10}(1+\delta_1)$ $R_2 = R_{20}(1+\delta_2)$

$R_3 = R_{30}(1+\delta_1)$ $R_4 = R_{40}(1+\delta_2)$

$A_v^- = -\frac{R_2}{R_1}$ $A_v^+ = \left(\frac{R_2}{R_1}+1\right)\frac{R_4}{R_3+R_4}$

$V_{out} = A_v^+ V_{in}^+ + A_v^- V_{in}^- = \frac{1}{2}\left(\frac{R_2+R_1}{R_4+R_3}\cdot\frac{R_4}{R_1} - \frac{R_2}{R_1}\right)V_d + \left(\frac{R_2+R_1}{R_4+R_3}\frac{R_4}{R_1} + \frac{R_2}{R_1}\right)V_c$

$CMRR = \frac{1}{2}\left|\frac{(R_1+R_2)R_4 + (R_4+R_3)R_2}{(R_1+R_2)R_4 - (R_4+R_3)R_2}\right| = \frac{1}{2}\left|\frac{\cdots}{\cdots}\right| \geq \frac{1}{4\delta_{max}}$

（见作业）

？

余课题。

KOKUYO

Date . .

三. 电容电感

1. 对偶关系：

E		H
V		I
C	$i=\frac{dq}{dt}=C\frac{dv}{dt}$	L
q		φ
并		串
结点		回路

(高中 $C=\frac{\varepsilon_0 S}{4\pi kd}$ $E=\frac{U}{d}=\frac{Q}{Cd}=\frac{4\pi kQ}{\varepsilon_0 S}$)

$$\begin{cases} C=\frac{Q_0}{V_0}=\frac{\sigma_0 S}{E_0 d}=\varepsilon\frac{S}{d} \\ L=\frac{N\Phi_0}{i_0}=\frac{NBS}{i_0}=\frac{N(\mu\frac{N}{l}i_0)S}{i_0}=N^2\mu\frac{S}{l} \end{cases}$$

2. f较低时. 寄生的小电容视为开路, 寄生的小电感 视为开路

高 效应不能忽略不计

3. 特性 (以电容为例)

①记忆性 $v(t)=v_0+\frac{1}{C}\int_0^t i(\tau)\,d\tau$

②连续性 $v(t)\in C(\mathbb{R})$ (除非加冲激)

③无损性：C吸收的能量，可以在下一个时间段完全释放出来

注：不能用 i-v 描述 C，要用 q-v

(∵ i-v 图受 w 影响)

4. C 串并联 ↔ G

L ↔ R

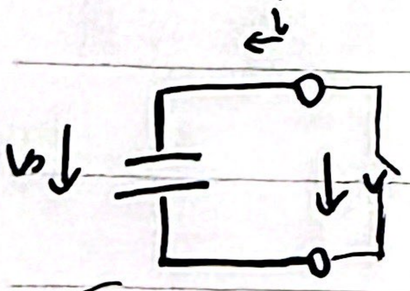
No.

Date

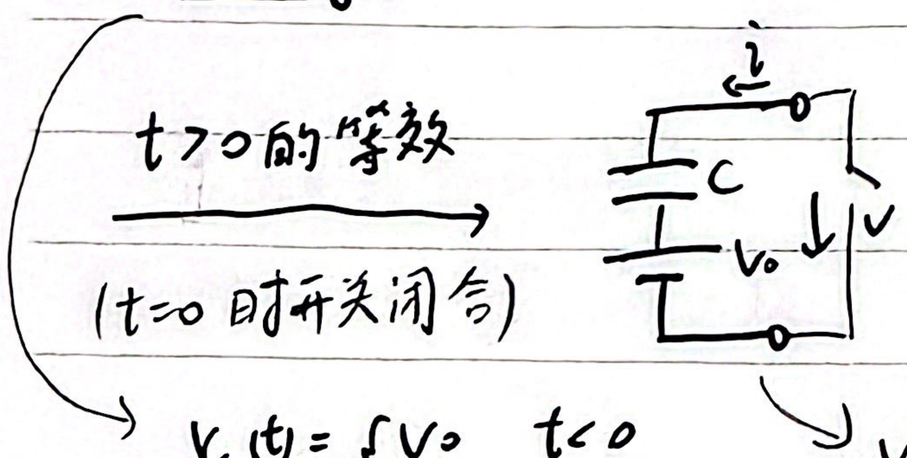
第七讲 状态初值

(回顾：有初 v_i 的 C 有源. 无~ 无源 i_L)

一. 有 V_0 的 C 的源等效



$$v(t) = v_0 + \frac{1}{C}\int_0^t i(\tau)\,d\tau$$

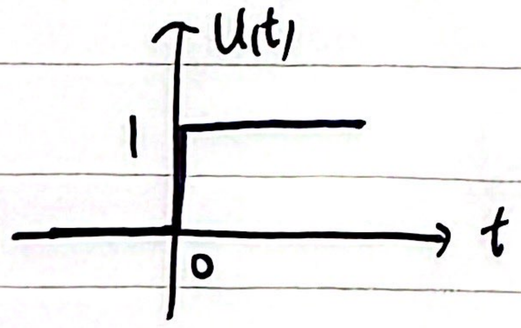


$$v_c(t) = \begin{cases} v_0 & t<0 \\ 0 & t=0 \end{cases} \qquad v_{ce}(t) = \begin{cases} 0 & t<0 \\ -v_0 & t>0 \end{cases}$$

$$i(t) = C\frac{du_c(t)}{dt}$$

引入单位阶跃函数

$$u(t) = \begin{cases} 0 & t<0 \\ 1 & t>0 \end{cases}$$ (t=0 时取啥都行. 不妨 $u(0)=1$)

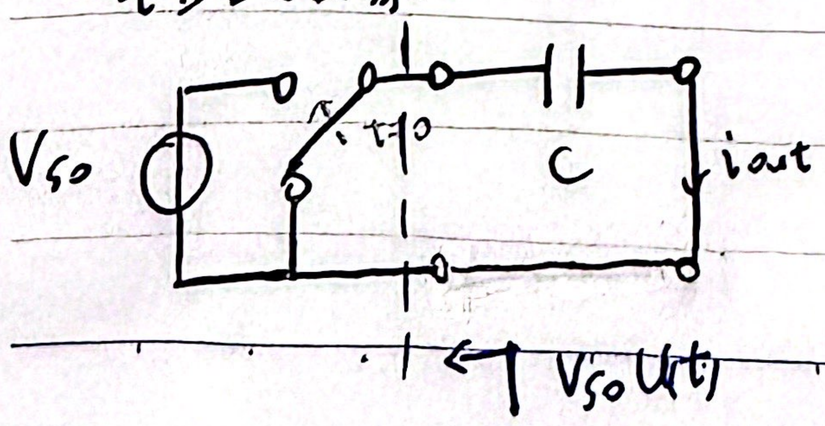


($u(t)$ 量纲 =1)

单位冲激函数 数学意义应在积分下理解；可看成一族 δ_Δ 的 lim

$\delta(t)$ 定义：$\begin{cases} ① \int_{-\infty}^{+\infty} \delta(t)\,dt = 1 \\ ② \ \delta(t) = 0 \quad \forall t \neq 0 \end{cases}$ (①②缺一不可) ($\delta(t)$ 量纲 s^{-1})

电路抽象：



$$q(t) = CV_{S0}u(t)$$

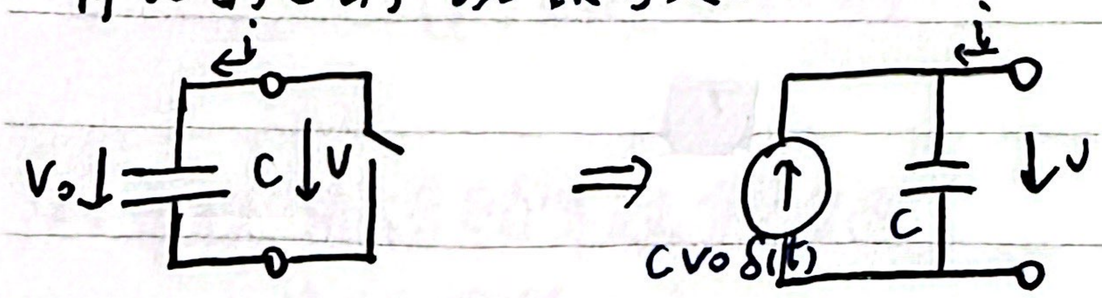
$$i_{out}(t) = \frac{dq(t)}{dt} = CV_{S0}\delta(t)$$

· 什 i 有界, C的V连续

· 当 $\underline{\text{日冲激}}$时, $\underline{\text{C的V可跳变}}$:

$$V_c(t_0^+) - V_c(t_0^-) = \frac{1}{C}\int_{t_0^-}^{t_0^+} i_C = V_0 \int_{t_0^-}^{t_0^+} \delta(t)\,dt = V_0$$

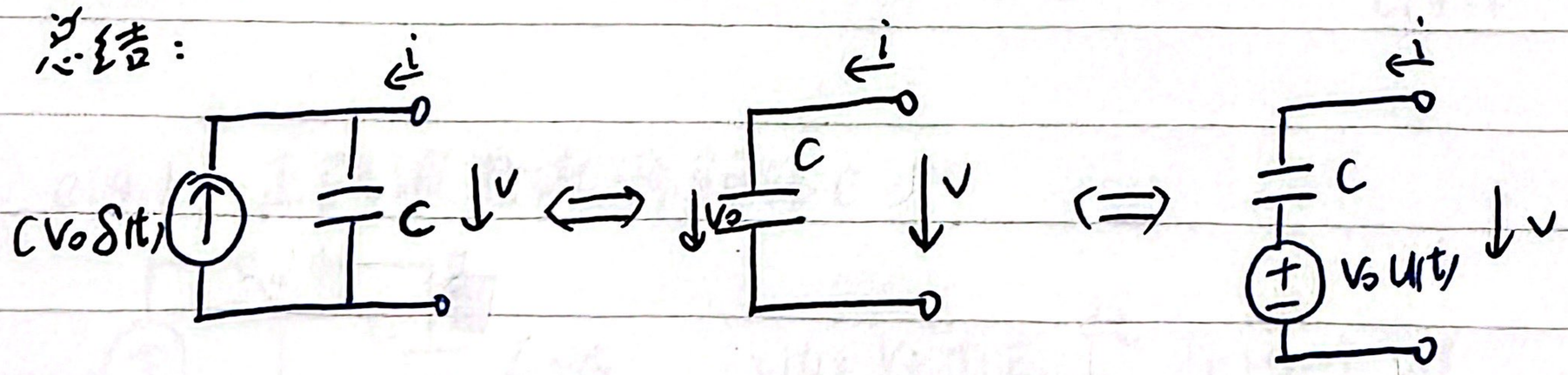
有 V_0 的 C 的 诺顿等效



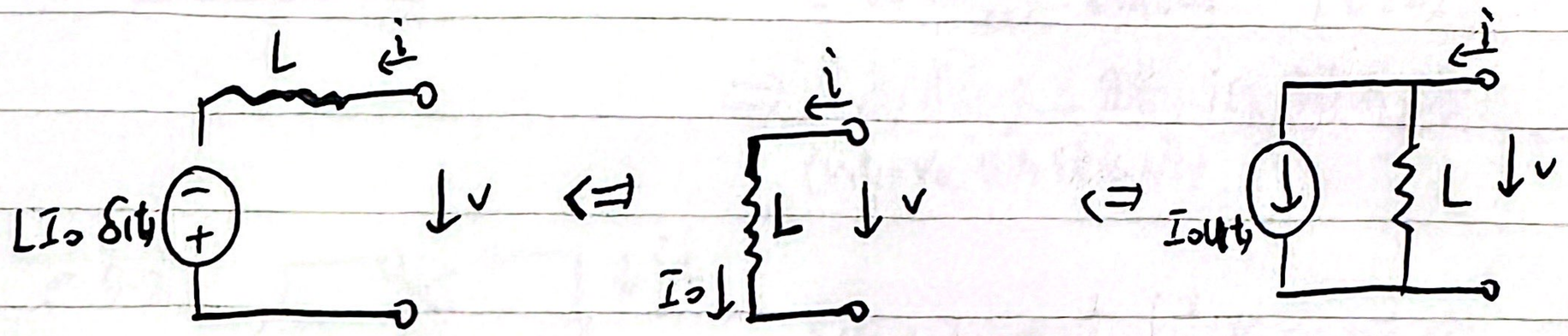
$$V_c(t) = V_0\,(1-u(t))$$

短路 $i = C\frac{dv_c}{dt} = -CV_0\delta(t)$

总结:



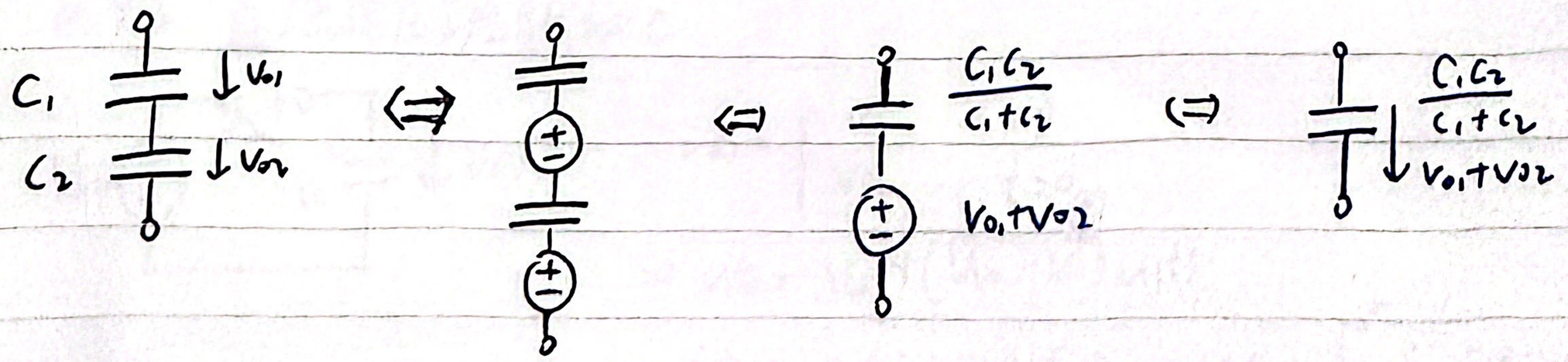
此等效只在t>0时才成立



(t>0时 v)

三. 有 V_0 的 C 的串并联

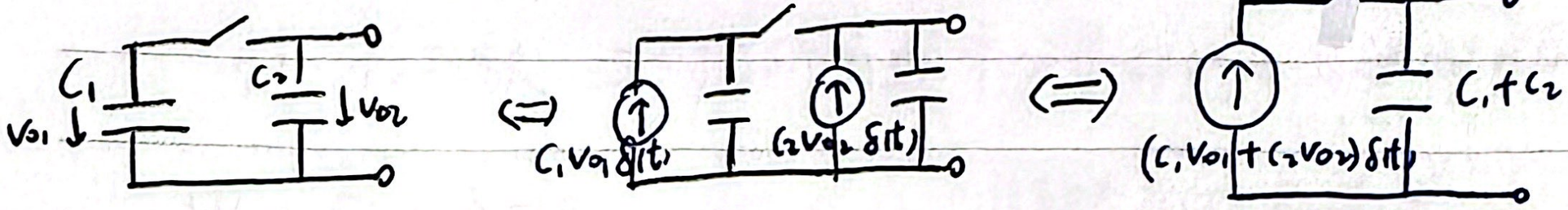
1.



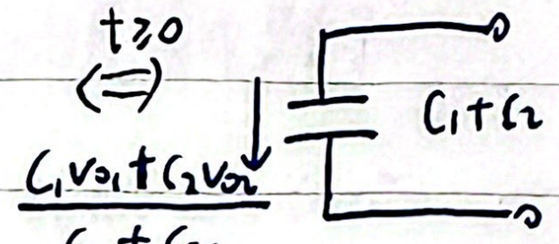
No.

Date

2. 并联关



(t>0时等效成立)



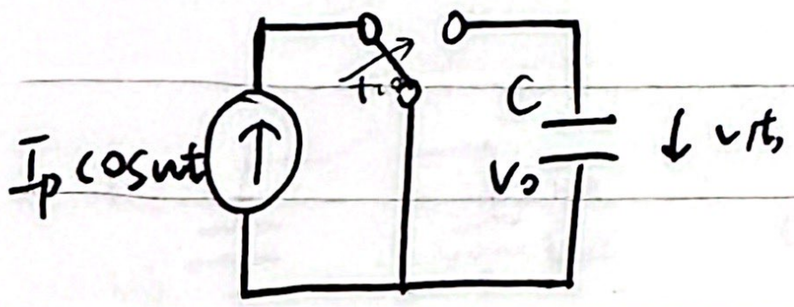
注：并联用诺顿等效较合适

(之前串联用戴一)

法二) 电荷守恒

$$C_1V_{o1}+C_2V_{o2}=C_1V_1+C_2V_2 \xrightarrow{V_1=V_2=V} V=\frac{C_1V_{o1}+C_2V_{o2}}{C_1+C_2}$$

3. e.9.1 正弦波电流源驱动C



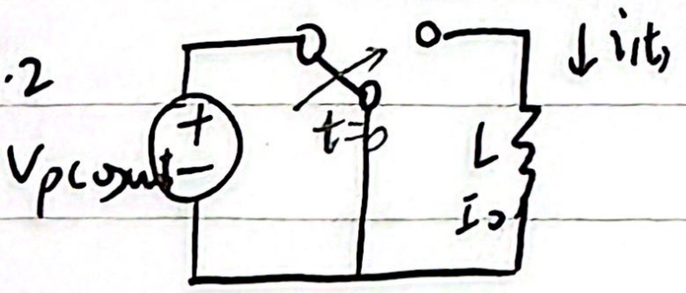
$$v(t)=V_0+\frac{1}{C}\int_0^t I_p\cos\omega t\,dt$$

$$=V_0+\frac{I_p}{\omega C}\sin\omega t \quad (t>0)$$

⇒ C高频短路，低频开路

($v(t)=V_0$，变为恒压源)

e.9.2

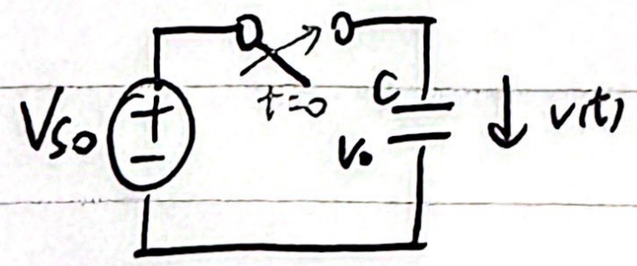


$$i(t)=I_0+\frac{1}{L}\int_0^t V_p\cos\omega t\,dt$$

$$=I_0+\frac{V_p}{\omega L}\sin\omega t \quad (t>0)$$

⇒ L高频开路

e.9.3 直流电压源驱动C



$$v(t)=\begin{cases}V_o & t<0\\ V_{s0} & t>0\end{cases}$$

$$\Rightarrow v(t)=V_0+(V_{s0}-V_0)u(t)$$

$$i(t)=C(V_{s0}-V_0)\delta(t) \qquad \Delta q=\int_{0^-}^{0^+} i=C(V_{s0}-V_0)$$

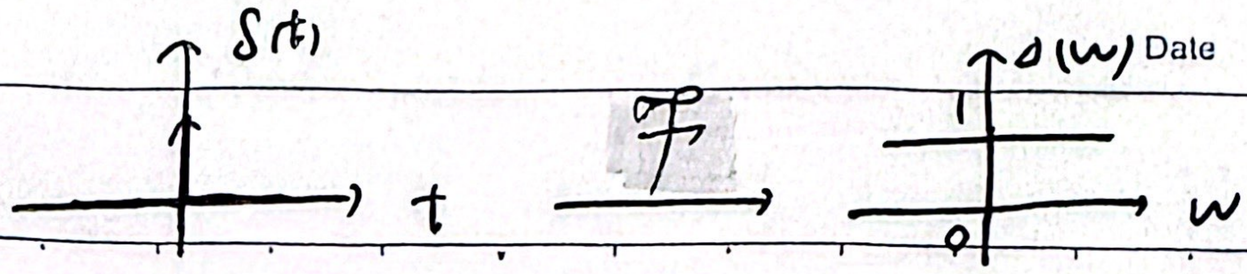
$$W_s=\Delta q\cdot V_{s0}=C(V_{s0}^2-V_0V_{s0})$$

$$\Delta E_C=\frac{1}{2}C(V_{s0}^2-V_0^2)$$

$$\Delta E=W_s-\Delta E_C=\frac{1}{2}C(V_{s0}-V_0)^2$$ 能量丢了

(开启KOKUYO)

原因：$\mathcal{F}\{\delta(t)\}=1$

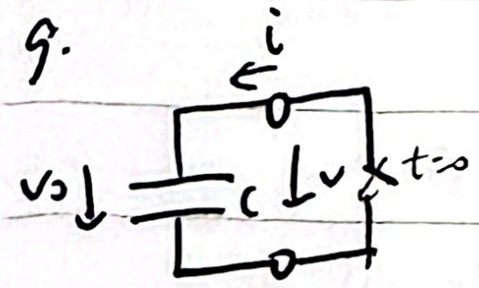


∴ 加一个冲激 ≃ 把所有频率分量都加上且这些频率密度=

$\omega \to \infty$，$\lambda \to 0$　$d_{AB} \ll \lambda$（准静态条件）不满足

不能被抽象成电路了

e.g.

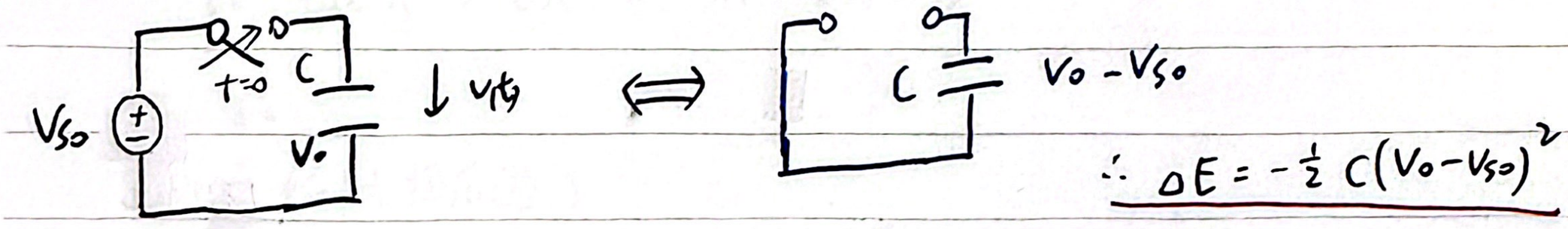


$v_c(0^-)=V_0$　$E_c(0^-)=\frac{1}{2}CV_0^2$

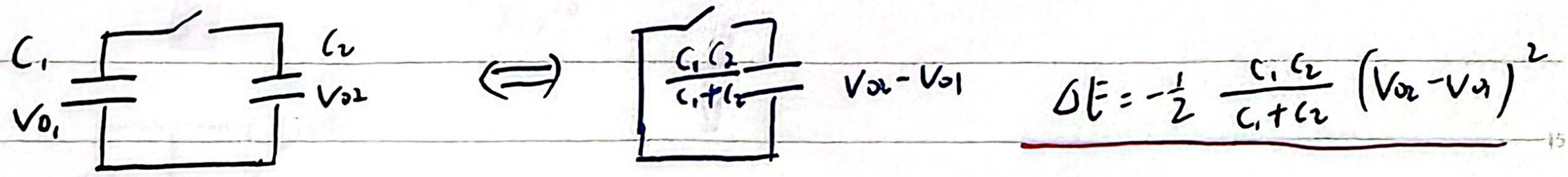
$v_c(0^+)=0$　$E_c(0^+)=0$

$\Delta E=-\frac{1}{2}CV_0^2$ 以电磁辐射形式被瞬间释放

人人等效电路看E损：



$\therefore \Delta E=-\frac{1}{2}C(V_0-V_{S0})^2$



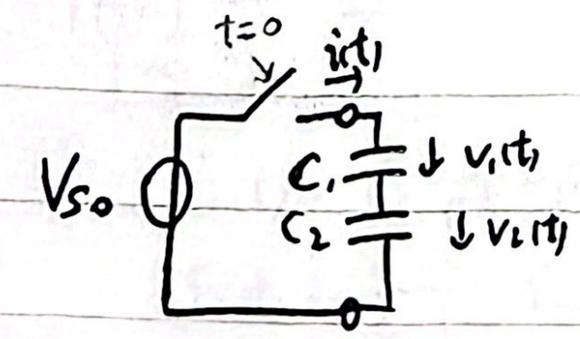
$\Delta E=-\frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_{02}-V_{01})^2$

No.

Date

第8讲 一阶时域

回顾：电容分压电路（无 v_0）



$$i(t) = C_{in}\frac{d}{dt}v_{in}(t) = \frac{C_1C_2}{C_1+C_2}\frac{d}{dt}V_{in}(t)$$

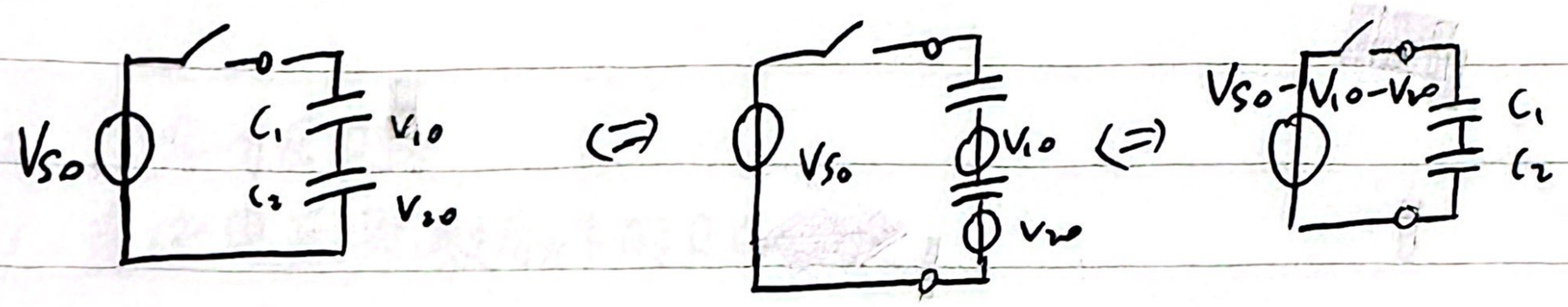
$$v_1(t) = v_1(0^-) + \frac{1}{C_1}\int_{0^-}^{t} i$$

$$= \frac{C_2}{C_1+C_2}V_{in}(t)$$

$$v_{2(t)} = \frac{C_1}{C_1+C_2}V_{in}(t)$$

一. 有初始电压的电容

1. 等效电路

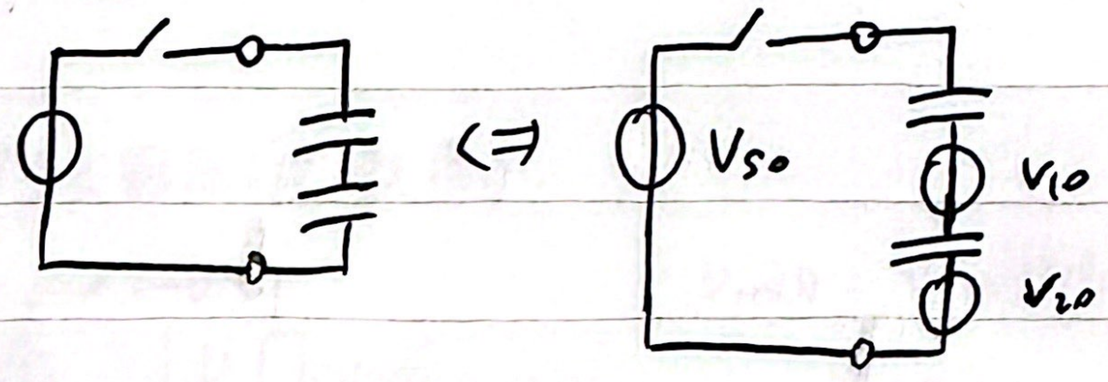


$$\Rightarrow V_{in0}(t) = (V_{S0} - V_{10} - V_{20})\,U(t)$$

$$V_1(t) = \frac{C_2}{C_1+C_2}V_{in0}(t) + V_{10} = \frac{C_2}{C_1+C_2}V_{S0} + \frac{C_1}{C_1+C_2}V_{10} - \frac{C_2}{C_1+C_2}V_{20}$$

$\Rightarrow$

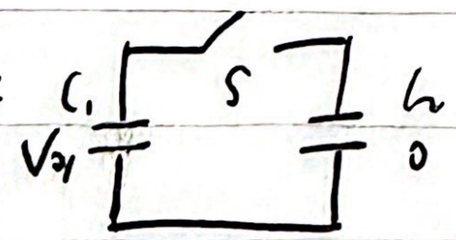
2. 叠加定理



三个源分别独立起作用，得…

$$\Rightarrow \text{叠加：}\quad V_1(t) = \frac{C_2}{C_1+C_2}V_{S0} + \frac{C_1}{C_1+C_2}V_{10} - \frac{C_2}{C_1+C_2}V_{20}$$

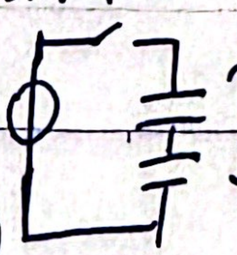
另例：



S闭，$v_1 = ?$

法一）电荷守恒 $C_1V_{01} = C_1v_1 + C_2v_2 \overset{v_1=v_2}{=\!=} (C_1+C_2)v_1 \Rightarrow v_1 = \frac{C_1V_{01}}{C_1+C_2}$

法二）等效 $\Leftrightarrow$



$$v_1 = -\frac{C_2}{C_1+C_2}v_{01} + v_{01}$$

$$= \frac{C_1}{C_1+C_2}v_{01}$$

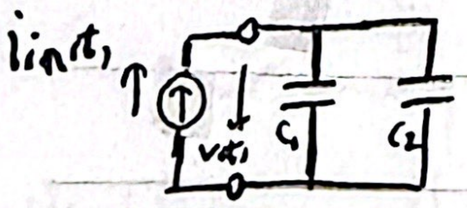
（要会调整正负向）

KOKUYO

No.

Date

电容分流电路：



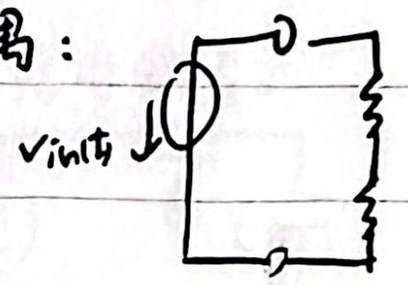
$$C_{in} = C_1 + C_2$$

$$v(t) = \frac{1}{C_{in}} \int_{-\infty}^{t} i$$

$$i_1(t) = C_1 \frac{d}{dt} v(t) = \frac{C_1}{C_1 + C_2} i_{in}(t)$$

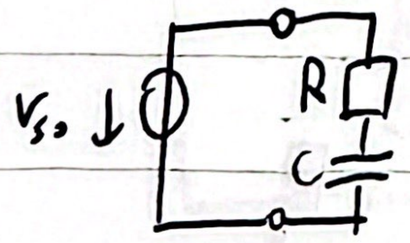
$$i_2(t) = \frac{C_2}{C_1 + C_2} i_{in}(t)$$

对偶：



二. 阻容分压电路

1. 直流电压源激励下的RC串联

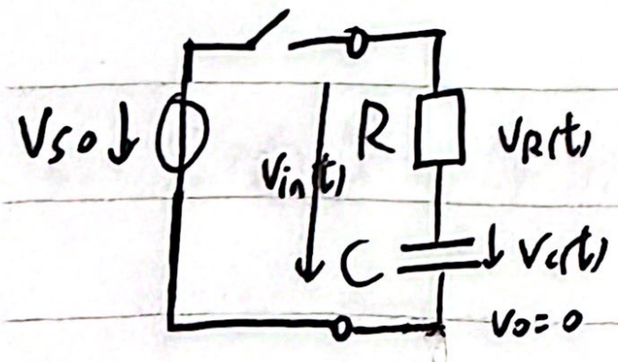


（直流信号可理解为 $t=-\infty$ 时加载的，（的 v_0 给 ∞ 长 $+C$ 已衰减至0，∴直流分析不考虑初值问题）

$$V_{in}(t) = V_{S0}, \quad V_R(t) = V_{R0} \quad \underline{V_c(t) = V_{c0} = V_{S0} - V_{R0}}$$

$$i(t) = C \frac{d}{dt} V_c(t) = 0 \quad (\text{直流开路})$$

2. 阶跃电压源激励



$$V_{in}(t) = V_{S0}\, U(t)$$

$$= R\, i(t) + \frac{1}{C} \int_{-\infty}^{t} i \qquad (t > 0)$$

$$\Rightarrow \quad i(t) + \frac{1}{RC} \int_{-\infty}^{t} i = \frac{V_{S0}}{R} U(t)$$

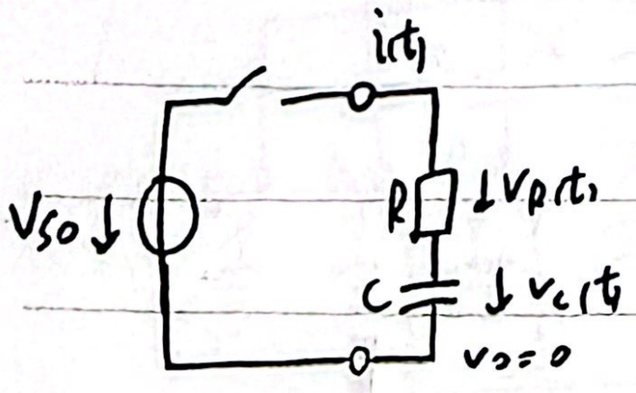
$$\Rightarrow \frac{d}{dt} i(t) + \frac{1}{\tau} i(t) = \frac{V_{S0}}{R} \delta(t)$$

$$\Rightarrow \quad i(t) = \frac{V_{S0}}{R} e^{-\frac{t}{\tau}} U(t) \qquad (\tau = RC)$$

No.

Date

3. 零状态响应 (指 $V_0=0$)



$v_{in}(t) = V_{S0}\,U(t)$

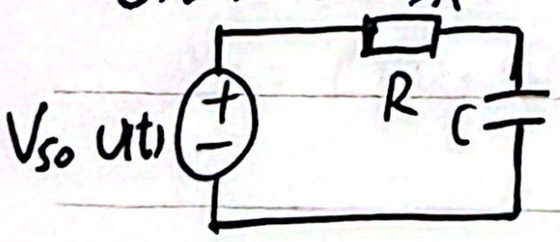
$i(t) = \frac{V_{S0}}{R} e^{-\frac{t}{\tau}} U(t)$

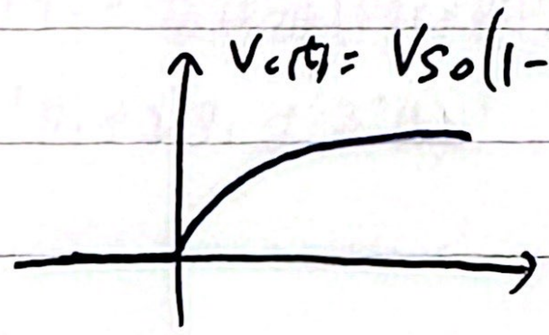
$v_R(t) = R\,i(t) = V_{S0} e^{-\frac{t}{\tau}} U(t)$

$v_c(t) = v_{in}(t) - v_R(t) = V_{S0}(1-e^{-\frac{t}{\tau}})\,U(t)$

(理解: $t=0$, 瞬间高频, 短路; 后来直流, 开路)

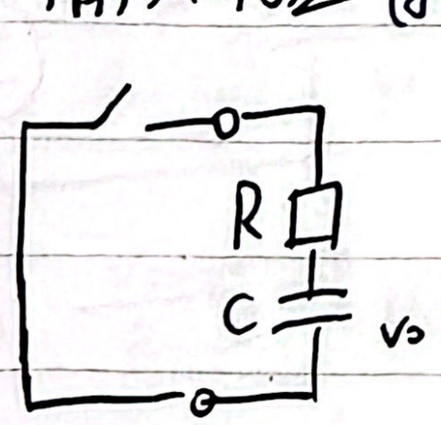
电容充电曲线:

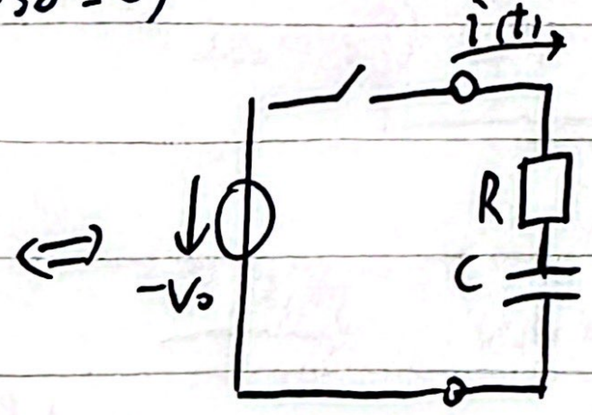




5τ后, 认为 C 充电结束, 进入稳态

4. 零输入响应 (指 $V_{S0}=0$)





$v_{in}(t) = -V_0\,U(t)$

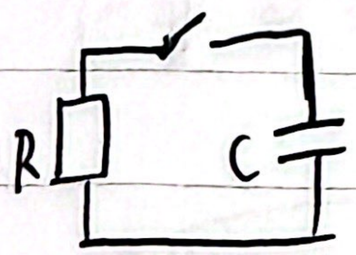
$i(t) = \frac{-V_0}{R} e^{-\frac{t}{\tau}} U(t)$

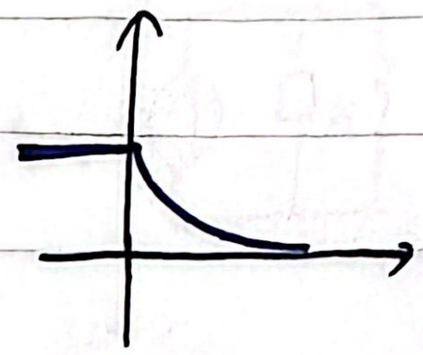
$v_R(t) = -V_0 e^{-\frac{t}{\tau}} U(t)$

$v_c(t) = -V_0(1-e^{-\frac{t}{\tau}})\,U(t) + V_0$

$= \begin{cases} V_0 & (t<0) \\ V_0 e^{-\frac{t}{\tau}} & (t>0) \end{cases}$

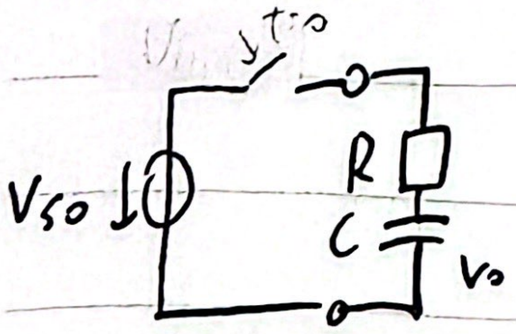
电容放电:





5τ后, ...

5. 全响应 = 零状态响应 + 零输入响应

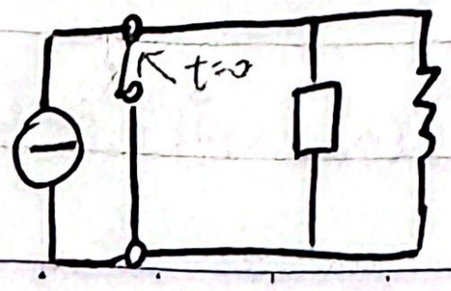


$v_{in}(t) = (V_{S0}-V_0)\,U(t) + V_0$

$v_R(t) = (V_{S0}-V_0)\,e^{-\frac{t}{\tau}} U(t)$

$v_c(t) = (V_{S0}-V_0)(1-e^{-\frac{t}{\tau}})\,U(t) + V_0$

又叫做:



(零状~: 电感充磁
零输~: 放)

No.

Date

第九讲 正弦稳态

复习：三角函数

(1) $\cos^3\alpha = \cos^2\alpha \cdot \cos\alpha = \frac{1+\cos 2\alpha}{2} \cdot \cos\alpha = \frac{\cos\alpha + \frac{\cos 3\alpha + \cos\alpha}{2}}{2}$

$= \frac{3\cos\alpha + \cos 3\alpha}{4}$ $\alpha = \omega t$ → 意义：计系统3次非线性，会产生激波分量($\cos\omega t$)和三次谐波分量($\cos 3\omega t$)

$\sin^3\alpha = \frac{3\sin\alpha - \sin 3\alpha}{4}$

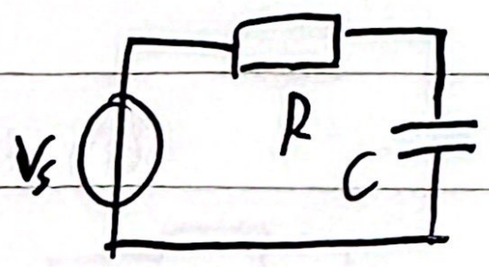
$\cos^2\alpha = \frac{1+\cos 2\alpha}{2}$ 产生直流分量和二次谐波分量

(2) 同频叠加

$A\cos\omega t + B\sin\omega t = \sqrt{A^2+B^2}\cos(\omega t - \varphi)$ $\varphi = \arctan\frac{B}{A}$ (默认$A>0$)

一. 相量法

1. RC分压：$t=-\infty$时加正弦激励



$V_{S1}(t) = V_{Sp}\cos(\omega t + \varphi_S)$ 设 $V_c(t) = V_{cp}\cos(\omega t + \varphi_c)$

电路方程：$RC\frac{dV_c(t)}{dt} + V_c(t) = V_S(t)$ 代入V_S、V_c求解，

$\Rightarrow V_{cp}\sqrt{1+(\omega RC)^2}\cos(\omega t + \varphi_c + \arctan(\omega RC)) = V_{Sp}\cos(\omega t + \varphi_S)$

$\Rightarrow V_{cp} = \frac{V_{Sp}}{\sqrt{1+(\omega RC)^2}}$ $\varphi_c = \varphi_S - \varphi$ $\varphi = \arctan(\omega RC)$

$v_c(t) = \frac{V_{Sp}}{\sqrt{1+(\omega RC)^2}}\cos\left(\omega t + \varphi_S - \arctan(\omega RC)\right)$

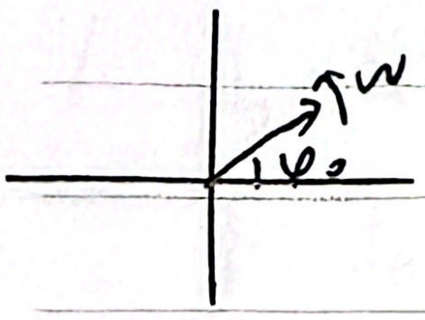
if 阶数更高、繁琐 ⇒ 引入相量法

KOKUYO

No.

Date

what is 相量,



$\varphi(t) = \omega t + \varphi_0$

$s(t) = x(t) + jy(t)$ $\quad x(t) = A_p \cos\varphi(t)$ $\quad y(t) = A_p \sin\varphi(t)$

$\Rightarrow s(t) = A_p e^{j(\omega t+\varphi_0)}$

$\frac{d}{dt}s = \omega A_p e^{j(\omega t+\varphi_0+\frac{\pi}{2})}$ (微分 ↔ 乘 $j\omega$)

$\int_{-\infty}^{t} s = \frac{A_p}{\omega} e^{j(\omega t+\varphi_0-\frac{\pi}{2})}$ (积分 ↔ 除以 $j\omega$)

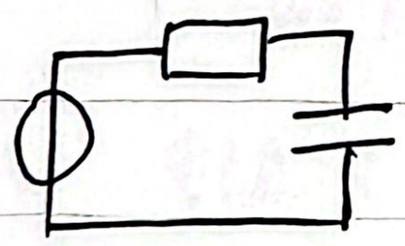
相量引入: (∵不产生新频率分量 对于线时不变)
有公共项 $e^{j\omega t}$, 可消掉

旋转矢量表示: $\vec{I}(t) = I_p e^{j(\omega t+\varphi_I)}$ $\quad \vec{V}(t) = V_p e^{j(\omega t+\varphi_V)}$

$= (I_p e^{j\varphi_I}) \cdot e^{j\omega t}$ $\quad = \underline{(V_p e^{j\varphi_V})} \cdot e^{j\omega t}$

相量电流 $\dot{I} = I_p e^{j\varphi_I} = I_p \angle \varphi_I$ $\quad \dot{V} = V_p e^{j\varphi_V} = V_p \angle \varphi_V$

应用: e.g.



$v_s(t) = \dot{V}_s e^{j\omega t}$ $\quad v_c(t) = \dot{V}_c e^{j\omega t}$

时域微分方程: $RC\frac{d}{dt}v_c(t) + v_c(t) = v_s(t)$ → 代入,

$\Rightarrow j\omega RC\dot{V}_c e^{j\omega t} + \dot{V}_c e^{j\omega t} = \dot{V}_s e^{j\omega t}$

$\Rightarrow$ 频域相量方程, $j\omega RC\dot{V}_c + \dot{V}_c = \dot{V}_s$ (⊛)

if 输入不是单频正弦波, $\dot{V}_s \to \dot{V}_s(j\omega)$, 有 $j\omega RC\dot{V}_c(j\omega) + \dot{V}_c(j\omega) = \dot{V}_s(j\omega)$

求解 ⊛: $\quad j\omega\tau\dot{V}_c + \dot{V}_c = \dot{V}_s$

$\Rightarrow \left(\sqrt{1+(\omega\tau)^2}\angle \arctan\omega\tau\right) \cdot (V_{cp}\cdot\angle\varphi_c) = V_{sp}\angle\varphi_s$

复数 × 复数 = (幅度相乘)∠(角度相加)

$\Rightarrow V_{cp}\sqrt{1+(\omega\tau)^2}\cdot\angle(\varphi_c + \arctan\omega\tau) = V_{sp}\angle\varphi_s$

$\Rightarrow V_{cp} = \frac{V_{sp}}{\sqrt{1+(\omega\tau)^2}}$ $\quad \varphi_c = \varphi_s - \arctan\omega\tau$

No.

Date

二. 电容. 电感的相量法表述

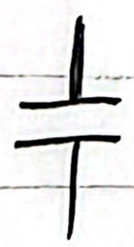
1.

$i_G = G \cdot v_G$

⇓

$\dot{I}_G = G\dot{V}_G$

电导: G



$i_C = C\frac{dv_C}{dt}$

⇓

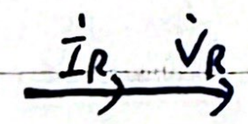
$\dot{I}_C = j\omega C\dot{V}_C = jB\dot{V}_C$

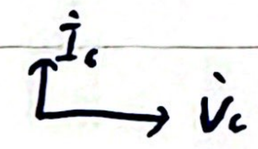
电纳: $B = \omega C$

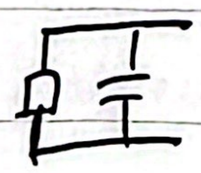
G与B统称为导纳 Y

$\frac{\dot{I}}{\dot{V}} = Y = G + jB$

$Re Y$ $I_m Y$

对于 —□— : $\dot{I} = G\dot{V}$ 

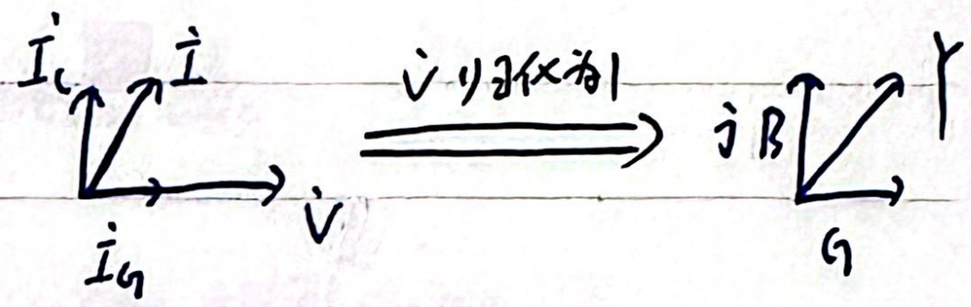
—| |— : $\dot{I} = jB\dot{V}$ 

RC 并联: 

$\dot{I} = \dot{I}_G + \dot{I}_C$

$= G\dot{V}_G + j\omega C\dot{V}_C = (G+jB)\dot{V} = Y\dot{V}$

$Y = G + j\omega C = \sqrt{G^2 + (\omega C)^2} \angle \arctan\frac{\omega C}{G}$



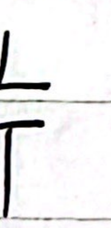
2.

$v_R = R i_R$

⇓

$\dot{V}_R = R\dot{I}_R$

电阻: R



$v_C = \frac{1}{C}\int i_C dt$

⇓

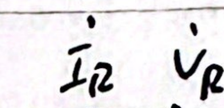
$\dot{V}_C = \frac{1}{j\omega C}\dot{I}_C = jX \cdot \dot{I}_C$

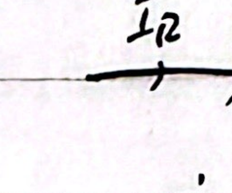
电抗 $X = \frac{-1}{\omega C}$

R. X 统称为阻抗 Z

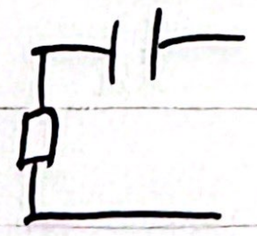
$\frac{\dot{V}}{\dot{I}} = Z = R + jX$

$Z = \frac{1}{Y}$

—□— : 

—| |— : 

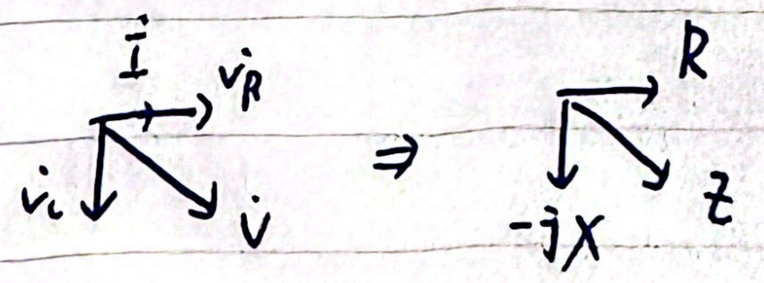
RC 串联:



$\dot{V} = \dot{V}_R + \dot{V}_C$

$= (G + jX)\dot{I} = Z\dot{I}$

$Z = R + \frac{1}{j\omega C} = \sqrt{R^2 + \frac{1}{(\omega C)^2}} \angle -\arctan\frac{1}{\omega RC}$



电感: 对偶即可

KOKUYO

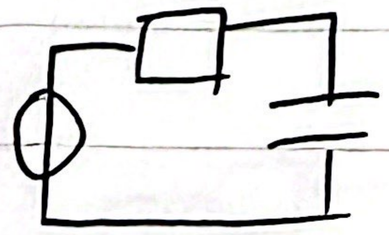
Date

2. 串联：$Z = \sum_k Z_k$

并联：$Y = \sum_k Y_k$

4. ∵ 微积分方程 → 代数方程

∴ 有"分压""分流"关系

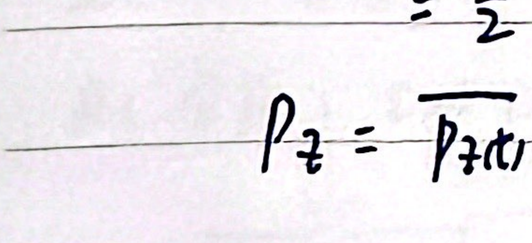


$$\dot{V}_C = \frac{jX}{R+jX}\dot{V}_S = \frac{1}{1+j\omega RC}\dot{V}_S$$

$$= \frac{e^{-j\arctan\omega RC}}{\sqrt{1+(\omega RC)^2}}\dot{V}_S$$

if $V_S = V_{Sp}\cos(\omega t+\varphi_S) \Rightarrow v_C = \frac{V_{Sp}}{\sqrt{1+(\omega RC)^2}}\cos(\omega t+\varphi_S-\arctan\omega RC)$

5. 功率分析



$$Z = \sqrt{R^2+\left(\frac{1}{\omega C}\right)^2}\, e^{-j\arctan\frac{1}{\omega RC}}$$

$$= |Z|e^{j\varphi_Z} = |Z|\cos\varphi_Z + j|Z|\sin\varphi_Z$$

$$\dot{I}_Z = \frac{\dot{V}_Z}{Z} = \frac{V_{SP}}{|Z|}e^{j(\varphi_S-\varphi_Z)} = I_{Zp}e^{j\varphi_I}$$

$$i_Z(t) = \frac{V_{SP}}{|Z|}\cos(\omega t+\varphi_S-\varphi_Z) = I_{Zp}\cos(\omega t+\varphi_I)$$

$$\Rightarrow p_Z(t) = i_Z(t)\cdot V_Z(t) = I_{Zp}\cos(\omega t+\varphi_I)\cdot V_{Zp}\cos(\omega t+\varphi_V)$$

$$= \frac{1}{2}V_{Zp}I_{Zp}\left[\cos(\varphi_V-\varphi_I)+\cos(2\omega t+\varphi_V+\varphi_I)\right]$$

$$P_Z = \overline{p_Z(t)} = \frac{1}{2}V_{Zp}I_{Zp}\cos(\varphi_V-\varphi_I) = \frac{1}{2}I_{Zp}^2\frac{|Z|\cos\varphi_Z}{R} = \frac{1}{2}I_R^2R$$

（合理∵电阻耗功率 电抗不耗）

{进一步考察，$P_Z = \frac{1}{2}V_{Zp}I_{Zp}\cos(\varphi_V-\varphi_I)$

功率因数角

（$S = \frac{1}{2}V_pI_p = V_{rms}I_{rms}$

视在功率 单位：VA伏安

不是有功率 ∵要考虑功率因数角）

频域上，定义复功率 $\hat{S} = \frac{1}{2}\dot{V}\dot{I}^*$

$$= \frac{1}{2}V_pI_p\angle(\varphi_V-\varphi_I) = Se^{j(\varphi_V-\varphi_I)}$$

$$= S\cos(\varphi_V-\varphi_I)+jS\sin(\varphi_V-\varphi_I)$$

$$= P+jQ$$

实功 虚功

(W) (Var 乏)

KOKUYO

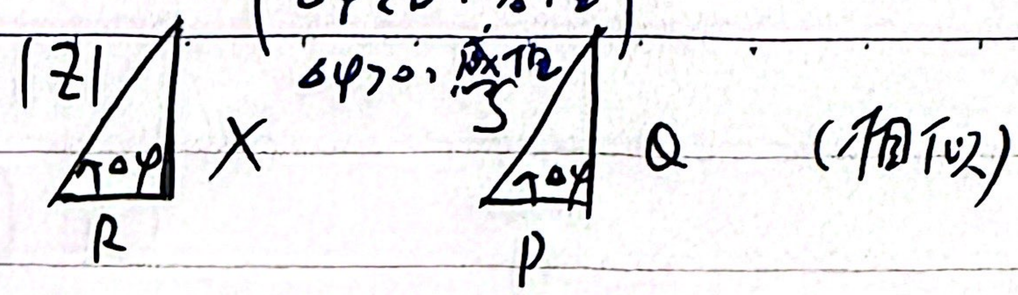
No.

Date

阻抗、功率三角形：

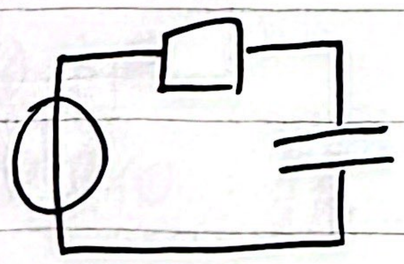
$$Z = \frac{\dot{V}}{\dot{I}} = \frac{V_p}{I_p} \angle(\varphi_V - \varphi_I) = R + jX$$

$$\hat{S} = S\angle(\varphi_V - \varphi_I) = P + jQ$$



$$\begin{cases} P = I_{rms}^2 R \\ Q = I_{rms}^2 X \\ S = I_{rms}^2 |Z| \end{cases}$$

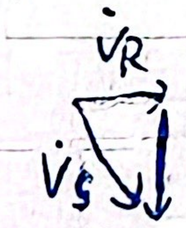
例：



$v_s(t) = 10\cos(2\pi f_0 t + \varphi_0)$ (V)　　$f_0 = 10^6$ Hz

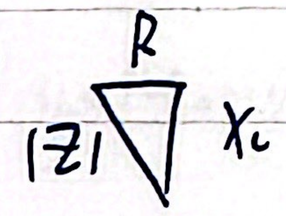
测得 $R = 600\Omega$，v_R 峰值 $= 6$V

求 v_c、C、S

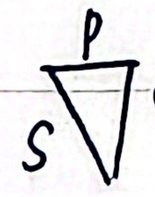
① 

（由I-V关系，$\dot{V}_C$ 比 $\dot{V}_S$ 滞后约37°）

$\therefore v_c(t) = 8\cos\left(2\pi f_0 t + \varphi_0 + \frac{37}{180}\pi\right)$

② 

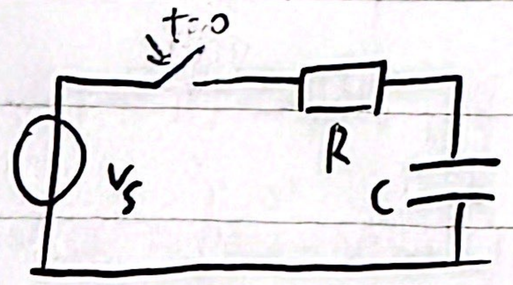
$\therefore X_C = -600\Omega = \frac{1}{-\omega_0 C} \Rightarrow C = 199$ pF

③ 

$P = \frac{|\dot{V}_R|^2}{R} = 30$mW $\Rightarrow$ $S = 50$mV/A　单位！

三、三要素法

RC 分压：t=0 时加正弦激励



一般理论形式：$v_{in}(t) = \begin{cases} V_0 & t<0 \\ V_{SP}\cos(\omega t + \varphi_S) & t>0 \end{cases}$

$t>0$：$\frac{d}{dt}v_c(t) + \frac{1}{\tau}v_c(t) = \frac{1}{\tau}v_S(t)$ …

$\Rightarrow$ 引入三要素法：对于 $\tau\frac{d}{dt}x(t) + x(t) = s(t)$　$(t>0)$

$$x(t) = \underbrace{(x_0 - x_{\infty 0})e^{-\frac{t}{\tau}}}_{\text{瞬态响应}} + \underbrace{x_\infty(t)}_{\text{稳态响应}} \quad (t>0)$$

要素1：时间常数 τ　　$\tau = RC$　　$\tau = \frac{G}{L}$

2：初值 x_0　　V_0　　I_0

2'：终值 稳态响应 $x_\infty(t)$　　$x_\infty(t) = \left(\int_t^t e^{\frac{\lambda - t}{\tau}} s(\lambda) d\frac{\lambda}{\tau}\right) U(t)$

KOKUYO

No.

Date

how 求: $\begin{cases} \tau = RC \ / \ \tau = GL \\ x_0 = v_{C0} = v_0 \ / \ x_0 = i_{L0} = I_0 \\ x_{\infty}(t) = \left(\int_{-\infty}^{t} e^{\frac{\lambda - t}{\tau}} s(\lambda) d\frac{\lambda}{\tau} \right) u(t) \end{cases}$

esp. ① 直流: C 开 / L 短 ⇒ 求 $x_\infty(t)$

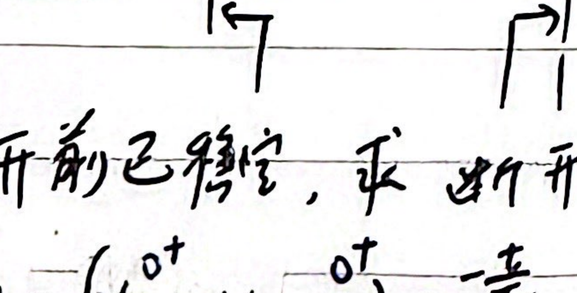
② 正弦波激励 相量法 (电容 → $j\omega C$，电感 → $j\omega L$，其他与纯电阻电路无区别，只是实数运算 → 复数运算)

求 $x_\infty(t)$ (导纳 阻抗)

③ 方波

④ 其他：猜

例1:



① $\tau = RC$

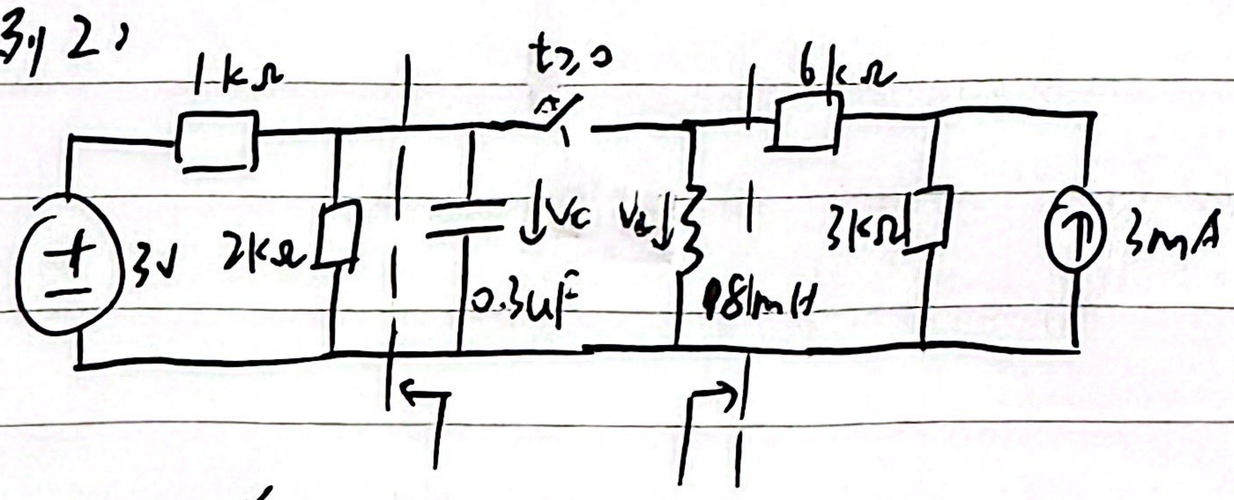
② $v_{C0} = V_0$

③ $v_{C\infty}(t) = \frac{V_{sp}}{\sqrt{1+(\omega\tau)^2}} \cos\left(\omega t + \varphi_s - \arctan(\omega\tau)\right)$

$v_C(t) = (v_{C0} - v_{C\infty 0}) e^{-\frac{t}{\tau}} + v_{C\infty}(t) \quad (t \geq 0)$

$= \cdots \quad (t \geq 0)$

例2:



断开前已稳定，求 断开后 $v_C(t)$ $v_L(t)$

$v_C(t) = (v_{C0}^{0^+} - v_{C\infty 0}^{0^+}) e^{-\frac{t}{\tau_C}} + v_{C\infty}(t)$ (严格证明)

$v_L(t) = (v_L(0^+) - v_{L\infty}(0^+)) e^{-\frac{t}{\tau_L}} + v_{L\infty}(t)$ (t>0) (后一律，取 0^+ 为初值，得 t>0 的结果)

问题求:

① C 开，L 短

$v_C(0^+) = 0 \quad i_L(0^+) = \frac{3V}{1k\Omega + 3k\Omega + 6k\Omega} 3mA = 1\,mA$

② C 开，L 短 $\quad v_{C\infty}(t) = 2V \quad i_{L\infty}(t) = 1mA$

③ $\tau_C = RC$ R 等效戴源 $\frac{1.2}{1+2} = \frac{2}{3}k\Omega$ $\tau_L = GL$ G 等效源阻 $\frac{1}{3+6} = \frac{1}{9}$

$$\Rightarrow v_c(t) = \begin{cases} 0 & t<0 \\ 2\left(1-e^{-\frac{t}{2\times10^{-4}}}\right)\ V & t\geq 0 \end{cases}$$

$$i_L(t) = \begin{cases} 4\ mA & t<0 \\ \left(1+3e^{-\frac{t}{2\times10^{6}}}\right) mA & t\geq 0 \end{cases}$$

$$v_L(t) = L\frac{d}{dt}i_L(t) = \begin{cases} 0 & t<0 \\ -27\,e^{\frac{-t}{2\times10^{6}}}\ V & t\geq 0 \end{cases}$$

也可直接求 $v_L(t)$:

① $v_L(0^-)=0$ $\quad v_L(0^+) = -4mA\times 6k\Omega + -(4-3)mA\times 3k\Omega = -27\ V$

② $v_L(\infty)=0$

③ τ_L = 同前

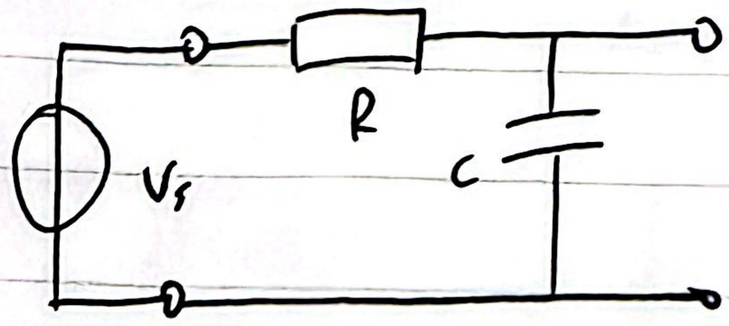
KOKUYO

Date

第十讲 一阶时频

一. 从二端口网络视角看分压

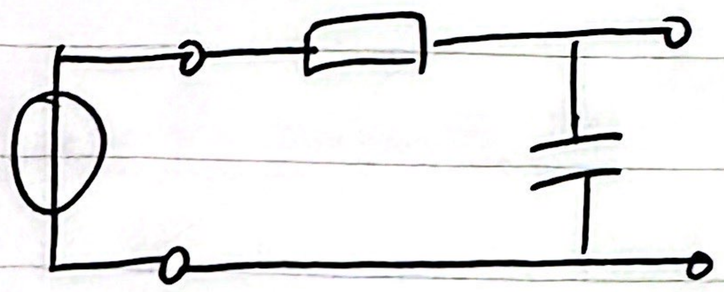
1. RC分压网络的电压传递函数（频域分析）



$$\underline{H_{VC}(j\omega)} = \frac{\dot{V}_C(j\omega)}{\dot{V}_R(j\omega)} = \frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}} = \frac{1}{1+j\omega RC}$$

$$= \frac{1}{\sqrt{1+(\omega RC)^2}} e^{-j\arctan\omega RC}$$

$$= \underline{A(\omega)e^{j\varphi(\omega)}}$$

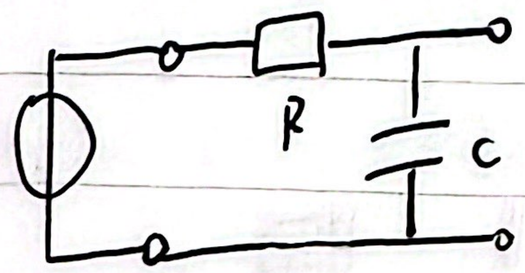


$$\underline{H_{VR}(j\omega)} = \frac{j\omega RC}{1+j\omega RC} = \frac{\omega RC}{\sqrt{1+(\omega RC)^2}} e^{j(\frac{\pi}{2}-\arctan\omega RC)}$$

$$= \underline{A(\omega)e^{j\varphi(\omega)}}$$

V_s可以不是单弦正弦波 $\xrightarrow{Fourier}$ 多个频率分量，$H_{总} = \sum H_k$

对上述电容分压：

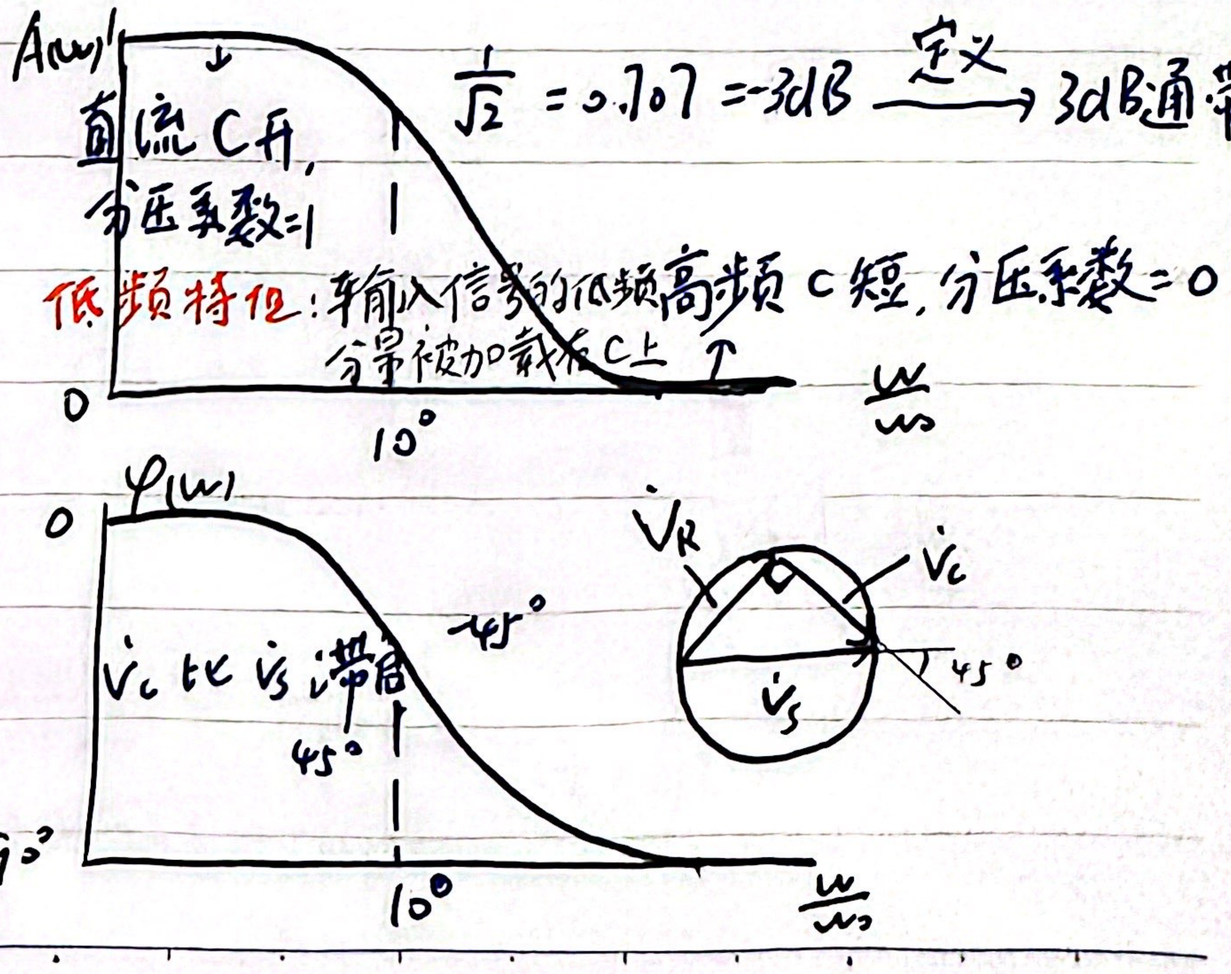


$$H(j\omega) = \frac{1}{\sqrt{1+(\omega RC)^2}} e^{-j\arctan\omega RC}$$

$$= \underbrace{A(\omega)}_{\text{幅频特性}} e^{j\underbrace{\varphi(\omega)}_{\text{相频特性}}}$$

$$A(\omega) = \frac{1}{\sqrt{1+(\omega RC)^2}} = \frac{1}{\sqrt{1+(\omega\tau)^2}}$$

$$= \frac{1}{\sqrt{1+(\frac{\omega}{\omega_0})^2}} \quad (\omega_0 = \frac{1}{\tau} = \frac{1}{RC})$$



$$\varphi(\omega) = -\arctan\omega RC = -\arctan\omega\tau$$

$$= -\arctan\frac{\omega}{\omega_0}$$

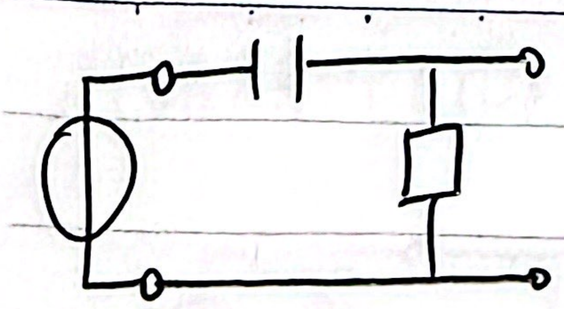
折线近似 → 伯特图 $\begin{cases} A_{dB}(\omega) \\ \tilde{\varphi}(\omega) \end{cases}$ 略

(含画)

No.

Date

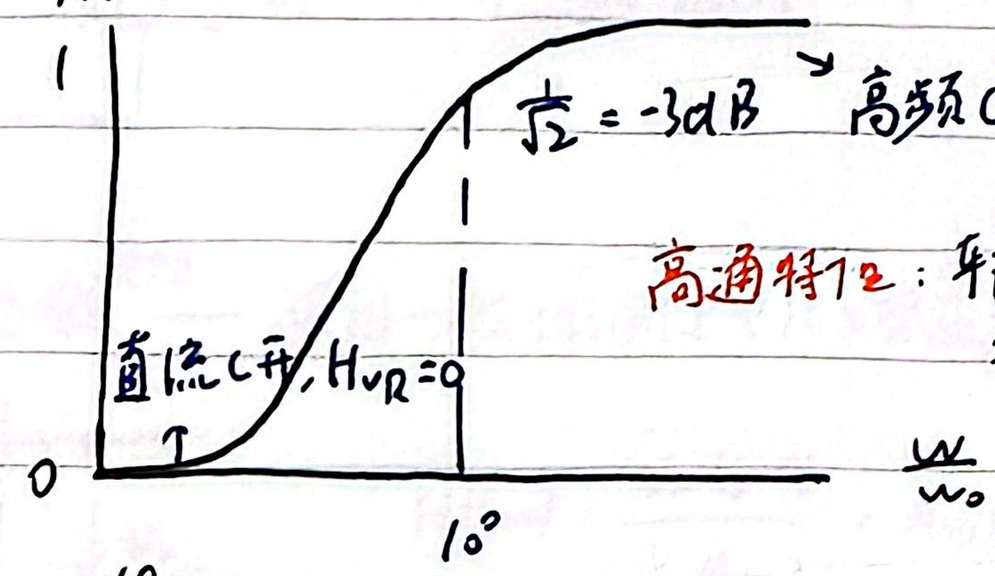
电阻分压：



$$H_{vR}(jw)=\frac{jwRC}{1+jwRC}=\frac{wRC}{\sqrt{1+(wRC)^2}}e^{j\left(\frac{\pi}{2}-\arctan(wRC)\right)}$$

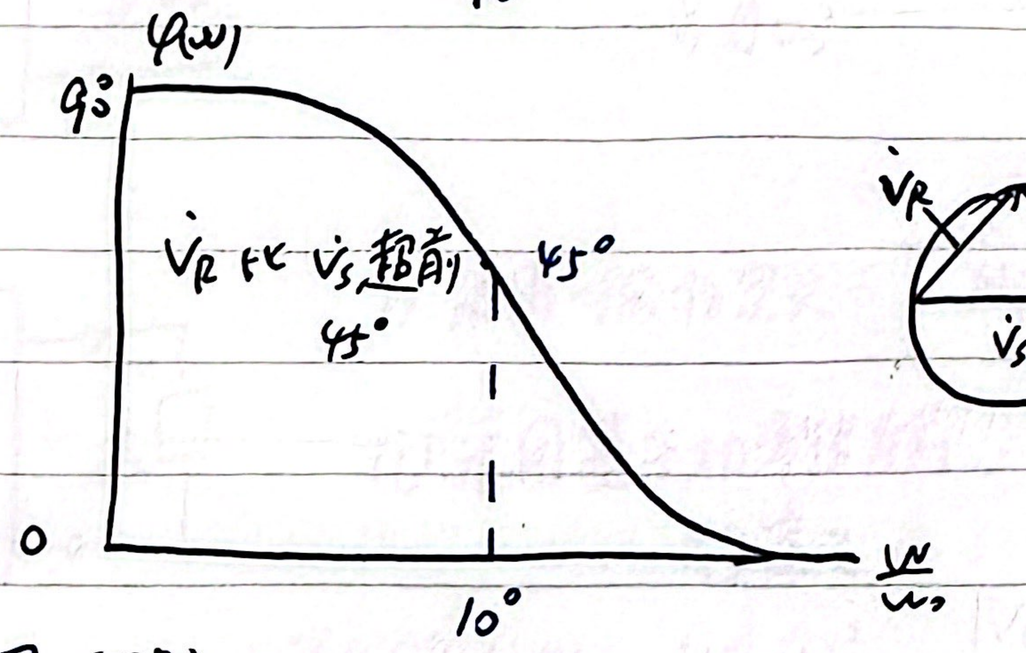
$$=A(w)e^{j\varphi(w)}$$

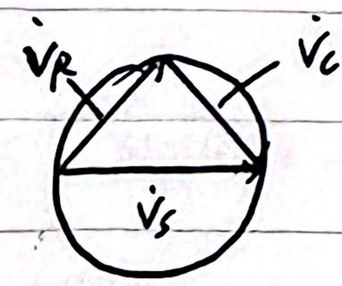
$$A(w)=\frac{\frac{w}{w_0}}{\sqrt{1+(\frac{w}{w_0})^2}}$$



高通特征：车输入信号中的高频分量被加载在电阻上

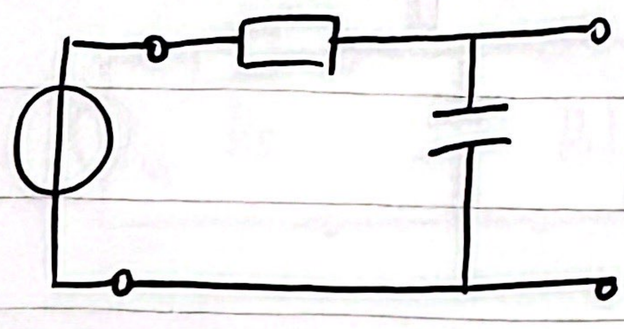
$$\varphi(w)=\frac{\pi}{2}-\arctan\frac{w}{w_0}$$





折线近似 → 伯特图 (略)

总结：RC 分压网络：一阶低通/高通滤波器



$$H_{vc}(jw)=\frac{1}{\sqrt{1+(wt)^2}}e^{-j\arctan wt}$$

$\tau=RC$, $w_0=\frac{1}{\tau}$

由低通定义

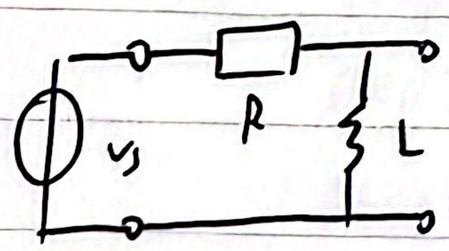
3dB 带宽 $BW_{3dB}=\frac{1}{2\pi\tau}=\frac{w_0}{2\pi}$

$$H_{vR}(jw)=\frac{wt}{\sqrt{1+(wt)^2}}e^{j(\frac{\pi}{2}-\arctan wt)}$$

3dB 步点：$f_{3dB}=\frac{1}{2\pi\tau}=\frac{w_0}{2\pi}$

由高通定义

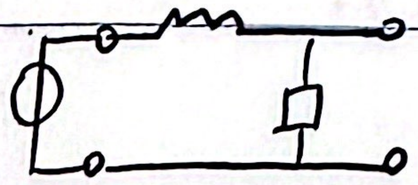
对偶地，



$$H_{vL}=\frac{jwL}{R+jwL}=\frac{jwt}{1+jwt}$$

$\tau=GL$

(高通) $f_{3dB}=\frac{1}{2\pi\tau}=\frac{w_0}{2\pi}$



$$H_{vR}=\frac{R}{R+jwL}=\frac{1}{1+jwt}$$

(低通) $BW_{3dB}=\frac{1}{2\pi\tau}=\frac{w_0}{2\pi}$

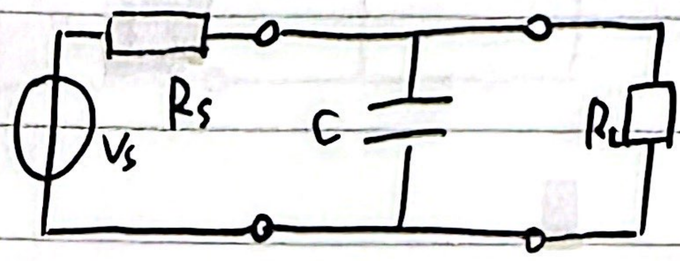
KOKUYO

No.

Date

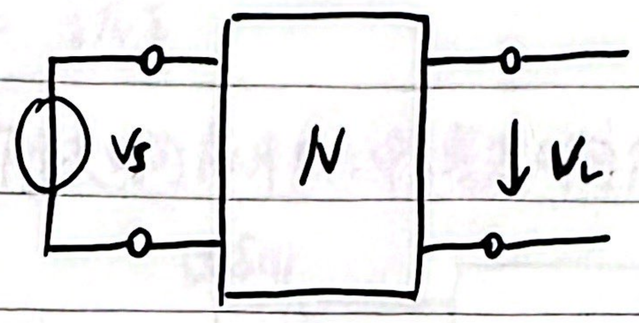
例：一阶系统传递函数的一般形式

（指只有一个 C(L)



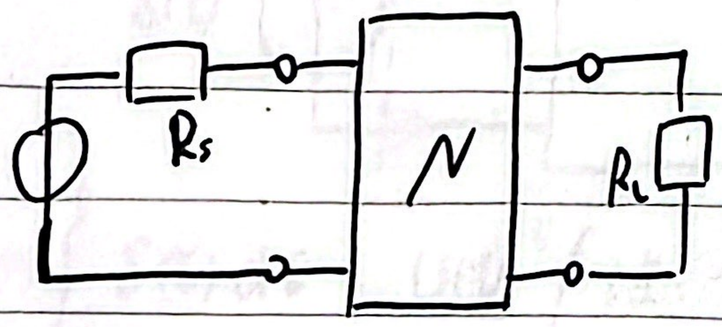
求 3dB 通带

分析：① 无信源内阻负载（电阻） —— 采用一般相(本征电压土增益)定义



$$H(j\omega)=\frac{\dot{V}_L(j\omega)}{\dot{V}_s(j\omega)}=A_{\omega}(j\omega)$$

② 有 ··



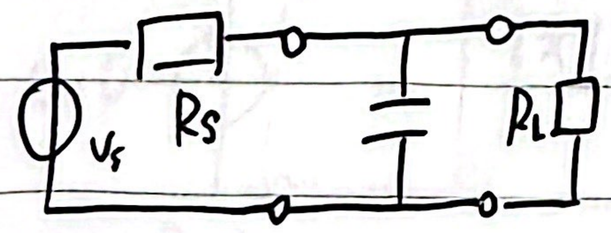
i) 采用一般相定义：$H(j\omega)=\frac{\dot{V}_L(j\omega)}{\dot{V}_s(j\omega)}$

ii) 采用基于功率传输的定义：$S_{21}=H(j\omega)=2\sqrt{\frac{R_s}{R_L}}\cdot\frac{\dot{V}_L(j\omega)}{\dot{V}_s(j\omega)}$

此定义的好处：$|H(j\omega)|^2=\frac{|\dot{V}_L(j\omega)|^2/2R_L}{|\dot{V}_s(j\omega)|^2/8R_s}=\frac{P_L(j\omega)}{P_{s,max}(j\omega)}$

$=G_T(j\omega)$ 功率土增益

回到本题：



$$H(j\omega)=\frac{\dot{V}_L(j\omega)}{\dot{V}_s(j\omega)}=\frac{\frac{1}{\frac{1}{R_L}+j\omega C}}{R_s+\frac{1}{\frac{1}{R_L}+j\omega C}}$$ 这么写方便些

$$=\frac{R_L}{R_s+R_L}\cdot\frac{1}{1+j\omega\frac{R_sR_L}{R_s+R_L}C}=H_0\frac{1}{1+j\omega\tau}$$

H_0：中、二步负点的传递系数（即求 $H(j0)$）

τ：RC　　$R=\frac{R_sR_L}{R_s+R_L}$（R：电容看出去的总电阻）（∵ ①短，R_s、R_L 并在C上 ∴ $R=\frac{R_sR_L}{R_s+R_L}$）

以后可直接写答案：step1：判断这是一阶低通 ∴形式为 $H(j\omega)=H_0\frac{1}{1+j\omega\tau}$

step2：$H_0=H(j0)=\frac{R_L}{R_L+R_s}$

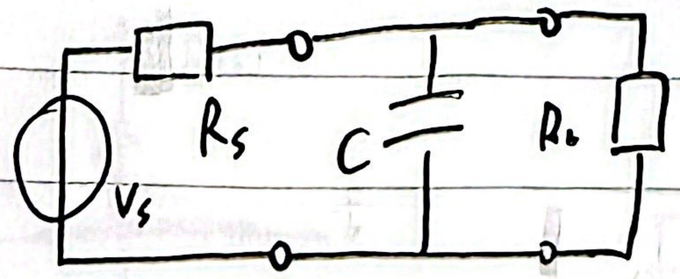
step3：$\tau=RC$　　$R=\frac{R_sR_L}{R_s+R_L}$

综上，$H(j\omega)=\cdots$

KOKUYO

Date

计对此图求基于功率传输的传递函数



$$H(j\omega) = 2\sqrt{\frac{R_s}{R_L}} \cdot \frac{\dot{V}_L(j\omega)}{\dot{V}_s(j\omega)}$$

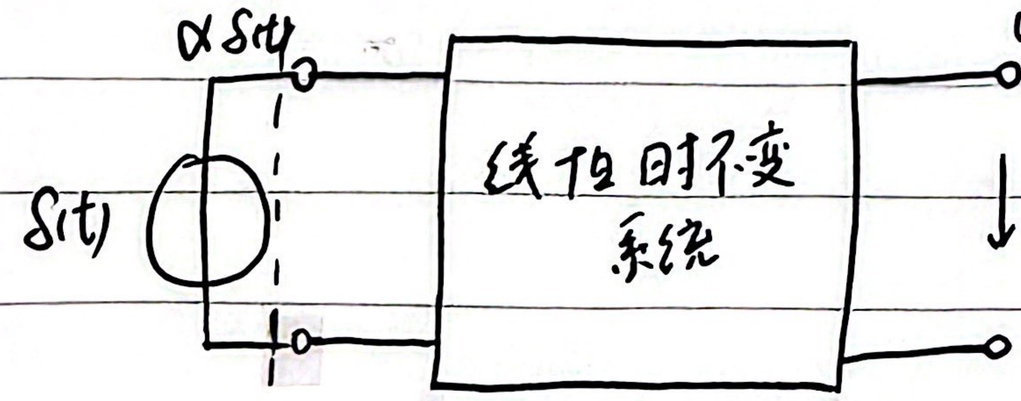
$$= \frac{2\sqrt{R_s R_L}}{R_s + R_L} \frac{1}{1+j\omega\tau} = H_0 \frac{1}{1+j\omega\tau}$$

H_0：中心频点的传递系数 $H(j0) = \frac{2\sqrt{R_s R_L}}{R_s + R_L}$

$\left(H(j0) = 2\sqrt{\frac{R_s}{R_L}} \cdot \frac{R_L}{R_L + R_s} = \frac{2\sqrt{R_s R_L}}{R_s + R_L}\right)$

$BW_{3dB} = \frac{1}{2\pi\tau}$

二. 阶跃响应和冲激响应



$\int_{-\infty}^{t} s(\tau)\,d\tau = u(t)$

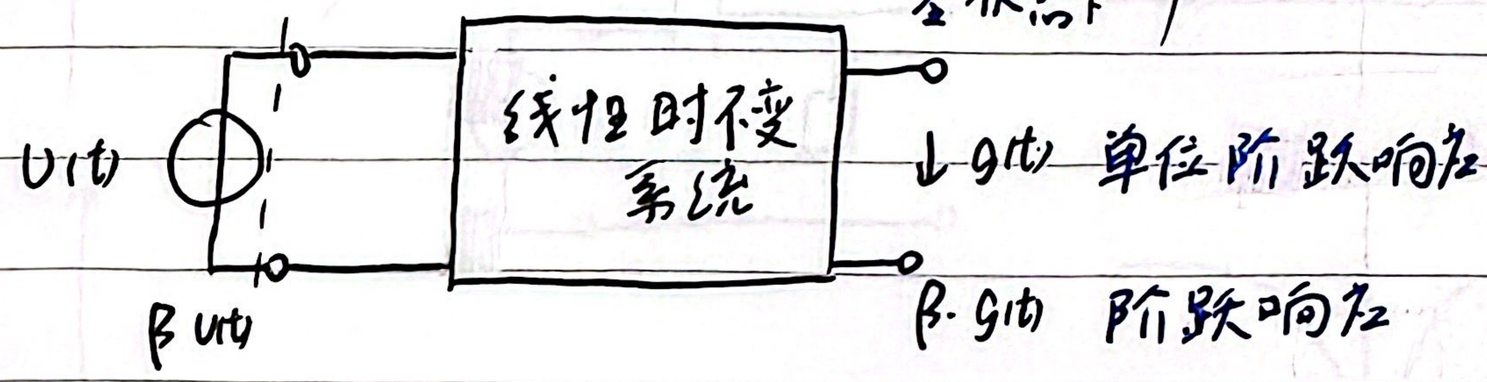
$\frac{d}{dt} u(t) = \delta(t)$

（注：分析阶跃和冲激激励加入和响应时，要使线性时不变系统在零状态下）

$\int_{-\infty}^{t} h(\tau)\,d\tau = g(t)$

微积分的关系

$\frac{d}{dt} g(t) = h(t)$

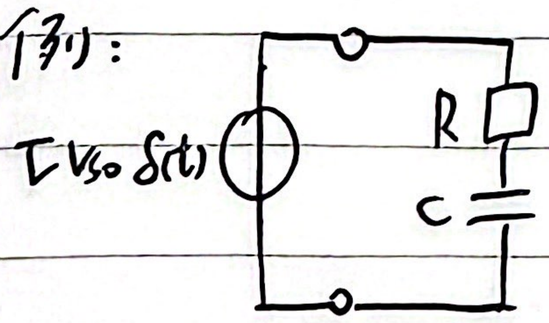


传递函数 & 冲激响应 是一对傅立叶变换对

$$H(j\omega) = \int_{-\infty}^{+\infty} h(t) e^{-j\omega t}\,dt$$

（频域）　　（时域）

例：

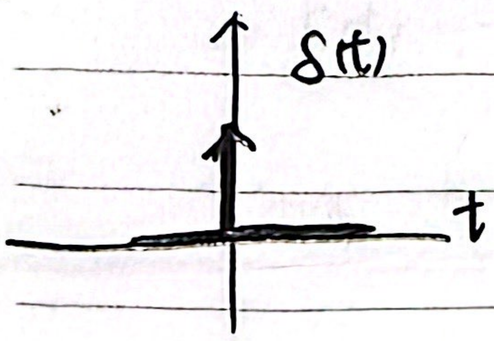


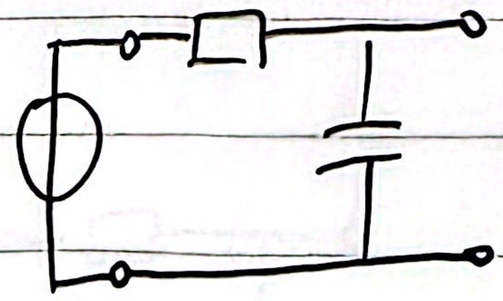
$\begin{cases} v_c(t) = \cdots \\ v_R(t) = \cdots \\ H_{v_c}(j\omega) = \cdots \end{cases}$ 单位响应 $\longrightarrow$ $\begin{cases} h_{v_c}(t) = \cdots \\ h_{v_R}(t) = \cdots \end{cases}$

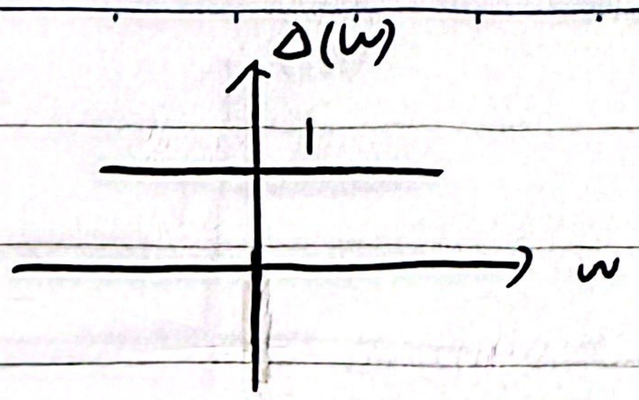
验证：$\mathcal{F}[h_{v_c}(t)] = H_{v_c}(j\omega)$

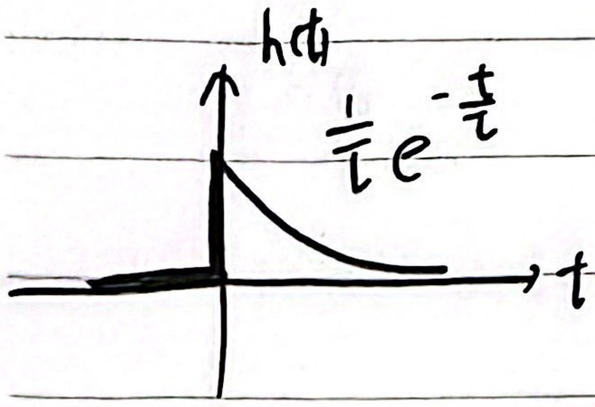
KOKUYO

一阶低通的时域及频域特性

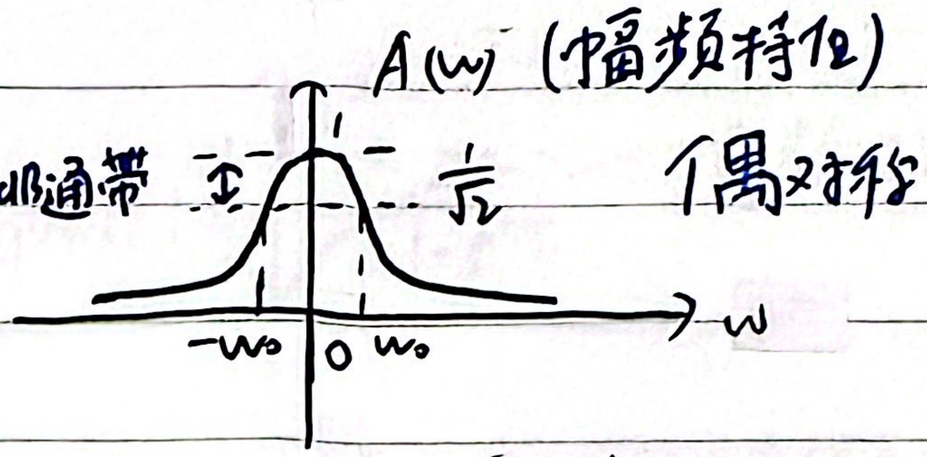


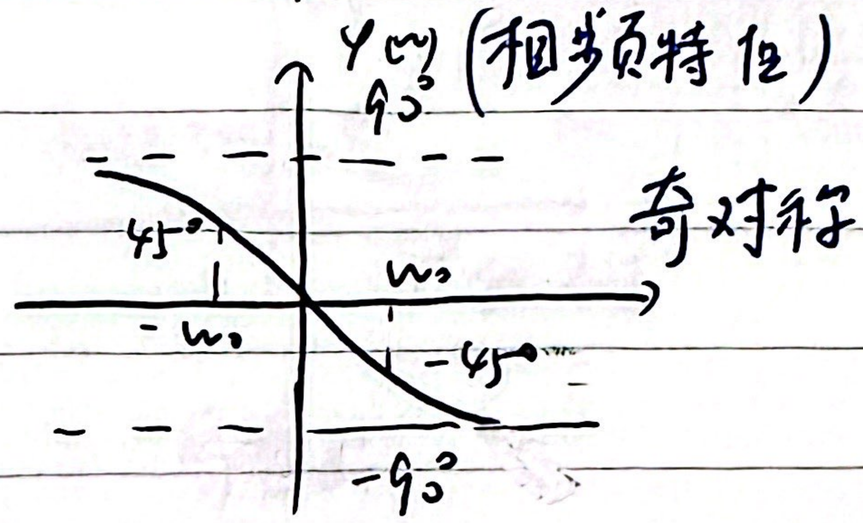




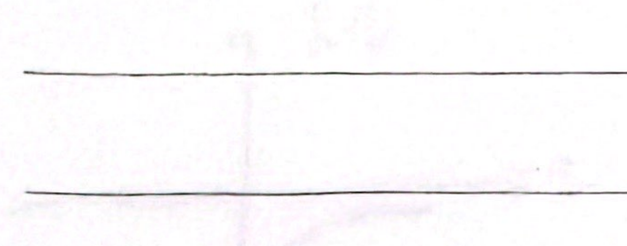


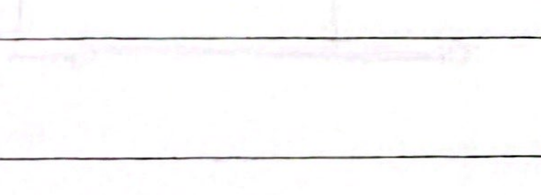
Fourier

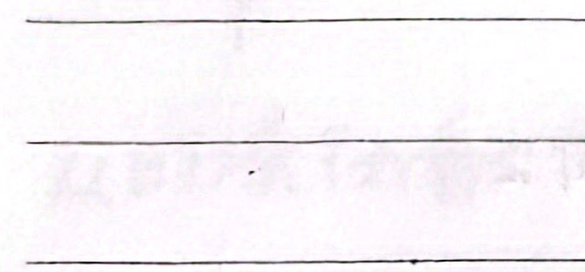


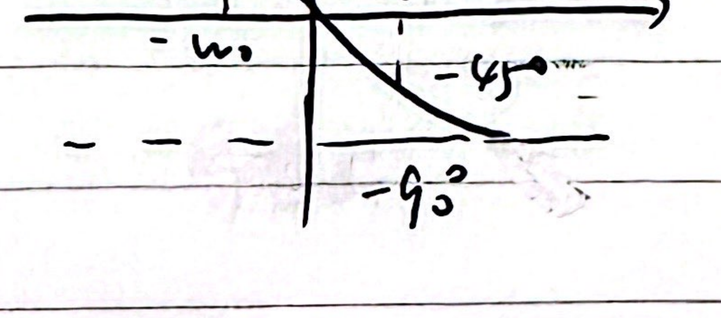


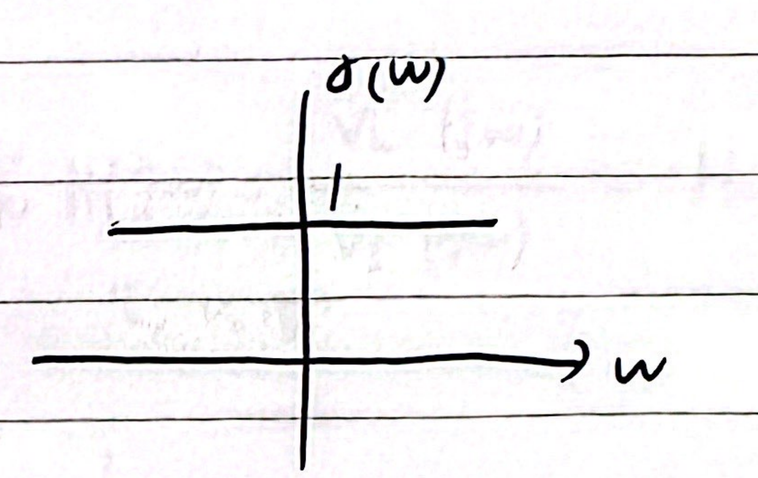
一阶高通









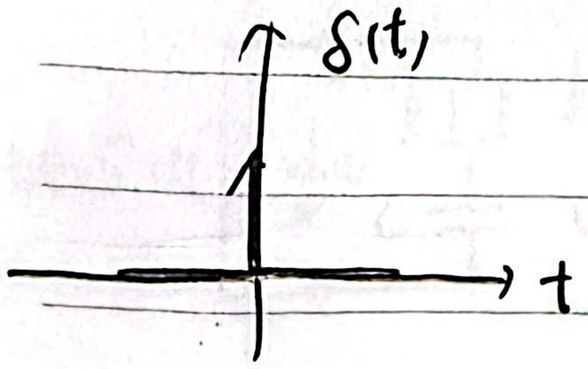


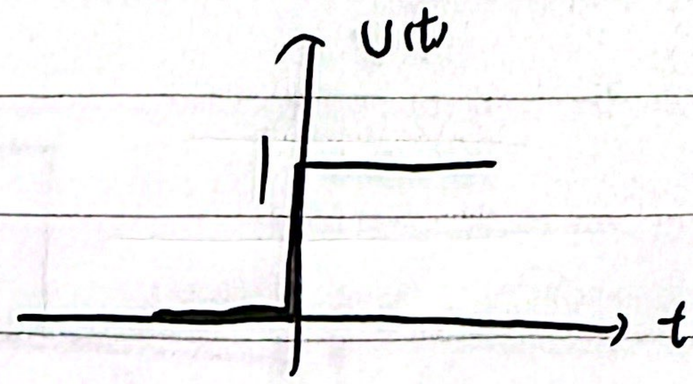
高通：中心频点为∞

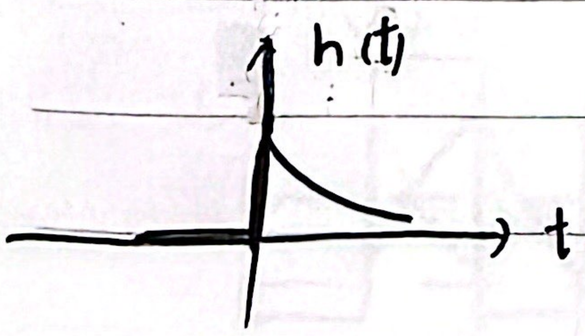
零频点相位有180°跳变

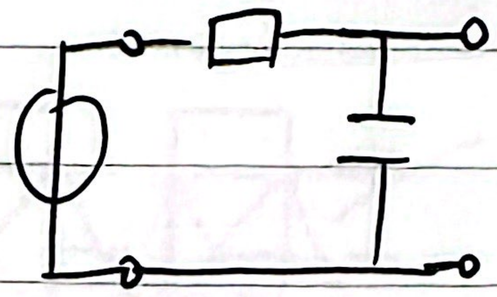
KOKUYO

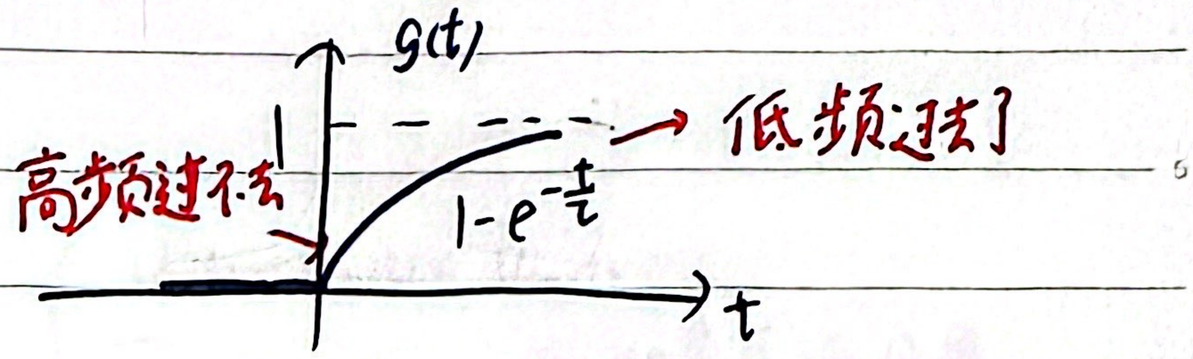
对比：冲激 / 阶跃

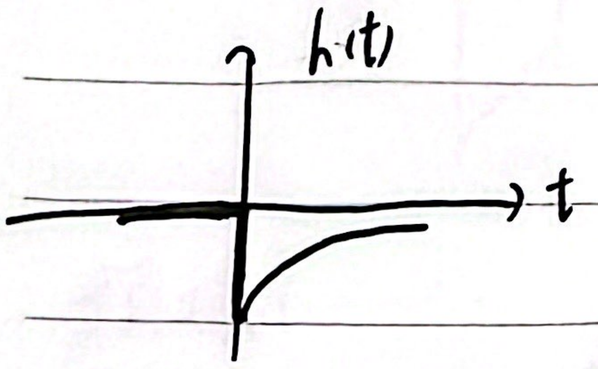


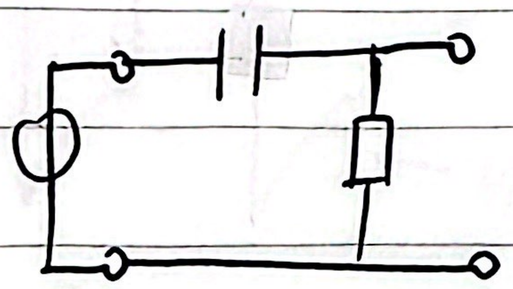


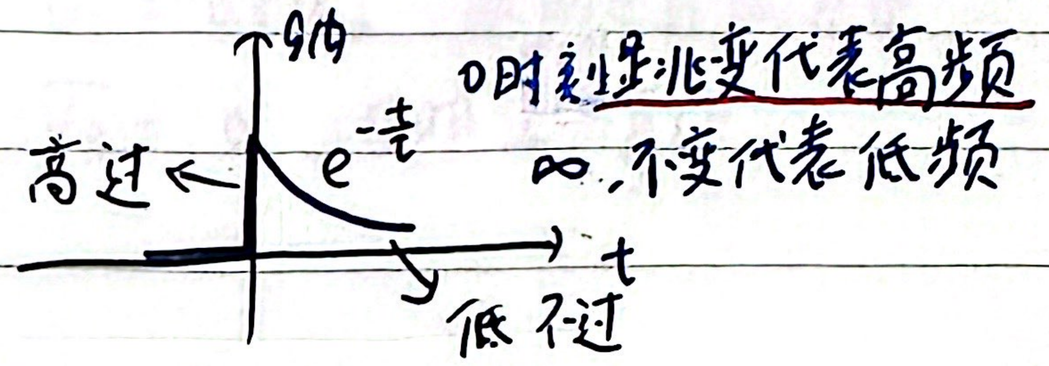








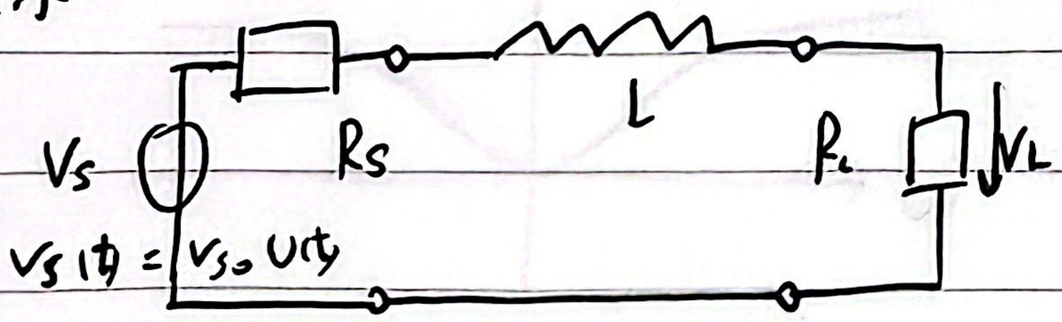




从时域不难看出低通 or 高通

易看出

练：



求 $H(j\omega) = \dfrac{\dot{V}_L(j\omega)}{\dot{V}_S(j\omega)} = H_0 \dfrac{1}{1+j\omega\tau}$

τ, $v_L(t)$

解：① $H_0 = \dfrac{R_L}{R_L+R_S}$ $\tau = GL$ $G = \dfrac{1}{R_L+R_S}$ ② $BW_{3dB} = \dfrac{1}{2\pi\tau}$

③ $v_{L\infty}(t) = \dfrac{R_L}{R_L+R_S} V_{S0}$

$\tau = \dfrac{L}{R_L+R_S}$

$V_{L\infty}(0^+) = \dfrac{R_L}{R_L+R_S} V_{S0}$

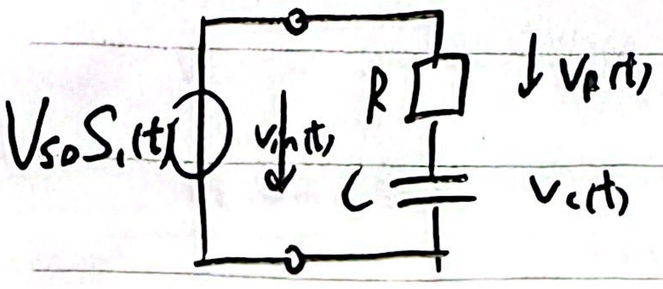
$V_L(0^+) = 0$

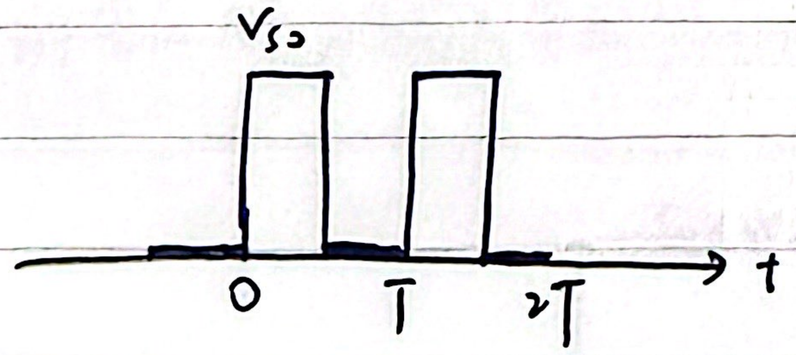
$\therefore V_L(t) = \left(V_L(0^+) - V_{L\infty}(0^+)\right) e^{-\frac{t}{\tau}} + V_{L\infty}(t)$

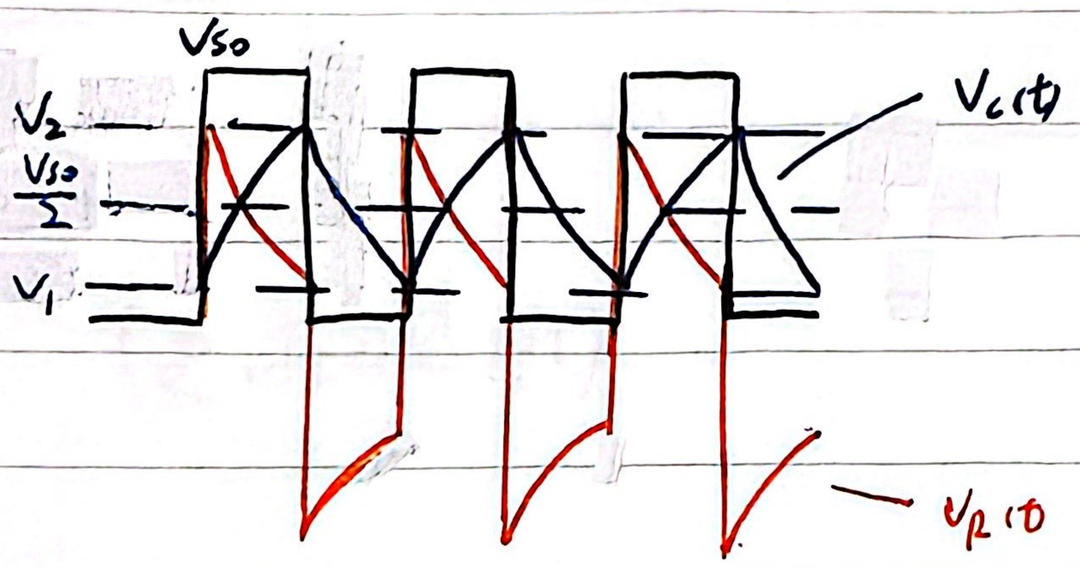
$= \dfrac{R_L}{R_L+R_S} V_{S0} \left(1 - e^{-\frac{R_L+R_S}{L}t}\right) \quad (t>0)$

$\Rightarrow V_L(t) = \dfrac{R_L}{R_L+R_S} V_{S0} \left(1 - e^{-\cdots t}\right) U(t)$

三. RC分压 — 方波信号激励

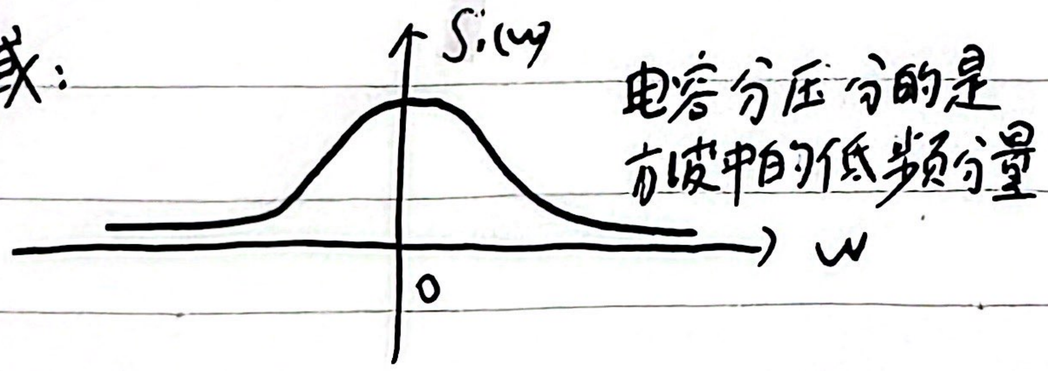


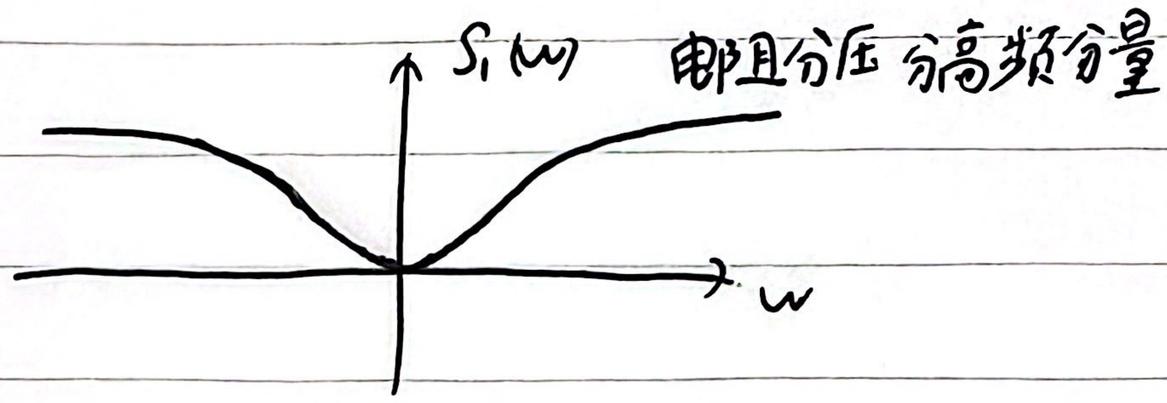




$$\begin{cases} V_C(t) + V_R(t) = V_{in}(t) \\ \overline{V_C} = 0.5V_{S0} \quad \text{取的是直流分量} \\ \overline{V_R} = 0 \quad \text{电阻上无直流} \end{cases}$$

频域：



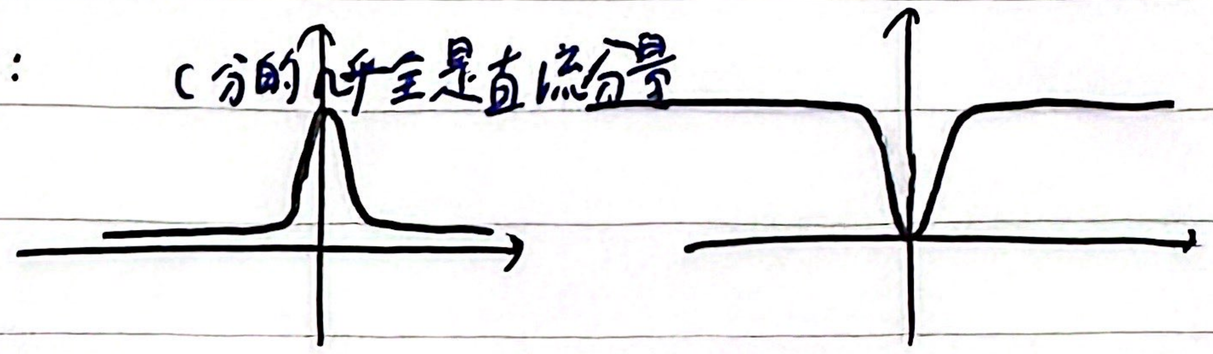


极端1：T很大，f_{3dB}很小　　（极大，C为耦合电容，隔直通交

时域：$V_1 \overset{\text{Taylor}}{\sim} \frac{V_{S0}}{2}\left(1 - \frac{0.5T_0}{2\tau}\right)$　　（V_1，V_2很接近）

$V_2 \overset{\text{Taylor}}{\sim} \frac{V_{S0}}{2}\left(1 + \frac{0.5T_0}{2\tau}\right)$

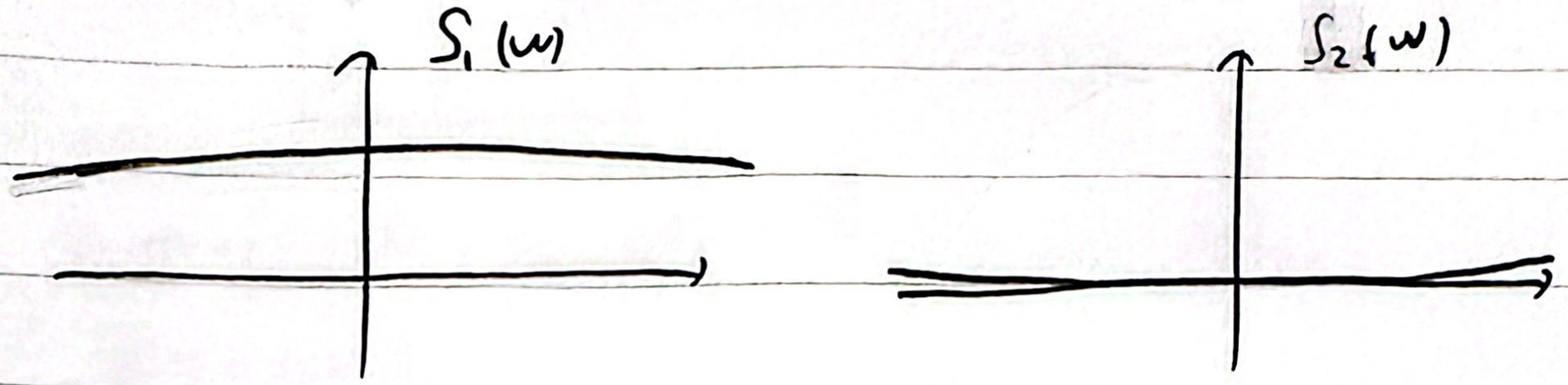
频域：



极端2：τ很小，f_{3dB} 很大

$V_1 \simeq 0$　　C 的电压几乎为输入电压

$V_2 \simeq V_{in}$



(更详细见 ppt)

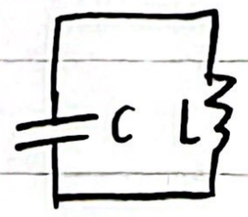
NO.

Date

第十一讲 二阶RLC

一. LC谐振腔

1.



$$v_C(t) = v_L(t) = L\frac{di_L(t)}{dt} = L\frac{di(t)}{dt}$$

$$v_C(t) = \frac{1}{C}\int i_L = \frac{1}{C}\int i(t)\,dt$$

$$\Rightarrow \frac{d^2i(t)}{dt^2} + \frac{i(t)}{LC} = 0$$

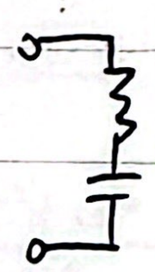
$\Rightarrow i(t) = I_0\cos(\omega_0 t + \varphi_0)$ $\quad \omega_0 = \frac{1}{\sqrt{LC}}$ LC谐振腔的自由振荡频率

$v(t) = -Z_0 I_0 \sin(\omega_0 t + \varphi_0)$ $\quad Z_0 = \omega_0 L = \frac{1}{\omega_0 C} = \sqrt{\frac{L}{C}}$ LC谐振腔的特征阻抗

$\Rightarrow E_C(t) + E_L(t) = \frac{1}{2}Cv^2(t) + \frac{1}{2}Li^2(t) = const$

即纯LC谐振腔内无能量损耗，L与C的电能相互转化

串联谐振腔：

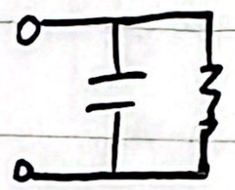


$$Z = Z_L + Z_C = j\omega L + \frac{1}{j\omega C} = j\sqrt{\frac{L}{C}}\left(\omega\sqrt{LC} - \frac{1}{\omega\sqrt{LC}}\right)$$

$$= jZ_0\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)$$

$\omega = \omega_0$ 时，$Z = 0$，相当于短路

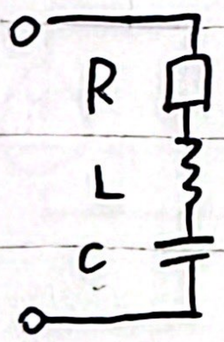
并联



$$Y = Y_C + Y_L = j\omega C + \frac{1}{j\omega L} = jY_0\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) \quad Y_0 = \sqrt{\frac{C}{L}}$$

$\omega = \omega_0$ 时，$Y = 0$，开路

串联RLC系统参量：



$$Z = R + jZ_0\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) = R\left[1 + jQ\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right] \quad (s = j\omega)$$

$$\begin{cases} \omega_0 = \frac{1}{\sqrt{LC}} & (\text{谐振频率}) \\ Z_0 = \sqrt{\frac{L}{C}} = \omega_0 L = \frac{1}{\omega_0 C} & (\text{特征阻抗}) \\ Q = \frac{Z_0}{R} = \frac{\text{虚功}}{\text{实功}} = \frac{\text{电抗}}{\text{电阻}} = \frac{\omega_0 L}{R} & (\text{品质因数}) \\ \zeta = \frac{1}{2Q} = \frac{R}{2Z_0} & (\text{阻尼系数}) \end{cases}$$

定义 $\zeta = \frac{R}{2Z_0}$

$\Rightarrow Z(s) = \cdots R\frac{s^2 + 2\zeta\omega_0 s + \omega_0^2}{2\zeta\omega_0 s}$

对偶，并联RLC：

$$\begin{cases} \omega_0 = \frac{1}{\sqrt{LC}} \\ Y_0 = \sqrt{\frac{C}{L}} \\ Q = \frac{Y_0}{G} \\ \zeta = \frac{1}{2Q} \end{cases}$$

KOKUYO

No.

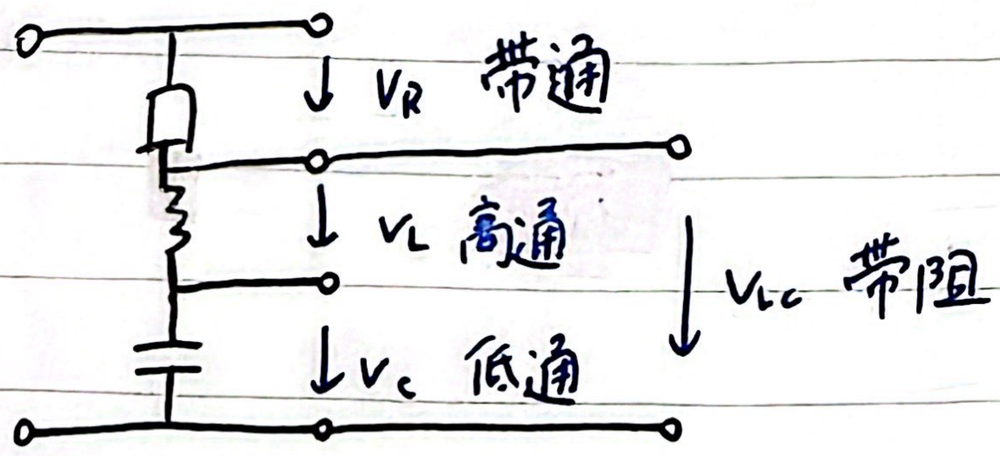
Date

2.

回顾：一阶：$H_{LP1}(j\omega) \xrightarrow{\text{低通}} H_0 \frac{\omega_0}{s+\omega_0}$　　$BW_{3dB} = \frac{1}{2\pi\tau}$

$H_{HP1}(j\omega) \xrightarrow{\text{高通}} H_0 \frac{s}{s+\omega_0}$　　$f_{3dB} = \frac{1}{2\pi\tau}$

二阶：



$H_{VC} = H_0 \frac{\omega_0^2}{s^2+2\zeta\omega_0 s+\omega_0^2}$　　中心频点：0 → $H_0 = 1$

$H_{VL} = H_0 \frac{s^2}{\cdots}$　　中…：∞ → $H_0 = 1$

$H_{VR} = H_0 \frac{2\zeta\omega_0 s}{\cdots}$　　中…：ω_0 → $H_0 = 1$

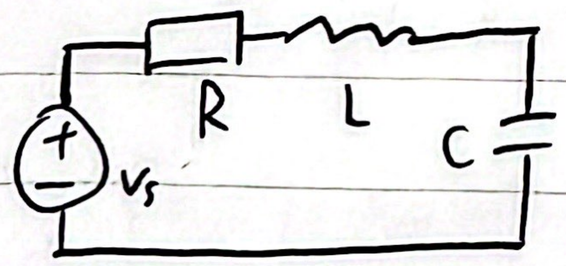
$H_{VLC} = H_0 \frac{s^2+\omega_0^2}{\cdots}$　　中：0或∞ → $H_0 = 1$

例：$H(s) = H_0 \frac{s^2+\omega_1^2}{s^2+2\zeta\omega_0 s+\omega_0^2}$　$(\omega_1 \gg \omega_0)$

$= \begin{cases} H_0 \left(\frac{\omega_1}{\omega_0}\right)^2 & s=0 \\ 0 & s=j\omega_1 \\ H_0 & s=\infty \end{cases}$　　$\because H(j\omega) \gg H(j\infty)$ ∴ 低通

（∴带阻不合适）

3. RLC 串联电路的一般性分析：状态方程法



$\begin{cases} C\frac{dv_c}{dt} = i_L = i_L \\ L\frac{di_L}{dt} = v_L = V_s - i_L R - v_c \end{cases}$（要状态变量的微分）

$\Rightarrow \frac{d}{dt}\begin{pmatrix} v_c \\ i_L \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{C} \\ -\frac{1}{L} & -\frac{R}{L} \end{pmatrix}\begin{pmatrix} v_c \\ i_L \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{V_s}{L} \end{pmatrix}$　⇒ 解出 v_c, i_L ⇒ 用恒压源 v_c 恒流源 i_L 替代

KOKUYO

No.

Date

LTI系统状态方程的一般形式：$\frac{d}{dt}\vec{x}(t) = A\vec{x}(t) + \vec{s}(t)$

一阶 $\frac{d}{dt}x = ax + s$

$\Rightarrow x(t) = x_\infty(t) + [x(t_0^+) - x_\infty(t_0^+)]e^{a(t-t_0)}$ ⟶ 高阶

$\vec{x}(t) = \vec{x}_\infty(t) + [\vec{x}(t_0^+) - \vec{x}_\infty(t_0^+)]e^{A(t-t_0)}$

(稳态响应) (自然态响应)

状态转移矩阵 e^{At} 求法同数学

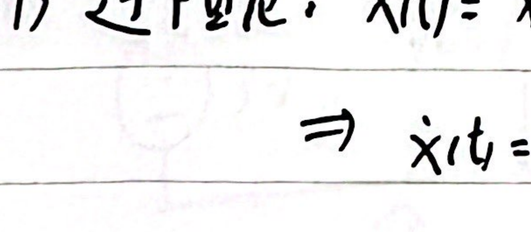
通常 $e^{At} = Q\begin{pmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{pmatrix}Q^{-1}$ $\Rightarrow$ 自然态响应为 $e^{\lambda_i t}$ 的线性叠加

how求 λ_i？ (注：特征根量纲 $= s^{-1}$) $\Rightarrow x(t) = x_\infty(t) + \sum b_i e^{\lambda_i t}$ (b_i 由初值及激励源确定)

法一 $|\lambda I - A| = 0 \Rightarrow \lambda_i =$

法二 略

指列电路方程，n=1为例 ⟹ $\frac{d}{dt}v_c(t) + \frac{1}{\tau}v_c(t) = \frac{1}{\tau}v_s(t)$ 解齐次：$\lambda + \frac{1}{\tau} = 0 \therefore \lambda =$



$\lambda_{1,2} = (-\zeta \pm \sqrt{\zeta^2 - 1})\omega_0$

$\zeta > 1$ 过阻尼

$\zeta = 1$ 临界阻尼

$0 < \zeta < 1$ 欠阻尼

$\zeta = 0$ 无阻尼

阻尼系数 $\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$ 是对电路中能量损耗的描述

$x(t) = x_\infty(t) + Ae^{\lambda_1 t} + Be^{\lambda_1 t}$ (待定系数法，求A,B)

(1) 过阻尼，$x(t) = x_\infty(t) + Ae^{\lambda_1 t} + Be^{\lambda_2 t} = x_\infty(t) + Ae^{-\frac{t}{\tau_1}} + Be^{-\frac{t}{\tau_2}}$

$\Rightarrow \dot{x}(t) = \dot{x}_\infty(t) + A\lambda_1 e^{\lambda_1 t} + B\lambda_2 e^{\lambda_2 t}$ 短暂期行为看短寿命项 (t小的) (即高频)

长 长

$\Rightarrow \begin{cases} x(0^+) = x_\infty(0^+) + A + B \\ \dot{x}(0^+) = \dot{x}_\infty(0^+) + A\lambda_1 + B\lambda_2 \end{cases} \Rightarrow \begin{cases} A = \frac{\lambda_2}{\lambda_2 - \lambda_1}[x(0^+) - x_\infty(0^+)] - \frac{1}{\lambda_2 - \lambda_1}[\dot{x}(0^+) - \dot{x}_\infty(0^+)] \\ B = (\lambda_2 \leftrightarrow \lambda_1) \end{cases}$

(2) 欠阻尼 $\lambda_{1,2} = (-\zeta \pm \sqrt{\zeta^2-1})\omega_0 = (-\zeta \pm j\sqrt{1-\zeta^2})\omega_0$

$x(t) = x_{ss}(t) + Ae^{\lambda_1 t} + Be^{\lambda_2 t}$

$= \cdots \xrightarrow{\text{重定义 A,B}} x(t) = x_{ss}(t) + e^{-\zeta\omega_0 t}\left(A\cos\sqrt{1-\zeta^2}\omega_0 t + B\sin\sqrt{1-\zeta^2}\omega_0 t\right)$

$\frac{d}{dt}x(t) = \cdots$

$\Rightarrow A = x(0^+) - x_{ss}(0^+)$

$B = \left(x(0^+) - x_{ss}(0^+) + \frac{\dot{x}(0^+) - \dot{x}_{ss}(0^+)}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}$

$\Rightarrow$ 五要素法

(1) $x(t) = x_{ss}(t) + (x_0 - x_{ss0})e^{-\zeta\omega_0 t}\cos\sqrt{1-\zeta^2}\omega_0 t$

$+ \left(x_0 - x_{ss0} + \frac{\dot{x}_0 - \dot{x}_{ss0}}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\sqrt{1-\zeta^2}\omega_0 t$ $(0<\zeta<1$ 欠阻尼)

(2) 临界阻尼 $\zeta=1$

$\Rightarrow x(t) = x_{ss}(t) + (x_0 - x_{ss0})e^{-\omega_0 t} + \left(x_0 - x_{ss0} + \frac{\dot{x}_0 - \dot{x}_{ss0}}{\omega_0}\right)\omega_0 t e^{-\omega_0 t}$ 取 $\lim_{\zeta\to 1}$

(3) 过阻尼 $\zeta>1$

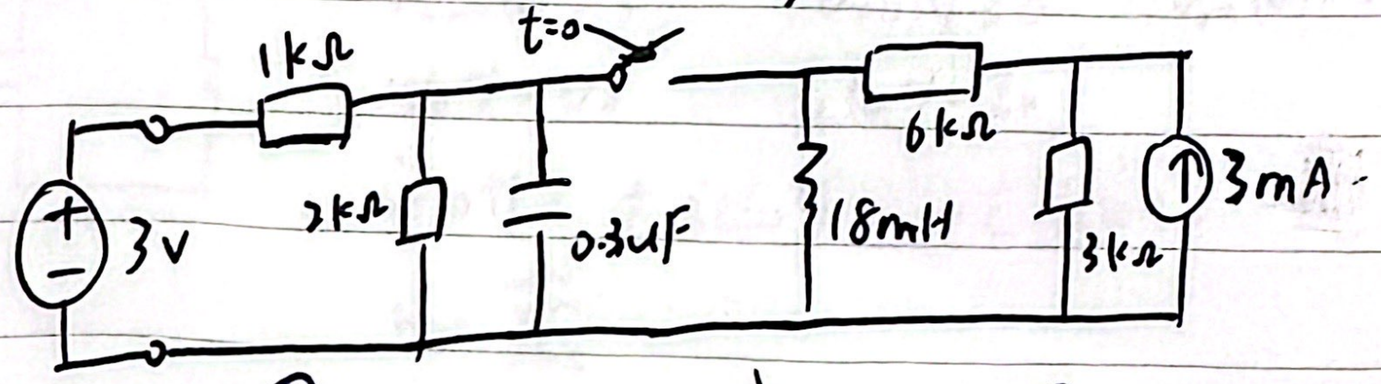
$x(t) = x_{ss}(t) + (x_0 - x_{ss0})e^{-\zeta\omega_0 t}\cosh\sqrt{\zeta^2-1}\omega_0 t$

$+ \left(x_0 - x_{ss0} + \frac{\dot{x}_0 - \dot{x}_{ss0}}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{\zeta^2-1}}e^{-\zeta\omega_0 t}\sinh\sqrt{\zeta^2-1}\omega_0 t$

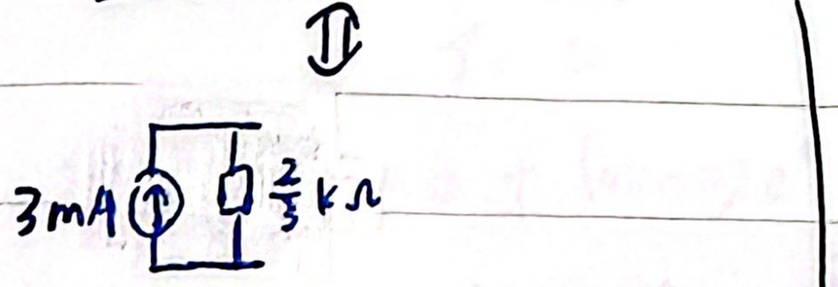
注(1)中 $\sqrt{1-\zeta^2} \to \sqrt{\zeta^2-1}$

$\sin,\cos \to \sinh,\cosh$

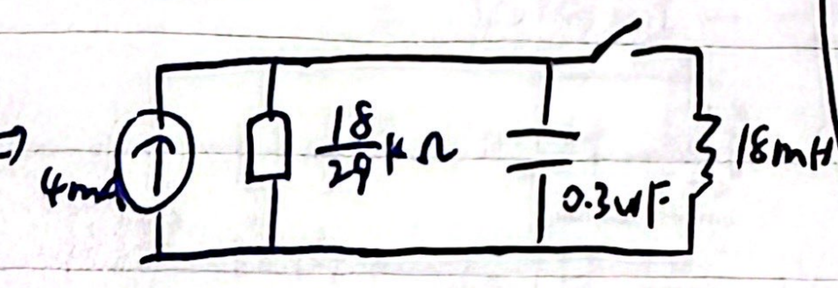
例:



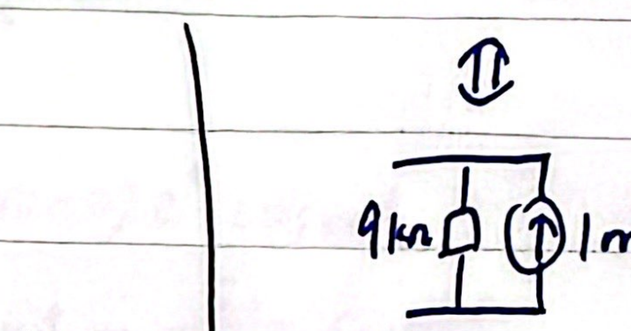
$\Rightarrow$



$\Leftrightarrow$



$\Rightarrow$



✓ $\omega_0 = \frac{1}{\sqrt{LC}}$ ① $\quad Y_0 = \sqrt{\frac{C}{L}} \quad Q = \frac{Y_0}{G}$.

✓ $\zeta = \frac{1}{2Q}$ ②

✓ $v_C(0^+) = 2$ ✓ ③ $\quad i_L(0^+) = 1mA \quad \frac{v_L(0^+)}{R}$

✓ $v_{ss}(t) = 0$ ④ ✓ $v_{ss}(0^+) = 0$

$v_0(0^+) = 0$

✓ $\dot{v}_C(0^+) = \frac{1}{C}i_C(0^+) = \frac{1}{C}\left(i_S(0^+) - i_R(0^+) - i_L(0^+)\right)$ ⑤

(用原图) (注: 此处初值用乘积)

对欠阻尼进一步讨论：

$$x(t) = x_{\infty}(t) + e^{-\zeta\omega_0 t}\left(A\cos\sqrt{1-\zeta^2}\,\omega_0 t + B\sin\sqrt{1-\zeta^2}\,\omega_0 t\right)$$

$$= x_{\infty}(t) + \sqrt{A^2+B^2}\,e^{-\frac{t}{\tau}}\cos(\omega_d t - \varphi_0) \qquad \tau = \frac{1}{\zeta\omega_0} \qquad \omega_d = \sqrt{1-\zeta^2}\,\omega_0$$

"振铃"

经历 Q 个 T，振铃衰减 43%　　$\left(Q = \frac{1}{2\zeta},\ T = \frac{2\pi}{\omega_d}\right)$

1.5Q	1%
2.2Q	0.1%

无阻尼

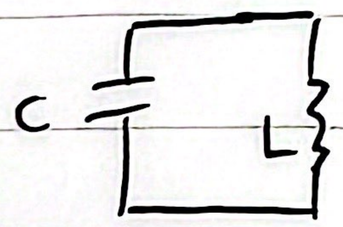
$$x(t) = x_{\infty}(t) + (x_0 - x_{\infty 0})e^{-\zeta\omega_0 t}\cos\sqrt{1-\zeta^2}\,\omega_0 t + \left(x_0 - x_{\infty 0} + \frac{\dot{x}_0 - \dot{x}_{\infty 0}}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\sqrt{1-\zeta^2}\,\omega_0 t$$

$\zeta = 0$ 代入，$x(t) = x_{\infty}(t) + (x_0 - x_{\infty 0})\cos\omega_0 t + \frac{\dot{x}_0 - \dot{x}_{\infty 0}}{\omega_0}\sin\omega_0 t$

$$= x_{\infty}(t) + \sqrt{(x_0 - x_{\infty 0})^2 + \left(\frac{\dot{x}_0 - \dot{x}_{\infty 0}}{\omega_0}\right)^2}\cos\left(\omega_0 t - \arctan\frac{\dot{x}_0 - \dot{x}_{\infty 0}}{\omega_0(x_0 - x_{\infty 0})}\right)$$

强迫响应　　自由响应　　无阻尼 ∴ 振荡幅度不变

无阻尼 LC 的自由振荡



$v_{C\infty}(t) = 0$　　$\dot{v}_{\infty}(t) = 0$　　$v_{\infty}(0^+) = 0$　　$\dot{v}_{\infty}(0^+) = 0$

$v_C(0^+) = V_0$　　$i_L(0^+) = I_0$

$\dot{v}_C(0^+) = \frac{1}{C}i_C(0^+) = -\frac{1}{C}i_L(0^+) = -\frac{I_0}{C}$

$\omega_0 = \frac{1}{\sqrt{LC}}$

$\zeta = 0$

代入，$v_C(t) = 0 + (v_0 - 0)e^0\cos\omega_0 t + \frac{-I_0/C - 0}{\omega_0}e^0\sin\omega_0 t$

$$= V_0\cos\omega_0 t - \frac{I_0}{\omega_0 C}\sin\omega_0 t$$

$$= V_0\sqrt{1 + \frac{LI_0^2}{CV_0^2}}\cos\left(\omega_0 t + \arctan\frac{Z_0 I_0}{V_0}\right)$$

同理，$i_L(t) = I_0\sqrt{1 + \frac{CV_0^2}{LI_0^2}}\sin\left(\omega_0 t + \arctan\frac{Z_0 I_0}{V_0}\right)$

有 $E(t) = E_C(t) + E_L(t) = \text{const}$

KOKUYO

注：简单串并联 ω_0, ζ, Z_0, Q 可消

其他：法一) 频域求 $H(S)=\dfrac{Y_0}{1\cdot S^2+a\cdot S+b}$

$a \updownarrow 2\zeta\omega_0$　　$b \updownarrow \omega_0^2$

法二) 时域列方程

$$1\cdot\frac{d^2}{dt^2}v_0+a\frac{d}{dt}v_0+bv_0=\cdots$$

$a \updownarrow 2\zeta\omega_0$　　$b \updownarrow \omega_0^2$

No.

Date

第12讲 二阶时频分析

LTI系统频域分析

$V_{mo}\cos(\omega_0 t+\varphi_0)$ → [LTI系统 $H(j\omega)=A(\omega)e^{j\varphi(\omega)}$] → $V_{mo}A(\omega_0)\cos(\omega_0 t+\varphi_0+\varphi(\omega_0))$

理想低通滤波特性：

$H(j\omega)=A_0 e^{-j\omega t_0}$

设 $in=x(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(\omega)e^{j\omega t}d\omega$

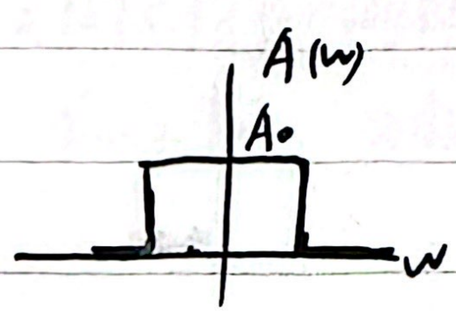
$\Rightarrow out=y(t)=\cdots=A_0 x(t-t_0)$

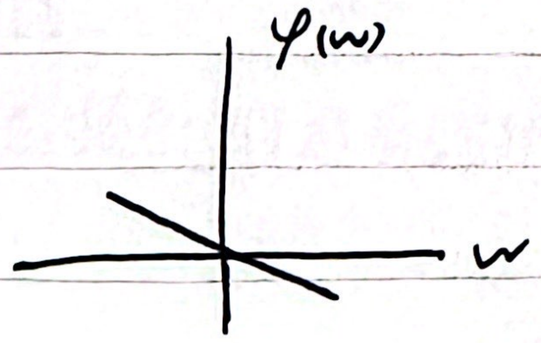
out仅是对in的比例放大和整体延时

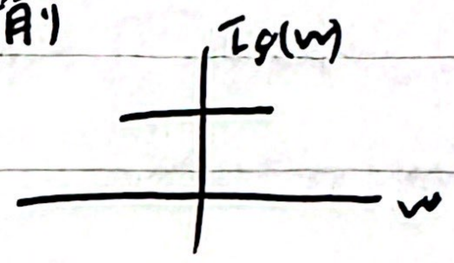
"无失真传输"

定义群延时 $\tau_g(\omega)=-\frac{d\varphi(\omega)}{d\omega}$

if , $\tau_g(\omega)=t_0$



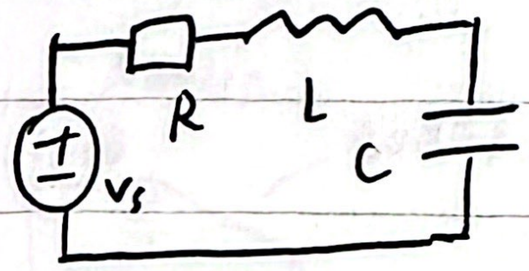




但实际滤波特性会产生失真

一. 二阶低通：C分压

1.



$$\frac{\frac{1}{j\omega C}}{R+j\omega L+\frac{1}{j\omega C}}=\frac{\frac{1}{LC}}{s^2+s\frac{R}{L}+\frac{1}{LC}}$$

$$H(j\omega)=\frac{\dot{V}_c}{\dot{V}_s}=H_0\frac{\omega_0^2}{s^2+2\xi\omega_0 s+\omega_0^2}$$

$s\to j\omega$

$H_0=1 \quad \omega_0=\frac{1}{\sqrt{LC}} \quad \xi=\frac{R}{2}\sqrt{\frac{C}{L}}$

$$H(j\omega)=\frac{\omega_0^2}{(\omega_0^2-\omega^2)+j2\xi\omega_0\omega}=A(\omega)e^{j\varphi(\omega)}$$

$$\Rightarrow A(\omega)=\frac{\omega_0^2}{\sqrt{(\omega_0^2-\omega^2)^2+(2\xi\omega_0\omega)^2}} \qquad \varphi(\omega)=-\arctan\frac{2\xi\omega_0\omega}{\omega_0^2-\omega^2}$$

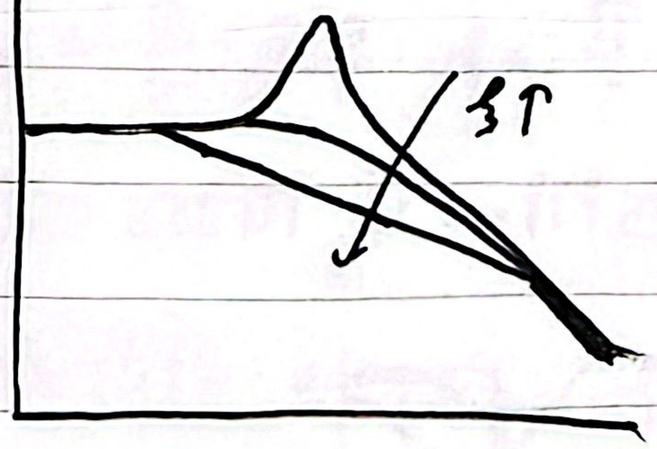
$$\tau_g(\omega)=\frac{d\varphi(\omega)}{d\omega}=\frac{2\xi\omega_0(\omega^2+\omega_0^2)}{\omega^4+2(2\xi^2-1)\omega_0^2\omega^2+\omega_0^4}$$

KOKUYO

$$H(s)=\frac{\omega_0^2}{s^2+2\zeta\omega_0 s+\omega_0^2}=\frac{\omega_0^2}{(s-\lambda_1)(s-\lambda_2)} \qquad \lambda_{1,2}=\left(-\zeta\pm\sqrt{\zeta^2-1}\right)\omega_0$$

幅频特性

$$A(\omega)=\frac{\omega_0^2}{\sqrt{(\omega_0^2-\omega^2)^2+(2\zeta\omega_0\omega)^2}}$$



$\zeta>1$　三段折线

$0.707\le\zeta\le1$　两段折线

$\zeta\ll0.707$　自由振荡频点. 附近有谐振峰出现

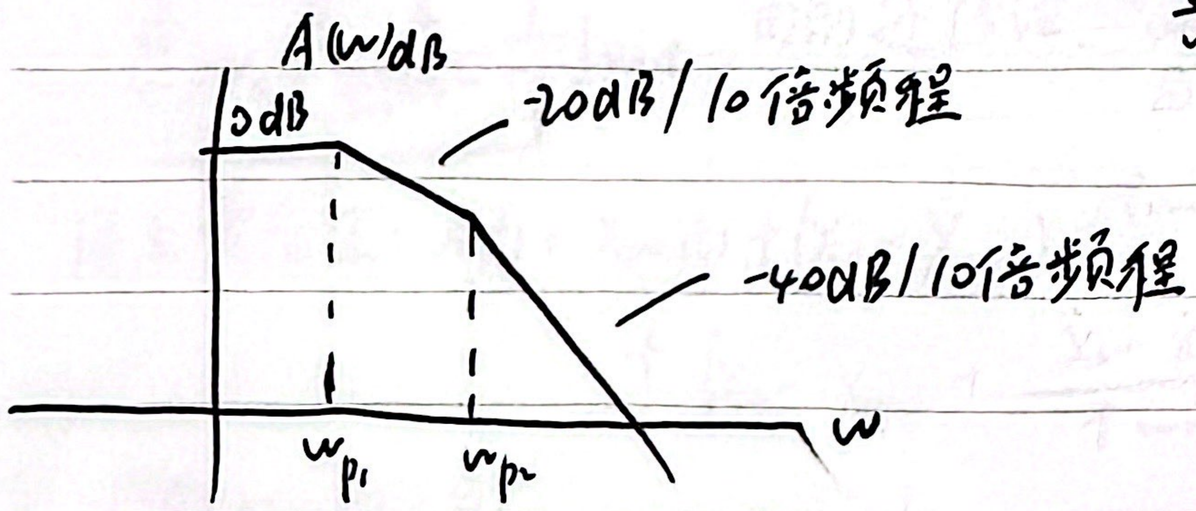
具体分析:

$$\lambda_{1,2}=\left(-\zeta\pm\sqrt{\zeta^2-1}\right)\omega_0=\begin{cases}-\left(\zeta-\sqrt{\zeta^2-1}\right)\omega_0=-\omega_{p1}\\-\left(\zeta+\sqrt{\zeta^2-1}\right)\omega_0=-\omega_{p2}\end{cases}$$

$$\omega_{p1}\cdot\omega_{p2}=\omega_0^2$$

过阻尼　$H(s)=\dfrac{\omega_0^2}{(s-\lambda_1)(s-\lambda_2)}$

$$=\frac{1}{\left(\frac{j\omega}{\omega_{p1}}+1\right)\left(\frac{j\omega}{\omega_{p2}}+1\right)}\cong\begin{cases}1 & \omega<\omega_{p1}\\ \dfrac{1}{\frac{j\omega}{\omega_{p1}}}=\dfrac{\omega_{p1}}{j\omega} & \omega_{p1}<\omega<\omega_{p2}\\ \dfrac{1}{\frac{j\omega}{\omega_{p1}}\cdot\frac{j\omega}{\omega_{p2}}}=\dfrac{\omega_0^2}{-\omega^2} & \omega>\omega_{p2}\end{cases}$$



两特征根对应两转折点.

欠阻尼:　$H(s)=\dfrac{\omega_0^2}{(s-\lambda_1)(s-\lambda_2)}$　$\lambda_{1,2}=\left(-\zeta\pm j\sqrt{1-\zeta^2}\right)\omega_0$　$(0<\zeta<1)$

$H(j0)=1$,　$H(j\infty)=0$　$H(j\omega_0)=-j\dfrac{1}{2\zeta}=\left(\dfrac{1}{2\zeta}\right)e^{-j\frac{\pi}{2}}$

$\zeta<0.5$时, $A(\omega)>1$ ⇒谐振峰

问: where is 谐振峰?

令 $\dfrac{dA(\omega)}{d\omega}=0\Rightarrow\omega_e=\sqrt{1-2\zeta^2}\,\omega_0$　$(\zeta<0.707$时$)$

$$A(\omega_e)=\frac{1}{2\zeta\sqrt{1-\zeta^2}}>1=A(0)$$

$\zeta\ll0.707$时, $\begin{cases}\omega_e\cong\omega_0\\ A(\omega_e)\cong\dfrac{1}{2\zeta}=Q\end{cases}$

No.

Date

背！ $\omega_{3dB}=\omega_0\sqrt{-2\zeta^2+1+\sqrt{(2\zeta^2-1)^2+1}}$ $BW_{3dB}=\frac{\omega_{3dB}}{2\pi}$

结论：（令 $A(\omega_{3dB})=\frac{1}{\sqrt{2}}$ 可解出）

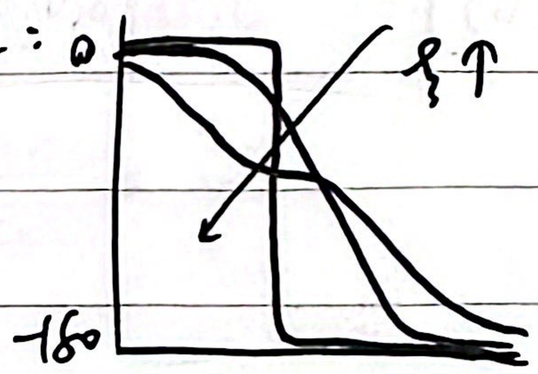
① 幅度最大平坦：$\zeta=0.707$ （↔对n阶系统，令 $\frac{dA(\omega)}{d\omega}=\frac{d^2A(\omega)}{d\omega^2}=\cdots=\frac{d^{(n)}A(\omega)}{d\omega^n}=0$）

② 3dB带宽：ω_{3dB} $\begin{cases}\omega_0 & \zeta=0.707\\ \frac{\omega_0}{2\zeta} & \zeta\gg1\end{cases}$

③ 群延时最大平坦：$\zeta=\frac{\sqrt{2}}{2}=0.866$ （↔ 对n阶系统 $\frac{dT_g(\omega)}{d\omega}=\cdots=\frac{d^{(n)}T_g(\omega)}{d\omega}=0$）

$\zeta<0.866$ 时，在 ω_0 附近，群延时也出现谐振峰现象

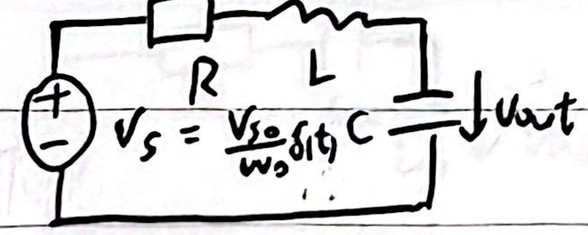
④ 相频特性：



综合①③，$\zeta\in(0.707,1)$ 时，为最优二阶低通系统

图见ppt

时域特性 e.g. 冲激响应



由前述结论，$\omega_0=\frac{1}{\sqrt{LC}}$ ① $Z_0=\sqrt{\frac{L}{C}}$ $\zeta=\frac{R}{2Z_0}=\frac{R}{2}\sqrt{\frac{C}{L}}$ ②

用五要素法：$x(t)=x_\infty(t)+(x_0-x_{\infty0})e^{-\zeta\omega_0 t}\cos\sqrt{1-\zeta^2}\omega_0 t$

$+\left(x_0-x_{\infty0}+\frac{\dot{x}_0-\dot{x}_{\infty0}}{\zeta\omega_0}\right)\frac{\zeta}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\cdot\sin\sqrt{1-\zeta^2}\omega_0 t\quad(t\geq0)$

③ $V_{c\infty}(t)=0$

④ $V_c(0^+)=V_c(0^-)=0$

⑤ $\frac{d}{dt}v_c(0^+)=\frac{1}{C}i_c(0^+)=\frac{1}{C}i_L(0^+)$ $\frac{V_{s0}}{\omega_0}\delta(t)$（冲激↔高频，∴L↔断路）

$i_L(0^+)=i_L(0^-)+\frac{1}{L}\int_{0^-}^{0^+}v_L(t)dt=\frac{V_{s0}}{\omega_0 L}$

$\Rightarrow \dot{v}_c(0^+)=\frac{V_{s0}}{\omega_0 LC}$

由③，$\dot{v}_{c\infty}(0^+)=0$

代入，$v_c(t)=v_{out}(t)=\frac{V_{s0}}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\sqrt{1-\zeta^2}\omega_0 t\cdot U(t)$

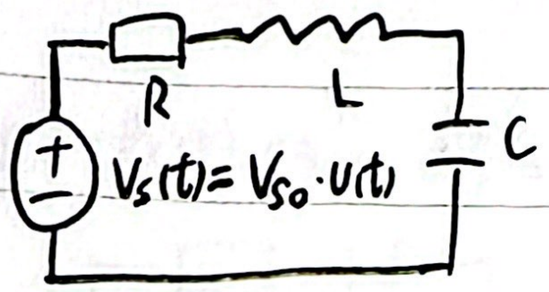
V_s 系数归化为1，$\Rightarrow h(t)=\frac{\omega_0}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin(\sqrt{1-\zeta^2}\omega_0 t)U(t)$ （图见ppt）

KOKUYO

No.

Date

e.g. 阶跃响应：



① $\omega_0=\frac{1}{\sqrt{LC}}$　② $\zeta=\frac{R}{2}\sqrt{\frac{C}{L}}$

③ $V_c(0^+)=V_c(0^-)=0$

④ $V_L(0^+)=\frac{i_c(0^+)}{C}=\frac{i_L(0^+)}{C}=0$

阶跃电压不能使 i_L、V_C 跳变

证：$\because \frac{1}{L}\int_{0^-}^{0^+} V_L(t)\,dt=\frac{1}{L}\int_{0^-}^{0^+} V_{s0}U(t)\,dt=$

$\therefore i_L(0^+)=i_L(0^-)=0$

⑤ $V_{c(\infty)}(t)=V_{s0}$

$$\Rightarrow V_{out}(t)=V_{s0}\left[1-e^{-\zeta\omega_0 t}\left(\cos\sqrt{1-\zeta^2}\,\omega_0 t+\frac{\zeta}{\sqrt{1-\zeta^2}}\sin\sqrt{1-\zeta^2}\,\omega_0 t\right)\right]U(t)$$

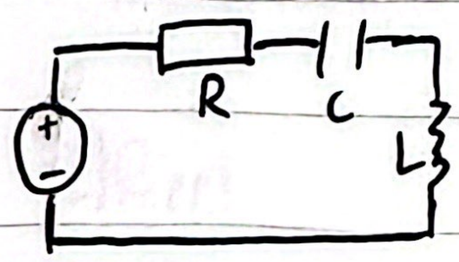
$$\Rightarrow g(t)=\frac{V_{out}(t)}{V_{s0}}=\cdots$$

（图见 ppt）

结论：$\zeta\in(0.707,1)$，系统进入稳态需要的时间最短，是最优二阶低通系统

（ζ 过大或过小，都不利于系统快速进入稳态）

二. 二阶高通：L 分压



$$H(\omega)=\frac{\dot V_L}{\dot V_s}=\frac{j\omega L}{R+j\omega L+\frac{1}{j\omega C}}=\frac{s^2}{s^2+s\frac{R}{L}+\frac{1}{LC}}$$

$$=H_0\frac{s^2}{s^2+2\zeta\omega_0 s+\omega_0^2}\qquad H_0=1,\ \omega_0=\frac{1}{\sqrt{LC}},\ \zeta=\frac{R}{2}\sqrt{\frac{C}{L}}$$

$$H(s)=A(\omega)e^{j\varphi(\omega)}$$

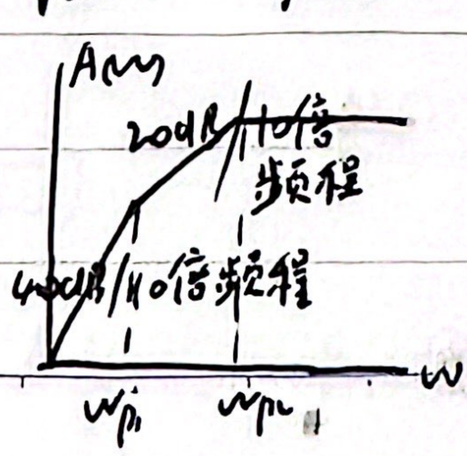
$$A(\omega)=\frac{\omega^2}{\sqrt{(\omega_0^2-\omega^2)^2+(2\zeta\omega_0\omega)^2}}\qquad \varphi(\omega)=\pi-\arctan\frac{2\zeta\omega_0\omega}{\omega_0^2-\omega^2}$$

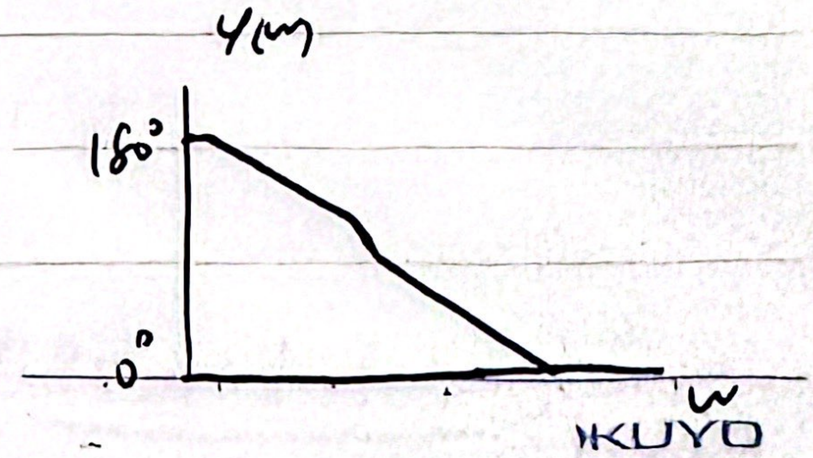
（图见 ppt）

过阻尼

$$H(j\omega)=\frac{-\omega^2}{(j\omega+\omega_{p1})(j\omega+\omega_{p2})}=\frac{1}{\left(1+\frac{\omega_{p1}}{j\omega}\right)\left(1+\frac{\omega_{p2}}{j\omega}\right)}$$

$$=\begin{cases}1 & \omega>\omega_{p2}\\ \frac{j\omega}{\omega_{p2}} & \omega_{p2}>\omega>\omega_{p1}\\ \frac{-\omega^2}{\omega_0^2} & \omega_{p1}>\omega\end{cases}$$

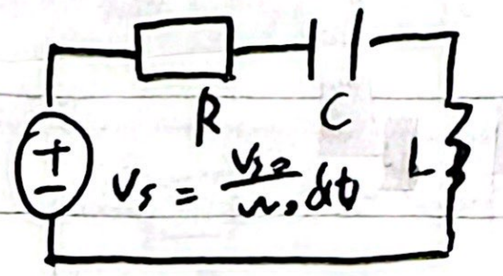




KUYO

欠阻尼：ζ很小时，谐振峰出现在ω_0附近，高度$\simeq \frac{1}{2\zeta}=Q$

时域特性：e.g. 冲激响应



① $\omega_0=\frac{1}{\sqrt{LC}}$ ② $\zeta=\frac{R}{2}\sqrt{\frac{C}{L}}$

③ $v_L(0^+)=v_s(0^+)-v_R(0^+)-v_C(0^+)$

$=0-i_L(0^+)\cdot R-0=-2\zeta V_{s0}$

④ $v_{L\infty}(t)=0$

⑤ $\dot{v}_L(0^+)=\dot{v}_s(0^+)-\dot{v}_R(0^+)-\dot{v}_C(0^+)$

$=0-R\cdot\frac{d}{dt}i_L(0^+)-\frac{i_L(0^+)}{C}$

$=0-R\cdot\frac{v_L(0^+)}{L}-\frac{i_L(0^+)}{C}=(4\zeta^2-1)\omega_0 V_{s0}$

$\Rightarrow v_{out}(t)=\cdots$

$\Rightarrow h(t)=\cdots$

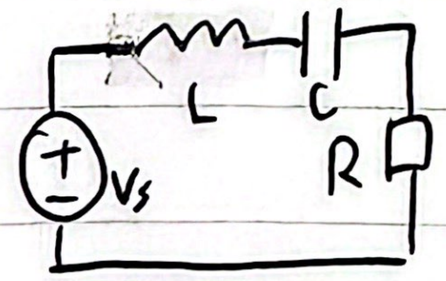
e.g. 阶跃响应 ① $\omega_0=\frac{1}{\sqrt{LC}}$ ② $\zeta=\frac{R}{2}\sqrt{\frac{C}{L}}$

③ $v_L(0^+)=V_{s0}$ ④ $v_{L\infty}(t)=0$

⑤ $\dot{v}_L(0^+)=0-R\frac{V_{s0}}{L}-0=-2\zeta\omega_0 V_{s0}$

图见ppt

三. 二阶带通：R分压



$H(s)=\frac{R}{R+sL+\frac{1}{sC}}=\frac{s\frac{R}{L}}{s^2+s\frac{R}{L}+\frac{1}{LC}}$

$=H_0\frac{2\zeta\omega_0 s}{s^2+2\zeta\omega_0+\omega_0^2}$ $H_0=1$ $\omega_0=\frac{1}{\sqrt{LC}}$ $\zeta=\frac{R}{2}\sqrt{\frac{C}{L}}$

习惯于用Q值分析 $Q=\frac{1}{2\zeta}=\frac{\omega_0 L}{R}$

$H(\omega)=A(\omega)e^{j\varphi(\omega)}$

$A(f)=\frac{1}{\sqrt{1+Q^2(\frac{f}{f_0}-\frac{f_0}{f})^2}}$ $(f_0=\frac{1}{2\pi\sqrt{LC}})$

结论：$BW_{3dB}=\frac{f_0}{Q}$ （令$A(f)=\frac{1}{\sqrt{2}}\Rightarrow f_1<f_2$ $BW_{3dB}=f_2-f_1=\frac{f_0}{Q}$）

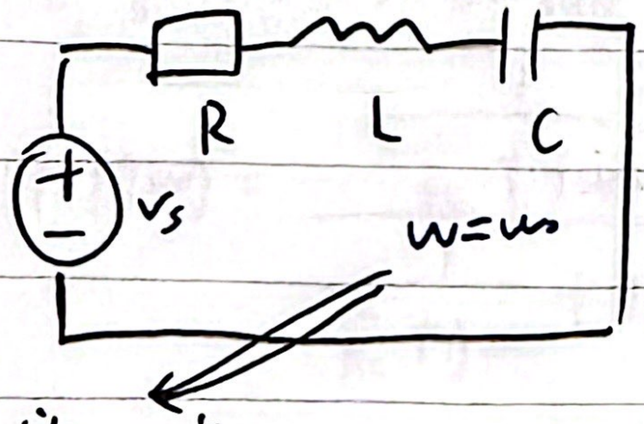
No.

Date

相频特性 $\varphi(\omega) = -\arctan Q\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)$

$$\tau_{g0} = -\left.\frac{d\varphi}{d\omega}\right|_{\omega=\omega_0} = \frac{2Q}{\omega_0}$$

$$\Rightarrow BW_{3dB} \cdot \tau_{g0} = \frac{1}{\pi} \quad \therefore BW_{3dB} \uparrow \longleftrightarrow \tau_{g0} \downarrow$$



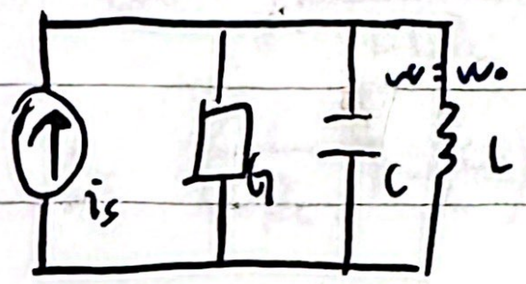
串联谐振：电压谐振

$$\dot{V}_R = \dot{V}_S$$

$$\dot{I} = \frac{\dot{V}_R}{R} = \frac{\dot{V}_S}{R}$$

$$\dot{V}_L = j\omega_0 L\dot{I} = jQ\dot{V}_S$$

$$\dot{V}_C = \frac{\dot{I}}{j\omega_0 C} = -jQ\dot{V}_S$$



并联谐振：电流谐振

$$\dot{I}_G = \dot{I}_S$$

$$\dot{V} = \frac{\dot{I}_G}{G} = \frac{\dot{I}_S}{G}$$

$$\dot{I}_C = j\omega_0 C\dot{V} = jQ\dot{I}_S$$

$$\dot{I}_L = \frac{1}{j\omega L}\dot{V} = -jQ\dot{I}_S$$

冲激响应与阶跃响应：见作业

(图见ppt)

伯特图画法总结：

$H(s)$ 因式分解 $A_0 \frac{(s+w_{z1})\cdots(s+w_{zm})}{(s+w_{p1})\cdots(s+w_{pn})}$ 零点… 极点 ↔ 特征根 (假设无复根)

$S_p = -w_{p1}, \cdots, -w_{pn} < 0$ (要求必<0)

$S_z = -w_{z1}, \cdots, -w_{zm}$

(排序为：$|w_{p1}| < |w_{p2}| < \cdots < |w_{pn}|$

$|w_{z1}| < |w_{z2}| < \cdots < |w_{zm}|$)

$$H(s) = H_0 \frac{(1+\frac{s}{w_{z1}})\cdots(1+\frac{s}{w_{zm}})}{(1+\frac{s}{w_{p1}})\cdots(1+\frac{s}{w_{pn}})}$$

幅频特性：每个极点都会导致 20dB/10倍频程↓，每个零点都会导致 20dB/10倍频程↑

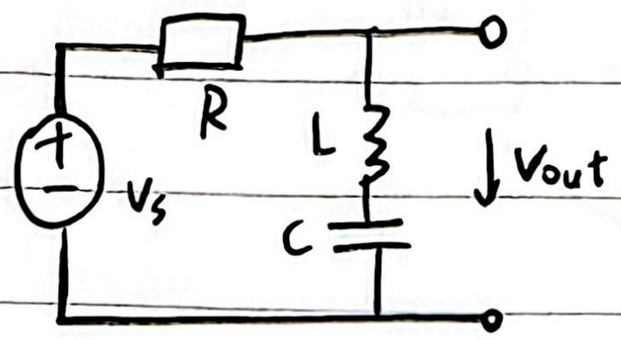
相频特性：极点：滞后90°

零点：$w_z > 0$：超前90°；$w_z < 0$：滞后90° (左超右滞) (针对的是 $w = -w_z$)

第十三讲 有源滤波

回顾：二阶低通.高通.带通 电路

一. 二阶带阻：LC分压

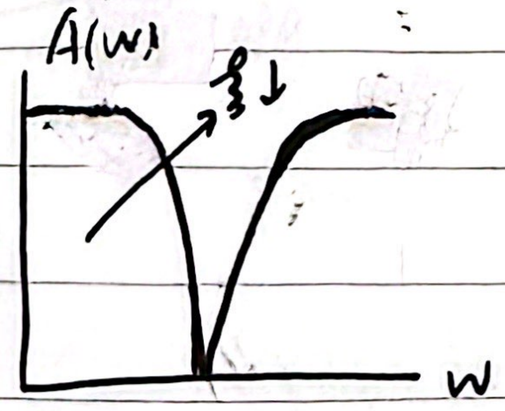


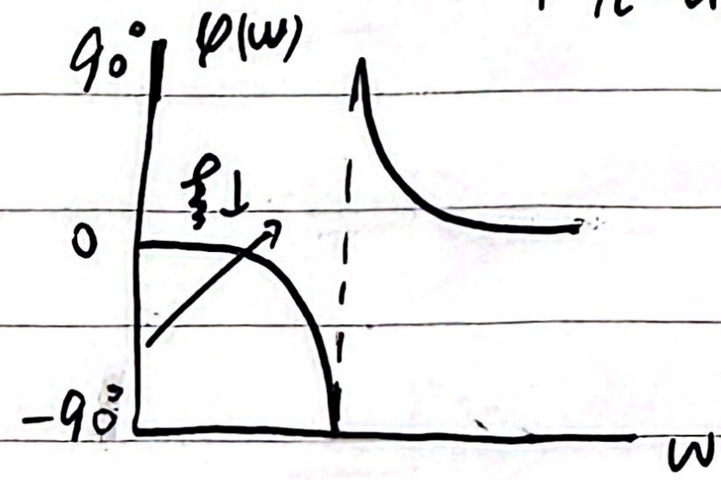
$$H(j\omega) = \frac{\dot{V}_L + \dot{V}_C}{\dot{V}_S} = H_0 \frac{s^2 + \omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

$$A(\omega) = H_0 \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\xi\omega_0\omega)^2}}$$

$$\varphi(\omega) = \begin{cases} -\arctan \dfrac{2\xi\omega_0\omega}{\omega_0^2 - \omega^2} & \omega < \omega_0 \\ \pi - \arctan \dfrac{2\xi\omega_0\omega}{\omega_0^2 - \omega^2} & \omega > \omega_0 \end{cases}$$

频域特性





时域特性

e.g.：冲激响应 （两种方法都可）

$$h_{BS}(t) = h_{LP}(t) + h_{HP}(t) = \delta(t) - h_{BP}(t)$$

（带阻） （电容） （电感） （电源） （电阻）

（阶跃响应同理）

⇒ 找规律（对应关系）

$h(t)$	$H(s)$
$\delta(t)$	1

……（略）

验证：冲激：$H_{BP,2}(s) = \cdots \xrightarrow{\text{对应}} h_{BP,2}(t) = \cdots$

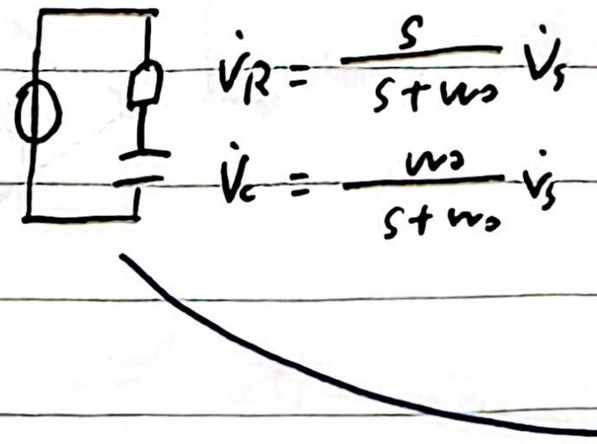
阶跃：$G_{LS,2}(s) = \frac{1}{s} H_{BS,2}(s) = \cdots \xrightarrow{\text{对应}} g_{BS,2}(t) = \cdots$

（这些五要素法都能求）

例：某一阶系统 $H(s)=\frac{as+b}{s+\omega_0}$，求冲激响应

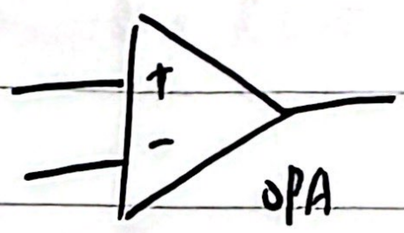
法一）

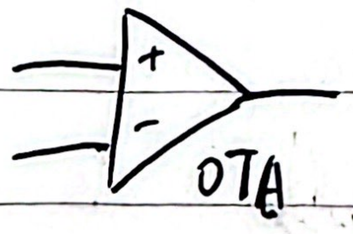
法二）考虑

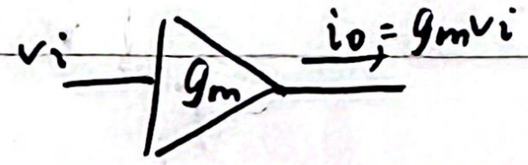


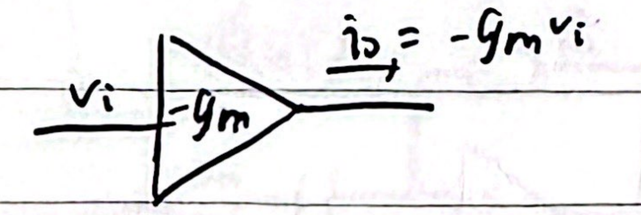
二、有源滤波器（电感替代法）

常用有源器件：

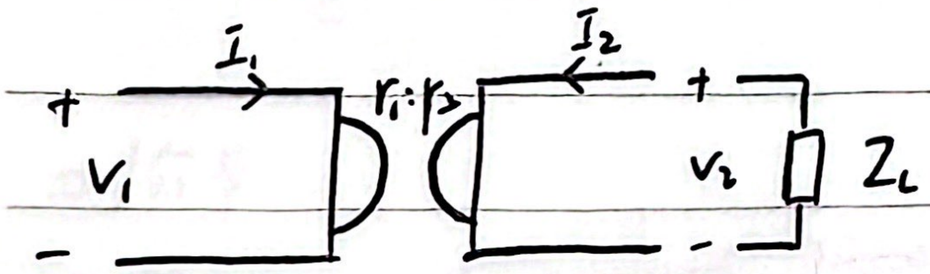








设计思路一：电容 回旋 电感



$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}=\begin{pmatrix} 0 & -r_{m_1} \\ r_{m_2} & 0 \end{pmatrix}\begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

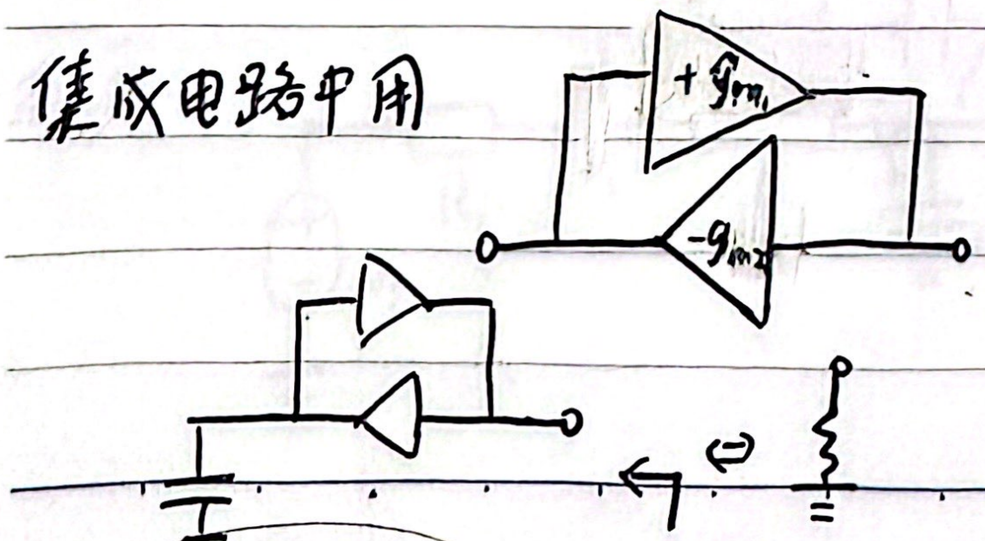
$$\begin{pmatrix} i_1 \\ i_2 \end{pmatrix}=\begin{pmatrix} 0 & g_{m_2} \\ -g_{m_1} & 0 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$Z_{in}=\frac{\dot{V}_1}{\dot{I}_1}=\frac{-r_{m_2}\dot{I}_2}{\dot{V}_2/r_{m_2}}=r_{m_1}r_{m_2}\frac{1}{-\dot{V}_2/\dot{I}_2}$$

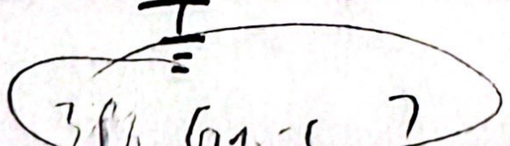
$$=\frac{r_{m_1}r_{m_2}}{Z_L}\quad \frac{\text{记 } Z_L \text{ 为电容}}{Z_L=\frac{1}{j\omega C}}\quad j\omega\, r_{m_1}r_{m_2}C$$

等效电感 $L_{eq}=r_{m_1}r_{m_2}C=\frac{C}{g_{m_1}g_{m_2}}$

集成电路中用



表示回旋器

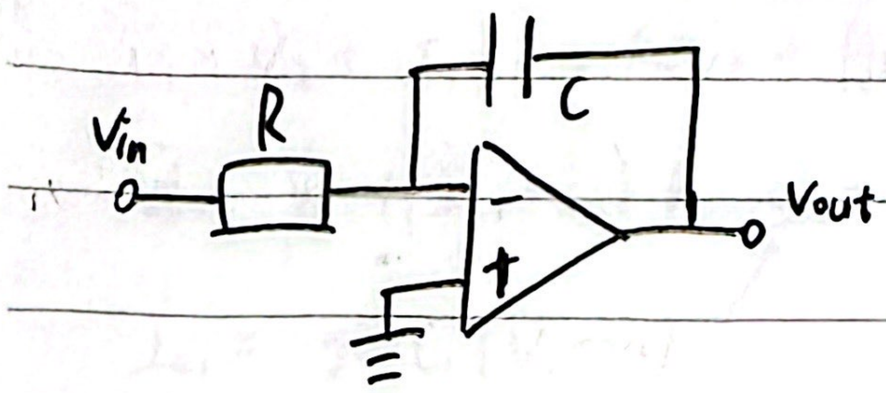


3阶 G_m-C 了 pic

No.

Date

设计思路二：积分器



$V_{out} = -\frac{1}{C}\int \frac{v_{in}(t)}{R}dt = -\frac{1}{RC}\int v_{in}(t)\,dt$ 这是理想积分（要求运放必须工作在线性区）

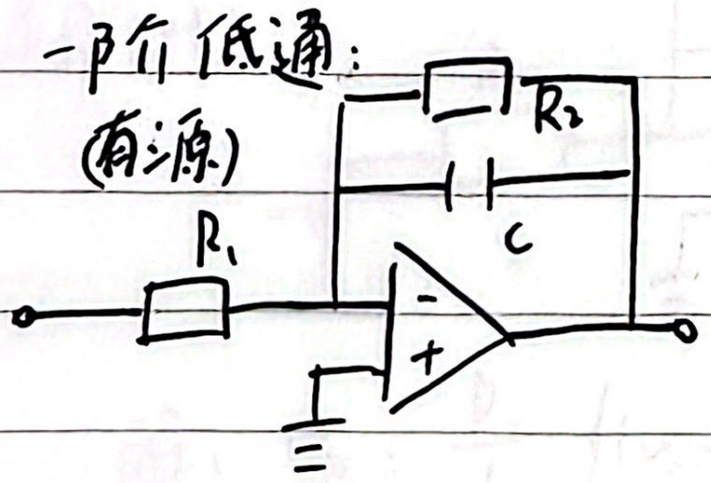
用此积分替代电感：

$i_L(t) = \frac{1}{L}\int v_L(t)\,dt$

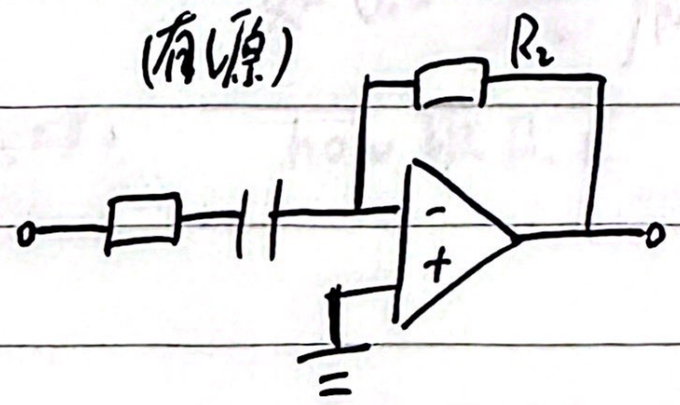
$v_{out}(t) = R i_L(t) = \frac{R}{L}\int v_L(t)dt$ → 合理设计 C 即可

设计思路三：一阶节二阶节设计

一阶低通：（有源）

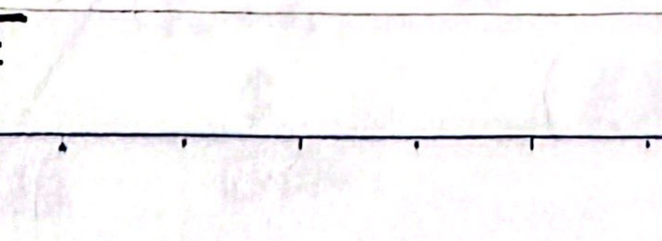


一阶高通：（有源）



二阶节

无源：

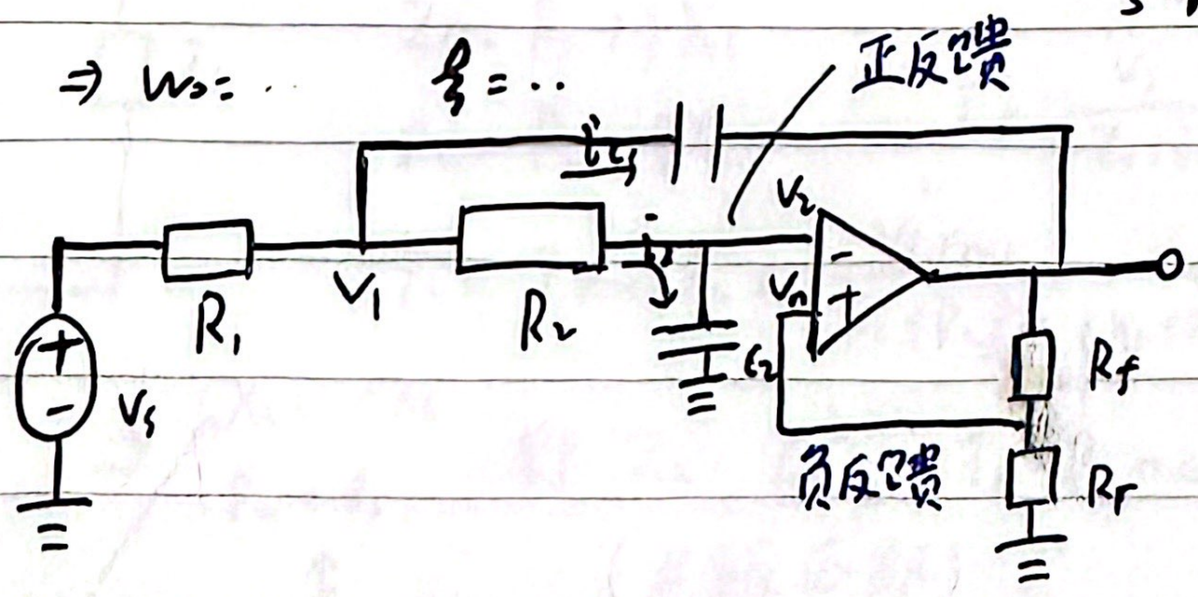


$H(s) = \frac{V_o}{V_s} = \frac{1}{A}$ $\begin{pmatrix} A & B \\ C & D \end{pmatrix} = (\ )(\ )(\ )(\ )$

$\Rightarrow H(s) = \dfrac{\frac{1}{R_1C_1R_2C_2}}{s^2 + \left(\frac{1}{R_2C_2} + \frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right)s + \frac{1}{R_1C_1R_2C_2}}$

$\Rightarrow w_0 = \ldots \quad \zeta = \ldots$

有源：



设 负反馈 > 正

二仍可用虚短、虚断分析

(倒推法)

$$\dot{V}_n = \frac{R_2}{R_1+R_f}\dot{V}_o = F_n\dot{V}_o$$

$\dot{V}_2$ //

$$\dot{V}_1 = \dot{V}_2 + \dot{I}_2 R_2 = F_n\dot{V}_o + (j\omega C_2 V_2)\cdot R_2 = (1+j\omega C_2 R_2)F_n\dot{V}_o$$

$$\dot{V}_s = \dot{V}_1 + (\dot{I}_{C_1}+\dot{I}_2)R_1 = F_n\dot{V}_o\left[1+s\left(R_2C_2+R_1C_1+R_1C_2-\frac{R_1C_1}{F_n}\right)+s^2R_1C_1R_2C_2\right]$$

$$\dot{I}_{C_1} = j\omega C_1(\dot{V}_1-\dot{V}_o)$$

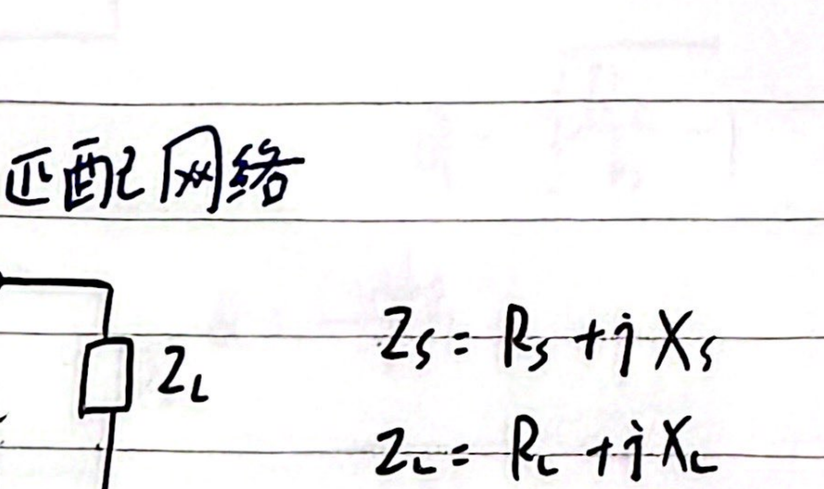
$$\Rightarrow H(s) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{\frac{1}{F_n}}{s^2R_1C_1R_2C_2+s\left(R_2C_2+R_1C_1+R_1C_2-\frac{R_1C_1}{F_n}\right)+1}$$

$$\Rightarrow \omega_0 = \cdots \quad f = \frac{1}{2}\left(\sqrt{\quad}+\sqrt{\quad}+\sqrt{\quad}-\frac{1}{F_n}\sqrt{\quad}\right)$$

↑ 正反馈便于L；正反馈不能过大，否则变成负阻尼，变成振荡器

续：(思路二 积分器)

(详见ppt)



知 $V_o = \frac{R_0}{L}\int(V_s-2V_o)\,dt$

how 取 R、C

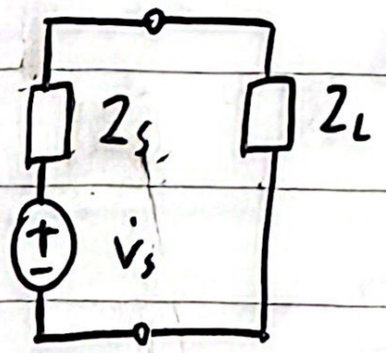
解：$V_o = \frac{R_0}{L}\int(V_s-2V_o)\,dt$

$$= -\frac{1}{C}\int i\,dt = -\frac{1}{C}\int\left(\frac{V_s}{R_0}+\frac{V_o}{R}\right)dt$$

∴ 取 $R=\frac{1}{2}R_0$ ✓

$C=\frac{1}{R_0^2}$

三. LC 匹配网络



$Z_s = R_s + jX_s$

$Z_L = R_L + jX_L$

$\dot{I} = \frac{V_s}{Z_s+Z_L} = \cdots$

$P_L = I_{rms}^2 R_L = \frac{V_{s,rms}^2}{(R_s+R_L)^2+(X_s+X_L)^2}R_L$

$\Rightarrow \begin{cases} X_L = -X_s \\ R_L = R_s \end{cases}$ 即 $Z_L = Z_s^*$ 时，P_L max

(共轭匹配)

↕

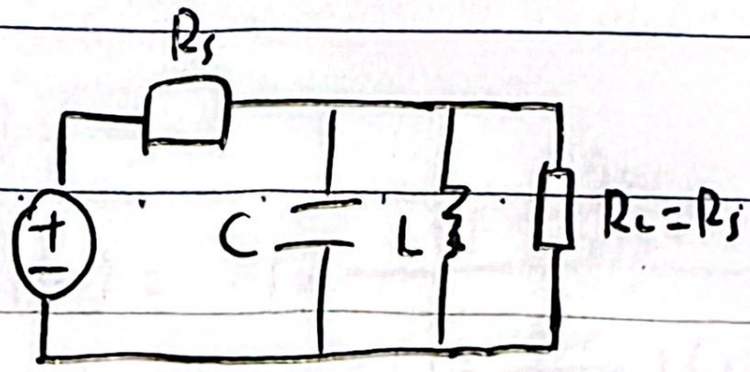
匹配

增振

No.

Date

例：



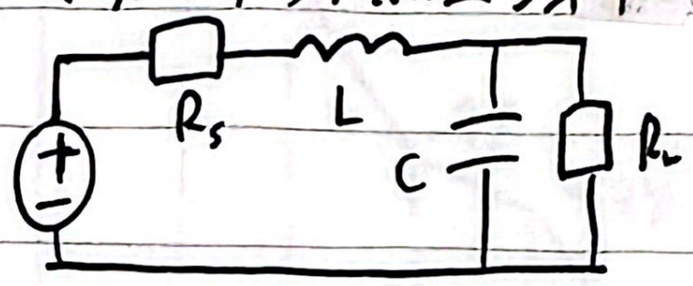
使负载获得最大功率传输匹配，C&L？

令 $-\frac{1}{\omega_r L_L} + \omega_r C_s = 0 \quad \Rightarrow L_L = \frac{1}{\omega_r^2 C_s}$

问：if $R_L \neq R_s$, how ？

思路1：谐振峰频点上强制匹配

思路2：



共轭匹配

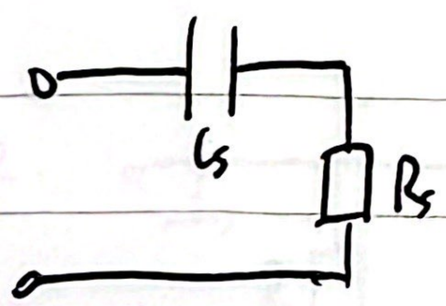
$R_s + j\omega_r L$ 与 $\frac{R_L}{1+j\omega_r R_L C}$ 互为共轭

$\Rightarrow C = \frac{1}{\omega_r R_L}\sqrt{\frac{R_L}{R_s}-1} \qquad L = \cdots$

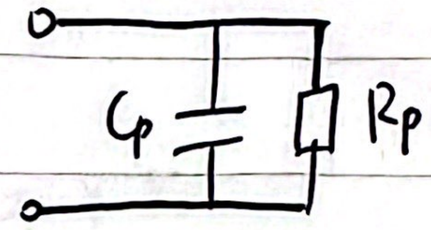
what if $R_L < R_s$?

思路3：谐振角度 （解决）

总结口诀：并大串小 Q 相等

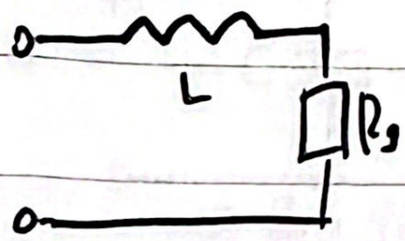


$Q = \frac{\text{串联电抗}}{\text{串联电阻}} = \frac{\frac{1}{\omega_r C_s}}{R_s}$

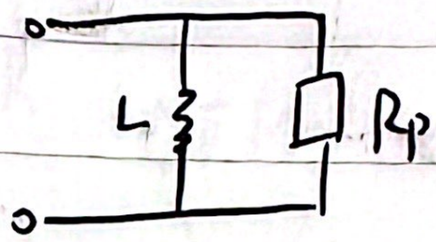


$Q = \frac{\text{并联电纳}}{\text{并联电导}} = \frac{\omega C_p}{G_p}$

$Q = \sqrt{\frac{R_p}{R_s}-1} \quad (R_p > R_s)$

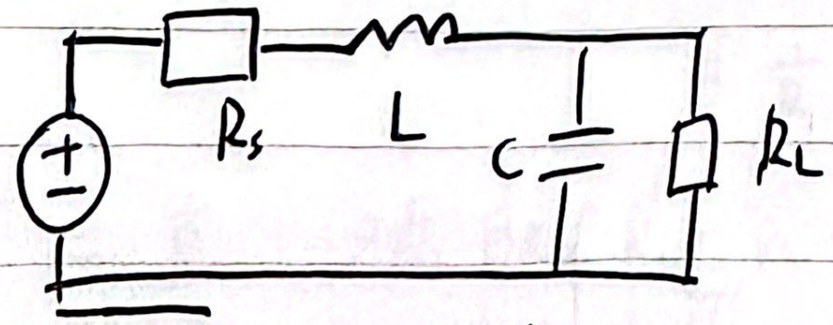


$Q = \frac{\omega L_s}{R_s}$



$Q = \frac{1}{\omega L_p}{G_p}$

例：



$Q = \sqrt{\frac{R_L}{R_s}-1} \qquad Q = \frac{\omega_r L}{R_s} \qquad Q = \frac{\omega_r C}{G_L}$

$\Rightarrow L = \frac{R_s}{\omega_r}Q \qquad C = \frac{G_L}{\omega_r}Q \qquad \Rightarrow L = \cdots \quad C = \cdots$

KOKUYO

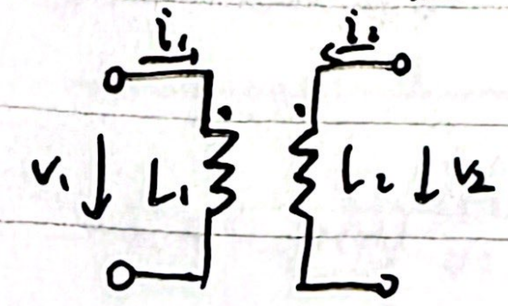
No.

Date

第十四讲 二阶典型应用

一. 互感变压器 (前几页回答)

端口约束方程：



二端口电感

$L_1 = N_1^2 \Xi$ $\quad L_2 = N_2^2 \Xi$

$M = kN_1N_2\Xi = k\sqrt{L_1L_2} = kM_0$

$k = \frac{M}{M_0}$ $\quad k$：耦合系数（磁通链连接百分比） $0 \le k \le 1$

$k=0$：无耦合 $\quad k=1$：全耦合

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \frac{d}{dt} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

二端口电感 $\dot{V} = Z\dot{I}$

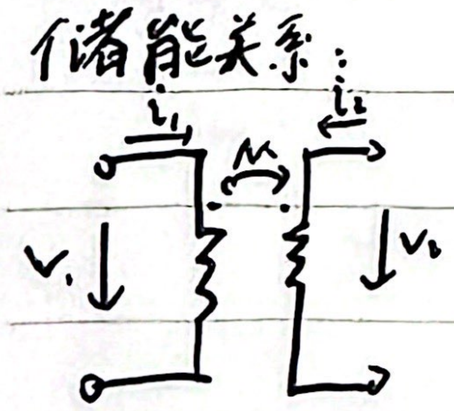
$$\begin{pmatrix} \dot{V}_1 \\ \dot{V}_2 \end{pmatrix} = \begin{pmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{pmatrix} \begin{pmatrix} \dot{I}_1 \\ \dot{I}_2 \end{pmatrix}$$

同名端依一：

$$\begin{pmatrix} \dot{V}_1 \\ \dot{V}_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \begin{pmatrix} \dot{I}_1 \\ \dot{I}_2 \end{pmatrix} \qquad M \begin{cases} \text{同名端} > 0 \\ \text{异名端} < 0 \end{cases}$$

$$-1 \le k = \frac{M}{\sqrt{L_1L_2}} \le 1$$

储能关系：



$$p = v_1i_1 + v_2i_2$$

$$= \left(L_1\frac{d}{dt}i_1 + M\frac{d}{dt}i_2\right)i_1 + \left(L_2\frac{d}{dt}i_2 + M\frac{d}{dt}i_1\right)i_2$$

$$= \frac{1}{2}L_1\frac{d}{dt}i_1^2 + M\frac{d}{dt}i_1i_2 + \frac{1}{2}L_2\frac{d}{dt}i_2^2$$

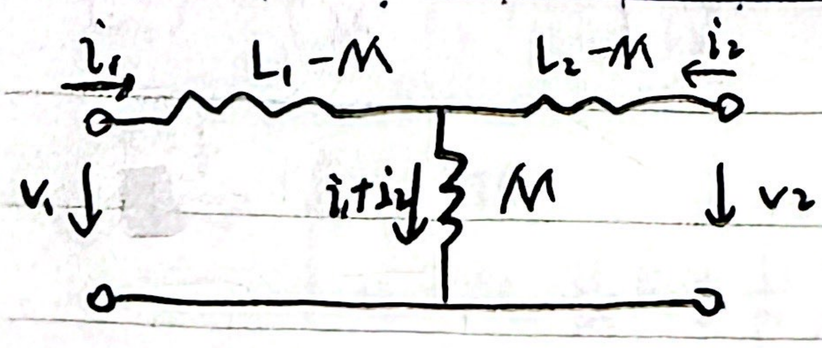
$$E(t) = \int_{-\infty}^{t} p = \frac{1}{2}L_1i_1^2 + Mi_1i_2 + \frac{1}{2}L_2i_2^2 \qquad (\text{设 } E(-\infty)=0)$$

正负不定

No.

Date

T型等效 (↔互易网络2参量) ✓

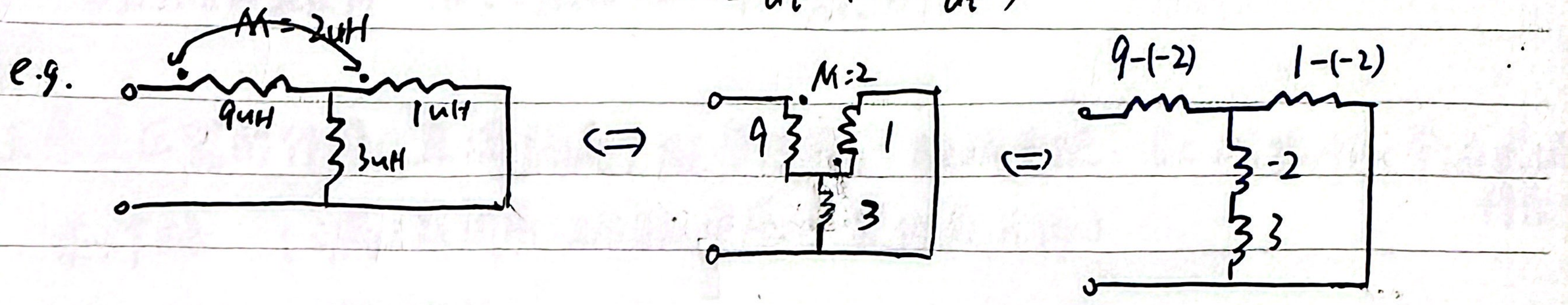


$$V_1 = (L_1 - M)\frac{di_1}{dt} + M\frac{d}{dt}(i_1+i_2) = L_1\frac{di_1}{dt} + M\frac{di_2}{dt}$$

$$V_2 = (L_2 - M)\frac{di_2}{dt} + M\frac{d}{dt}(i_1+i_2) = L_2\frac{di_2}{dt} + M\frac{di_2}{dt}$$

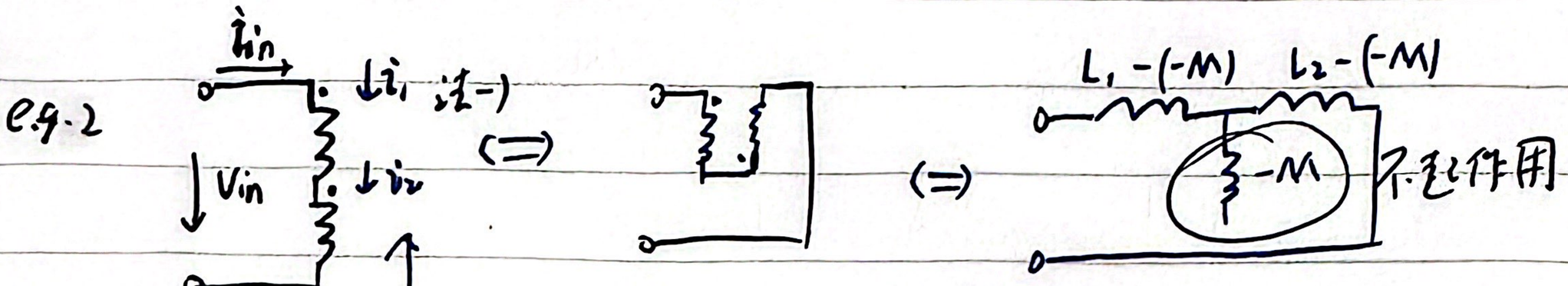
⟺ 互感变压器

e.g.



$$L = 11 + (3 \,//\, (3-2)) = 11.75\ \mu H$$

e.g.2



$L = L_1 + L_2 + 2M$

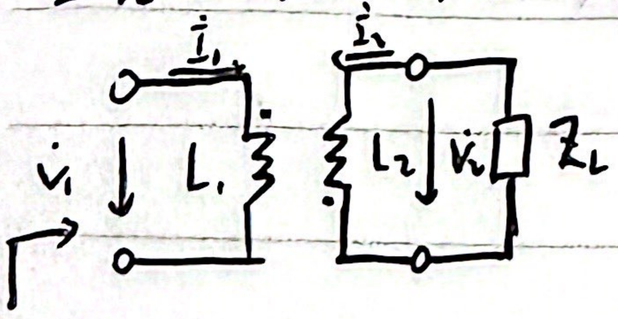
法3 加压求流

$\dot{V}_{in} = \dot{V}_1 + \dot{V}_2 = (j\omega L_1\dot{I}_1 + j\omega M\dot{I}_2) + (j\omega L_2\dot{I}_2 + j\omega M\dot{I}_1)$

$= j\omega(L_1 + L_2 + 2M)\dot{I}$ $(\dot{I}_1 = \dot{I}_2 = \dot{I})$

L

互感的阻抗变换作用



$$\begin{pmatrix}\dot{V}_1\\ \dot{V}_2\end{pmatrix} = \begin{pmatrix}j\omega L_1 & j\omega M\\ j\omega M & j\omega L_2\end{pmatrix}\begin{pmatrix}\dot{I}_1\\ \dot{I}_2\end{pmatrix}$$

由公式 $Z_{in} = Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22}+Z_L}$ $\Rightarrow Z_{in} \overset{\text{设}Z_L=R_L}{=\!=\!=} \frac{k^2}{1+Q_L^2}n^2R_L + j\omega\left(1 - \frac{k^2}{1+Q_L^2}\right)$

$n = \sqrt{\frac{L_1}{L_2}} = \frac{N_1}{N_2}$ $k = \frac{M}{\sqrt{L_1L_2}}$ $Q_L = \frac{R_L}{\omega L_2}$ $\left(Q_L = \frac{\text{并联电阻}}{\text{并联电感}}\right)$

$k=1$, $Q_L=0$ 时: $Z_{in} = n^2R_L$

互感变压器变为理想变压器

KOKUYO

No.

Date

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} L_1 & M \\ M & L_2 \end{pmatrix} \frac{d}{dt} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = M_0 \begin{pmatrix} n & k \\ k & \frac{1}{n} \end{pmatrix} \frac{d}{dt} \begin{pmatrix} i_1 \\ i_2 \end{pmatrix}$$

$\xLeftrightarrow{k=1}$ $\begin{cases} v_1 = nv_2 \\ v_2 = M_0 \frac{d}{dt}(i_1 + \frac{i_2}{n}) \end{cases}$ $\xRightarrow{M_0 \to \infty}$ $i_1 + \frac{i_2}{n} = \text{const } I_0$ $\xRightarrow[\text{的情况}]{\text{排除直流}}$ $I_0 = 0$

$\Rightarrow$ 互感 $\to$ 理想变压器

$\begin{cases} v_1 = nv_2 \\ i_1 = -\frac{1}{n} i_2 \end{cases}$

理想抽象条件：$M_0 = \sqrt{L_1 L_2} \to \infty$ $k = \frac{M_0}{M} \to 1$

互感变压器的作用：直流隔离；能量传输；阻抗变换；与C谐振形成带通滤波特性

乘小因各（漏磁电感，励磁电感；器件及元件）